GeMINI:

Cost Effective, Real time Machine Learning Powered GermaniuM Detector System with Enhanced DIrectioNality CapabIlity

PI: Dr. Fabrizio Napolitano

Laboratori Nazionali di Frascati, INFN

Audizione per bando
Grant Giovani INFN n. 25559/2023

Roma, 13/11/23

Outline of the presentation

Project's goal

State of the art & GeMINI Innovative key-features

GeMINI activities & timeline

Participating Institutions

Financial requests

PI short presentation

Impact

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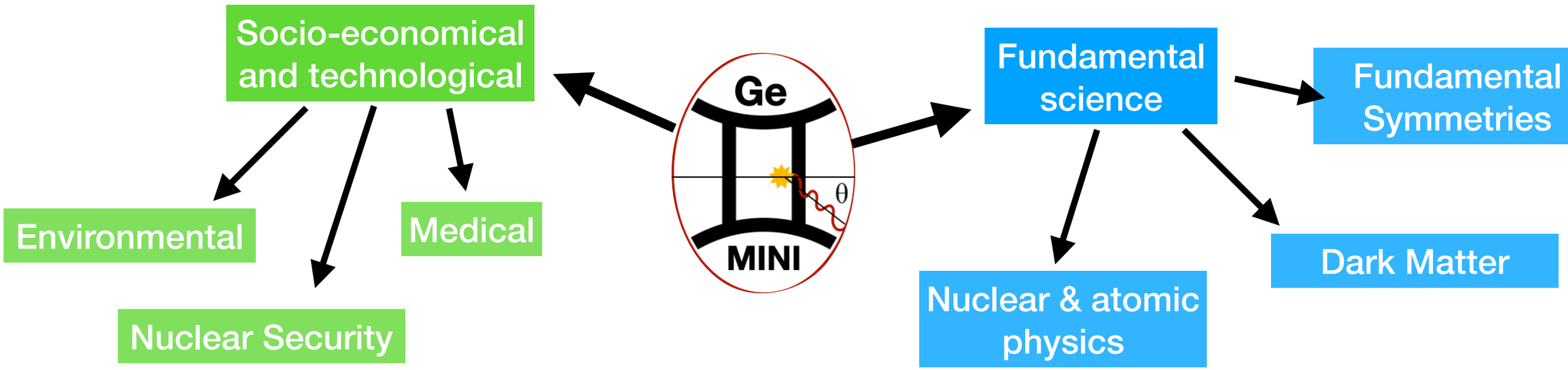
Project's goal

- Germanium detectors, particularly coaxial High Purity Germanium (HPGe) detectors, are frequently used in fundamental science and societal applications, thanks to their unparalleled energy resolution and high detection efficiency in a broad energy range.
- Typical monolithic HPGe detectors **lack** however the ability to measure the direction of the incident radiation
- Incident gamma direction is **crucial** in various fields, including nuclear & atomic physics, dark matter, environmental and medical physics



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GeMINI's goal: develop, characterize and deploy a cost-effective real-time high-rate ML technique with enhanced performance for estimating the direction of gamma radiation implemented on a HPGe detector system, significantly extending the physics and technological reach of these detectors. This will be achieved by a precise study and characterization of the detector's pulse shapes and real-time ML on FPGA.

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State of the art & GeMINI Innovative key-features

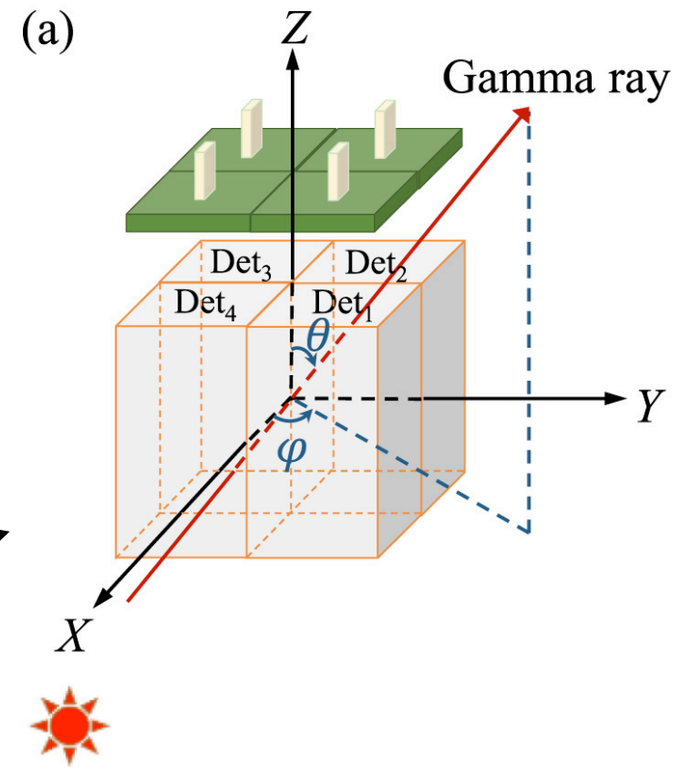
AGATA Segmented HPGe

Segmented readout electrodes

- Better spectroscopic performance
  - But capacitative coupling
- Very expensive in terms of
  - Cost, complexity, operation interpretation
- Relegated to large collaborations

Gamma ray direction

2D CsI(Tl) array [3]

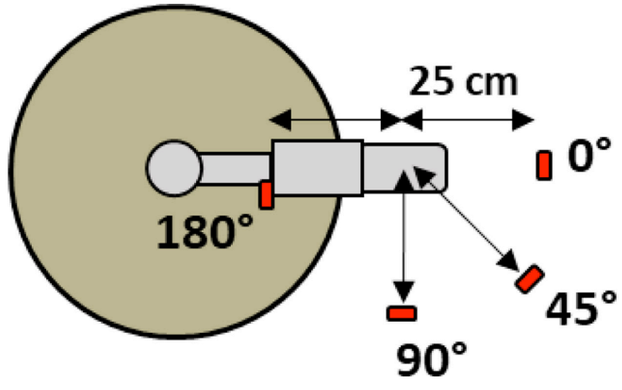


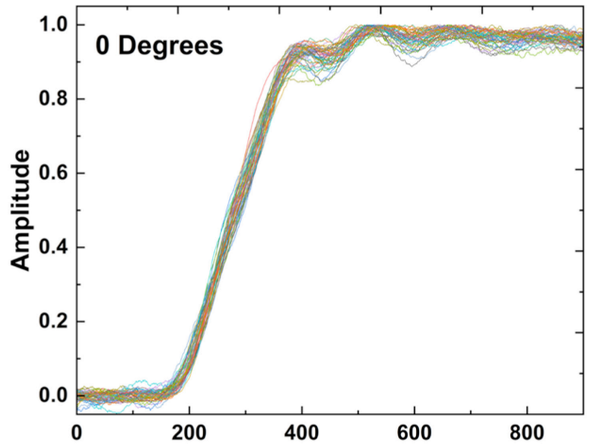
Separate active volumes

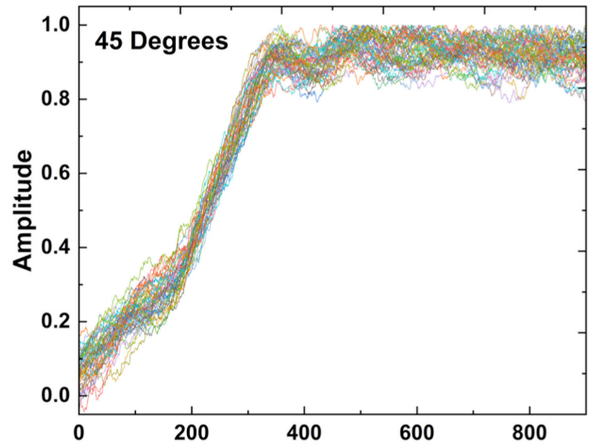
- Can use inexpensive scintillators
- But poor spectroscopic performance

State of the art & GeMINI Innovative key-features

- Idea of inferring the gamma radiation direction from pulse shapes in HPGe's pioneered by Gladen et. al. [12]







- They use a hybrid algorithmic-ML approach, claiming a 70% efficiency in direction classification.

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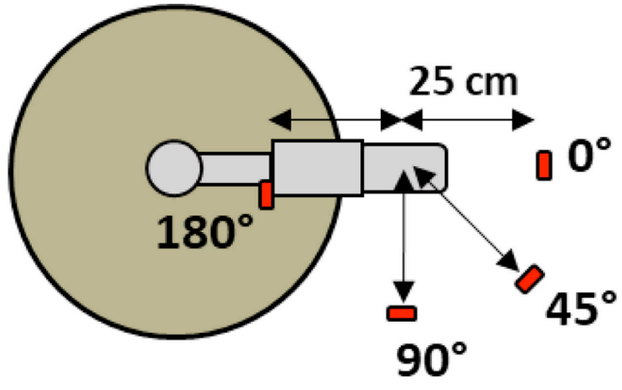

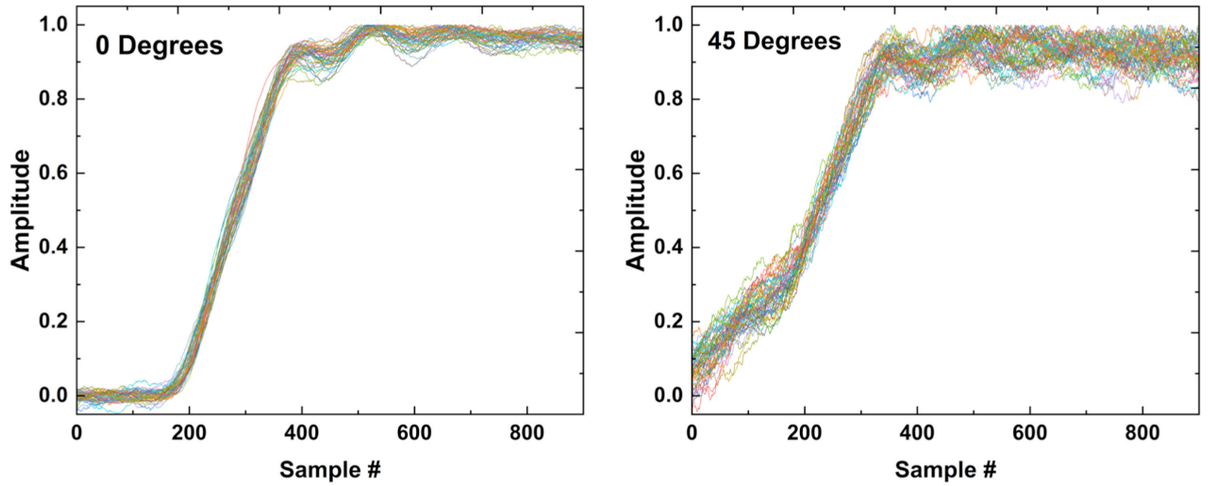

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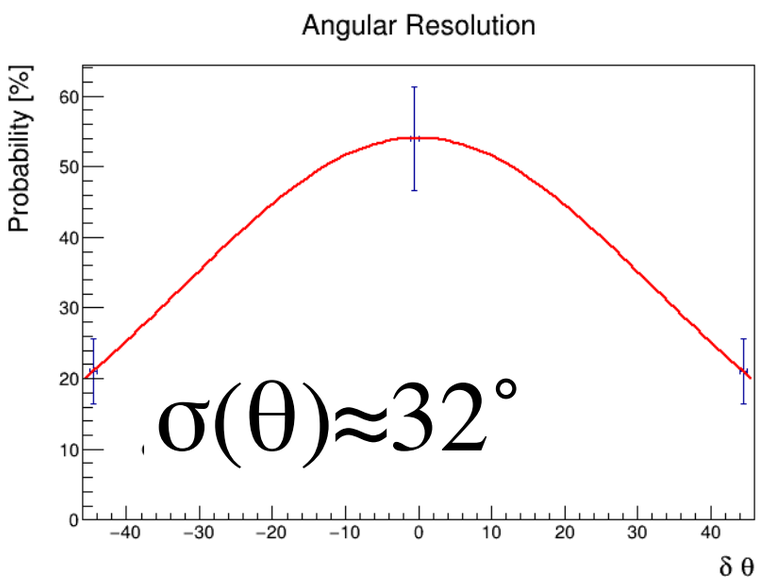


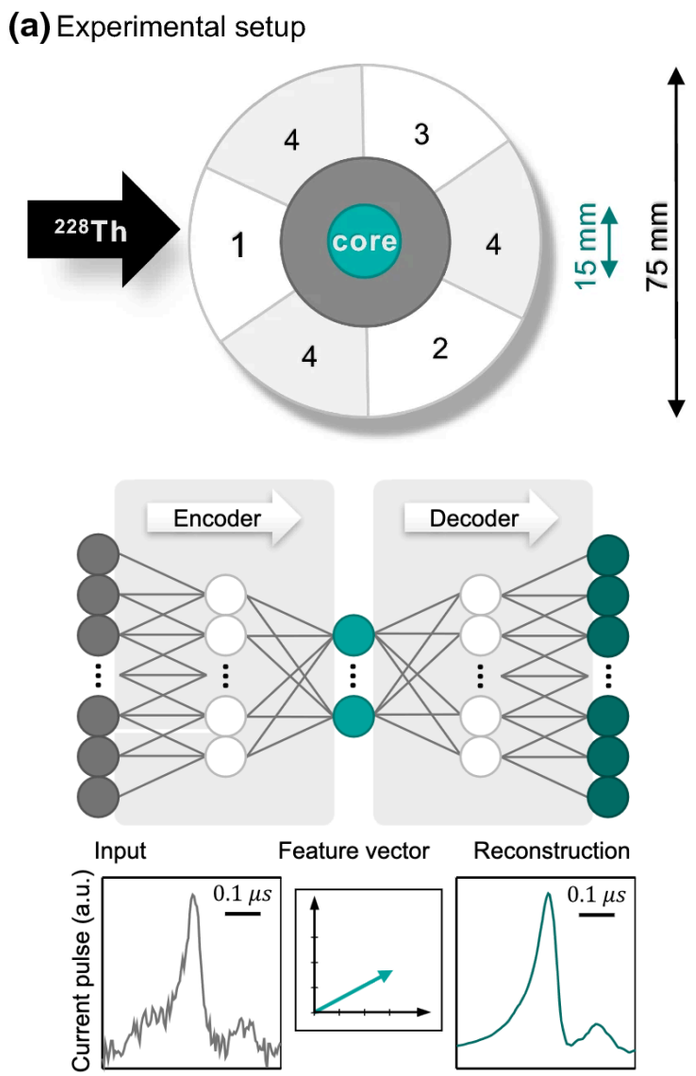
However

- **Missing**: general method to infer direction
  - As a **function** of gamma's **energies**
  - As a function of detector **rates**
- Missing **angular resolution**
- Can exploit modern ML approach to pulse shape analysis (CNN, (V)AEs) [13-15]
- Have to run the analysis **offline**

State of the art & GeMINI Innovative key-features

- **GeMINI** will provide a real-time, ML on-the-edge powered HPGe detector system with directionality capabilities leveraging the state-of-the-art of two worlds:

Underground Physics



Pulse shape analysis with deep CNNs and VAEs

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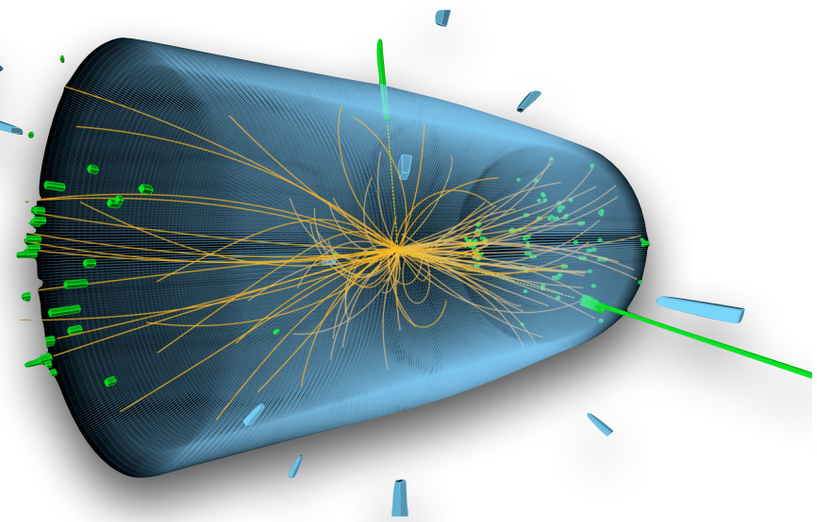

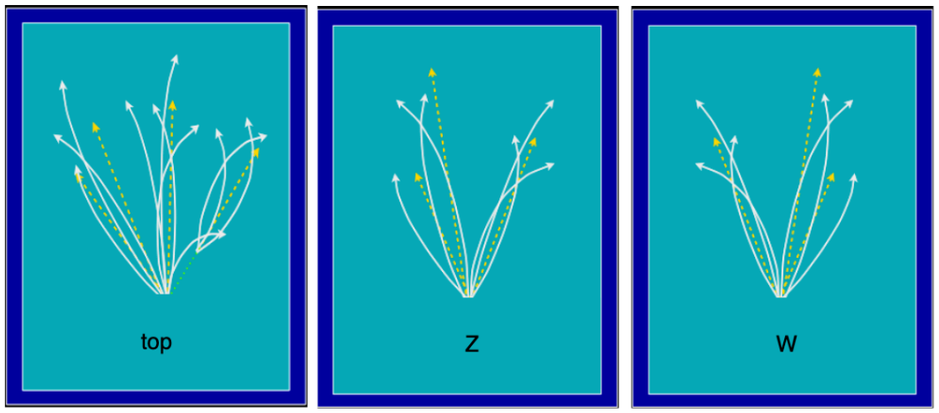

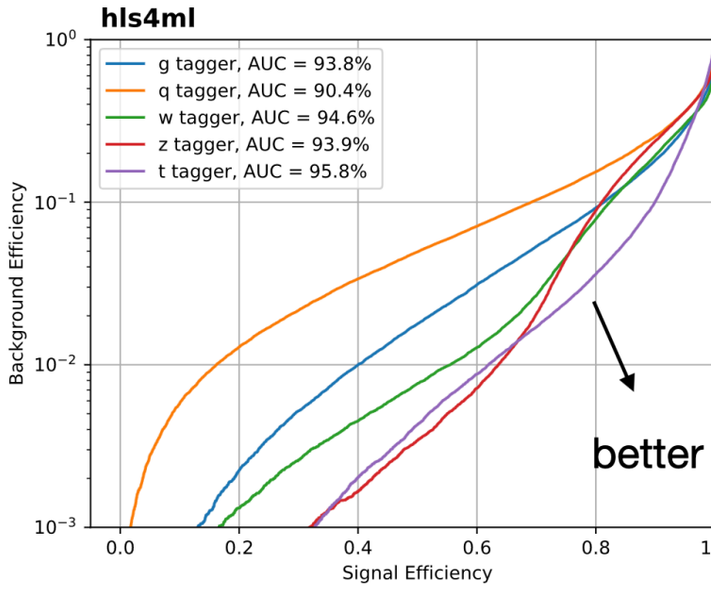

- Leverage already **rich environment** from LHC experiments:
  - ML to FPGA framework hls4ml
  - Quantisation aware model (QKeras Pruning)

High Energy Physics

FPGA for triggering at HL-LHC

hls4ml

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GeMINI activities & timeline

		YEAR 1						YEAR 2					
Work Packages	Task	1-2	3-4	5-6	7-8	9-10	11-12	13-14	15-16	17-18	19-20	21-22	23-24
WP1	T1.1		M1.1	D1.1									
	T1.2			M1.2	D1.2								
	T1.3						M1.3						
	T1.4												
	T1.5												
	T1.6												
	T1.7												D1.3
WP2	T2.1				M2.1								
	T2.2					M2.2							
	T2.3						D2.1						
	T2.4											M2.3	
WP3	T3.1												
	T3.2												
	T3.3							M3.1/D3.1					
	T3.4										M3.2		
WP4	T4.1											M4.1	D4.1
	T4.2												D4.2

WP1: Detector setup and operation

WP2: Front-end and FPGA setup

WP3: Simulation and ML

WP4: Dissemination

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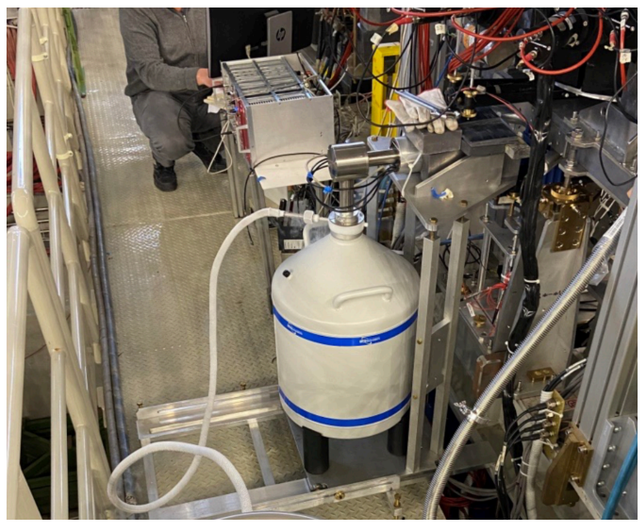
WP1	T1.1		M1.1	D1.1									
	T1.2			M1.2	D1.2								
	T1.3						M1.3						
	T1.4												
	T1.5												
	T1.6												
	T1.7												D1.3

Setup test bench
[M1.1,D1.1]

Data taking with Ba-133 radioactive source
[M1.2,D1.2]

Test full system at DAΦNE or BTF [D1.3]

Data taking at DAΦNE complex [M1.3]
Beam allocated until at least June



WP1: Detector setup and operation

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Work Packages	Task	1-2	3-4	5-6	7-8	9-10	11-12	13-14	15-16	17-18	19-20	21-22	23-24
WP2	T2.1				M2.1								
	T2.2					M2.2							
	T2.3						D2.1						
	T2.4											M2.3	

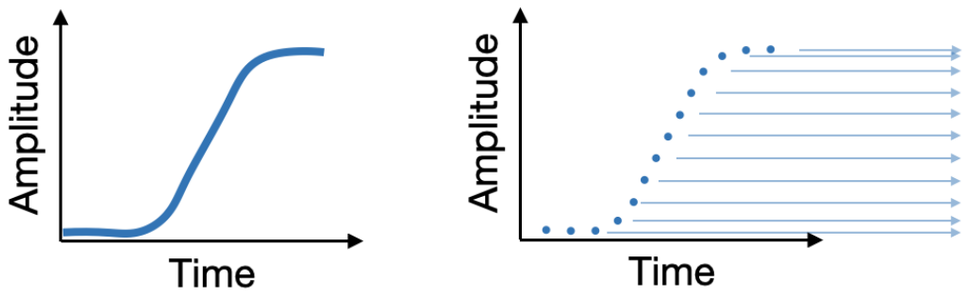
FPGA

Setup test bench

[M2.1]

- Virtex UltraScale+ VU9P (up to 200 kHz)

Full DAQ chain



[M2.2]

- Caen fast digital detector emulator DT5810
- Digitizer DT5725 (250 MS/s, 14 bits) 80MB/s optical link

Test inexpensive FPGA

[M2.3]

- Xilinx Zynq XC7Z020 SoC (up to 5 kHz)

WP2: Front-end and FPGA setup

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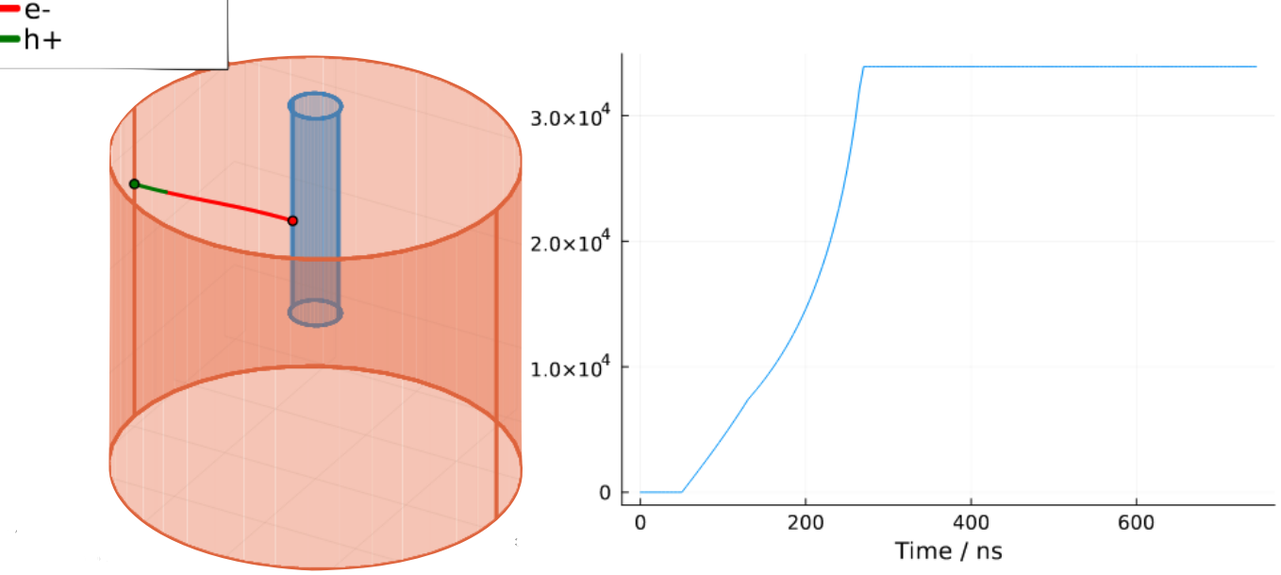
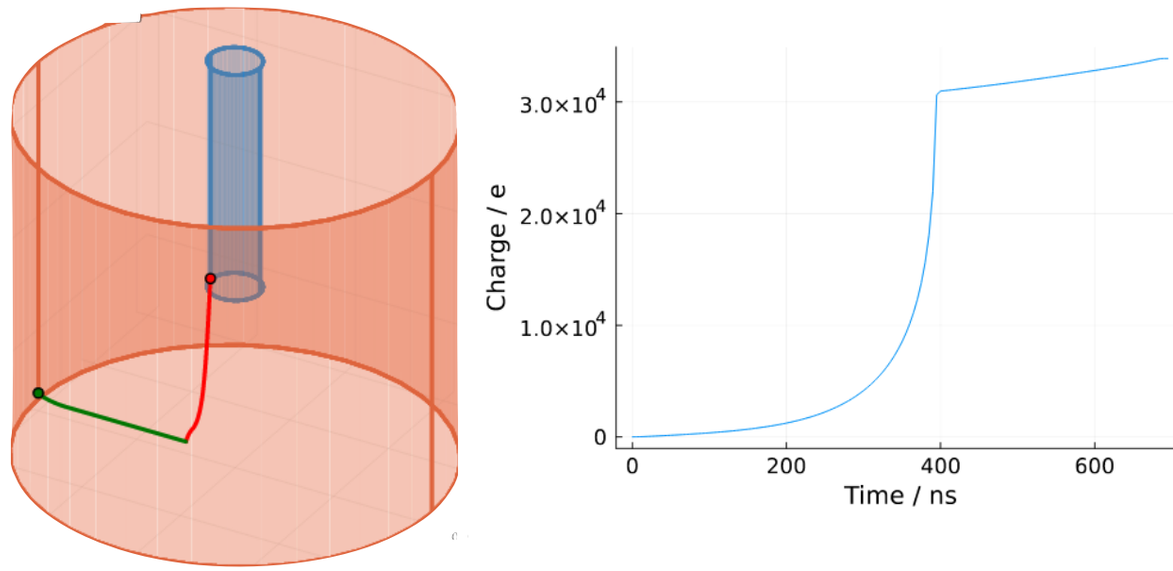
		YEAR 1						YEAR 2					
Work Packages	Task	1-2	3-4	5-6	7-8	9-10	11-12	13-14	15-16	17-18	19-20	21-22	23-24
WP3	T3.1												
	T3.2												
	T3.3							M3.1/D3.1					
	T3.4										M3.2		

ML architecture

Refined ML architecture

Simulations & offline ML





Simulation I made with SolidStateDetector.jl [20] of 100 KeV gamma hitting different regions of a coaxial HPGe

WP3: Simulation and ML

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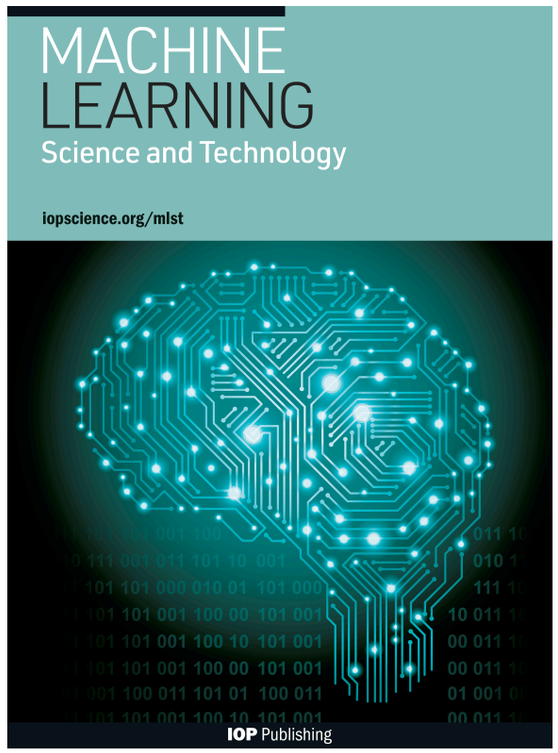
		YEAR 1						YEAR 2					
Work Packages	Task	1-2	3-4	5-6	7-8	9-10	11-12	13-14	15-16	17-18	19-20	21-22	23-24
WP4	T4.1											M4.1	D4.1
	T4.2												D4.2

Peer-review publication



Final Workshop

WP4: Dissemination

GeMINI activities & timeline

- At the end the **GeMINI** project **will deliver** a fully characterized system able to:
  - Measure real-time gamma direction
  - Angular resolution **better than 30°**
  - Operate to up **200 kHz**
  - Demonstrated in radiation hard environments
  - First-time ML on-the-edge for radiation detectors

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Participating Institutions

Coordinating Unit: **LNF-INFN**

Take part in: test rig assembly and test with source and at DAΦNE

University of Zagreb

Take part in: HPGe detector and front-end electronic

University of Zagreb

University of Zurich

Take part in: ML for pulse shape, simulations

Universität Zürich[UZH]

Participating Institutions

Name	Institute	FTE	Age
Fabrizio Napolitano	INFN LNF	**1.0**	32
Alessandro Scordo	INFN LNF	0.15	40
Alberto Clozza	INFN LNF	0.15	63
Massimiliano Bazzi	INFN LNF	0.15	44
Raffaele del Grande	INFN LNF	0.1	33
Elisabetta Pace	INFN LNF	0.2	62
Michelle Galloway	University of Zurich	0.15	51

P.I.

With a total of 1.9 FTE. The percentage of female involvement is 29%.

Groups involved:

Catalina Curceanu	INFN LNF
Laura Baudis	University of Zurich
Damir Bosnar	University of Zagreb

Experience in radiation detectors and front-end

Experienced el. engineer

Experience in HPGe operation

Experience in ML and HPGe pulse shapes

Rich team with decade-long experience dealing with HPGe at LNF and LNGS

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Timeline & Financial requests

Total cost : 214 k€ - external funding 120 k€

Type	Total	Year 1	Year 2
Instrumentation	52 k€	46 k€ (digitizers, FPGA, PC)	6 k€ (digitizers, 2nd FPGA)
Consumables	23 k€	12 k€ (sources, shield, SW license, cables & connectors)	11 k€ (sources, SW license, suspensions)
Missions	15 k€	5 k€ (To Zagreb for R/O Electronic and to Zurch for ML)	10 k€ (To Zagreb for R/O Electronic and to Zurch for ML)
Other costs	4 k€	1 k€ (Maintenance HPGe)	3 k€ (Maintenance HPGe, final workshop)
Total Request	94 k€	64 k€	30 k€

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PI: Fabrizio Napolitano

What qualifies me for GeMINI

- M.Sc. and PhD in particle physics at Universität Heidelberg (DE)
- Now assegnista senior @ INFN-LNF
- Coordinator of 2 laurea thesis
- Author of 350+ PR papers

Experiments:

ATLAS (2015-2020)

- L1Calo on-call
- Jet substructure

PhD: Dark Matter in extreme kinematic regimes

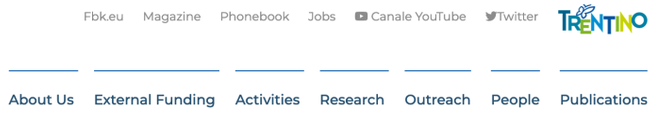
SIDDHARTA-2 (2020-)

- Co-responsible Luminometer
- Run Coordinator
- Resp. for online monitoring
- Commissioning of **SDDs** (installation, test, operation)
- Test of scintillators @ BTF
- **HPGe** operation @ DAΦNE

VIP-2 (2020-)

- **Responsible for data analysis**
- **Machine Learning**
- SDDs operation & monitoring

- Organizing a **workshop** on **Machine Learning** next week in Trento (financed by ECT* after competitive selection)



ALPACA: MODERN ALGORITHMS FOR MACHINE LEARNING AND DATA ANALYSIS: FROM MEDICAL PHYSICS TO RESEARCH WITH ACCELERATORS AND IN UNDERGROUND LABORATORIES

Organizers

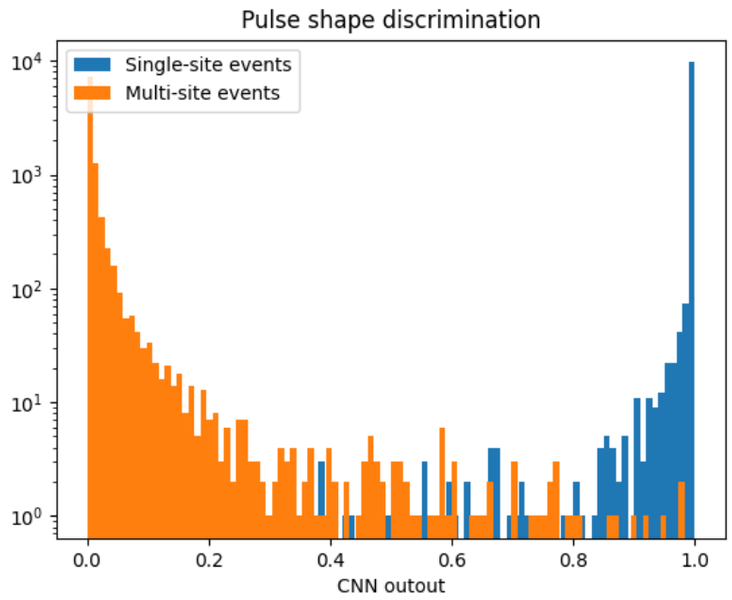
Fabrizio Napolitano (INFN - Laboratori Nazionali di Frascati, Italy)

Fabrizio.Napolitano@lnf.infn.it

PI: Fabrizio Napolitano

What qualifies me for GeMINI

- **Experience** with ML-based pulse shape discrimination of Germanium detector (GeMPI) at LNGS
- Experience with development of novel ML approaches



- **Experience with radiation detectors** (kaonic atoms and foundation of QM):

Measurement Science and Technology

Novel Machine Learning and Differentiable Programming techniques applied to the VIP-2 underground experiment

Fabrizio Napolitano[1], M Bazzi[2], M Bragadireanu[3], Michael Cargnelli[4], A Clozza[5], Luca De Paolis[2], Raffaele Del Grande[6], Carlo Fiorini[7], C Guaraldo[2], M Iliescu[8]

+ Show full author list

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- Germanium (HPGe and BEGe)
  - Operation and monitoring
  - Search for new physics

- Silicon Drift Detectors (SDD):
  - Installation and characterization
  - Front-end
  - Kaonic atoms measurements

PHYSICAL REVIEW LETTERS

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Strongest Atomic Physics Bounds on Noncommutative Quantum Gravity Models

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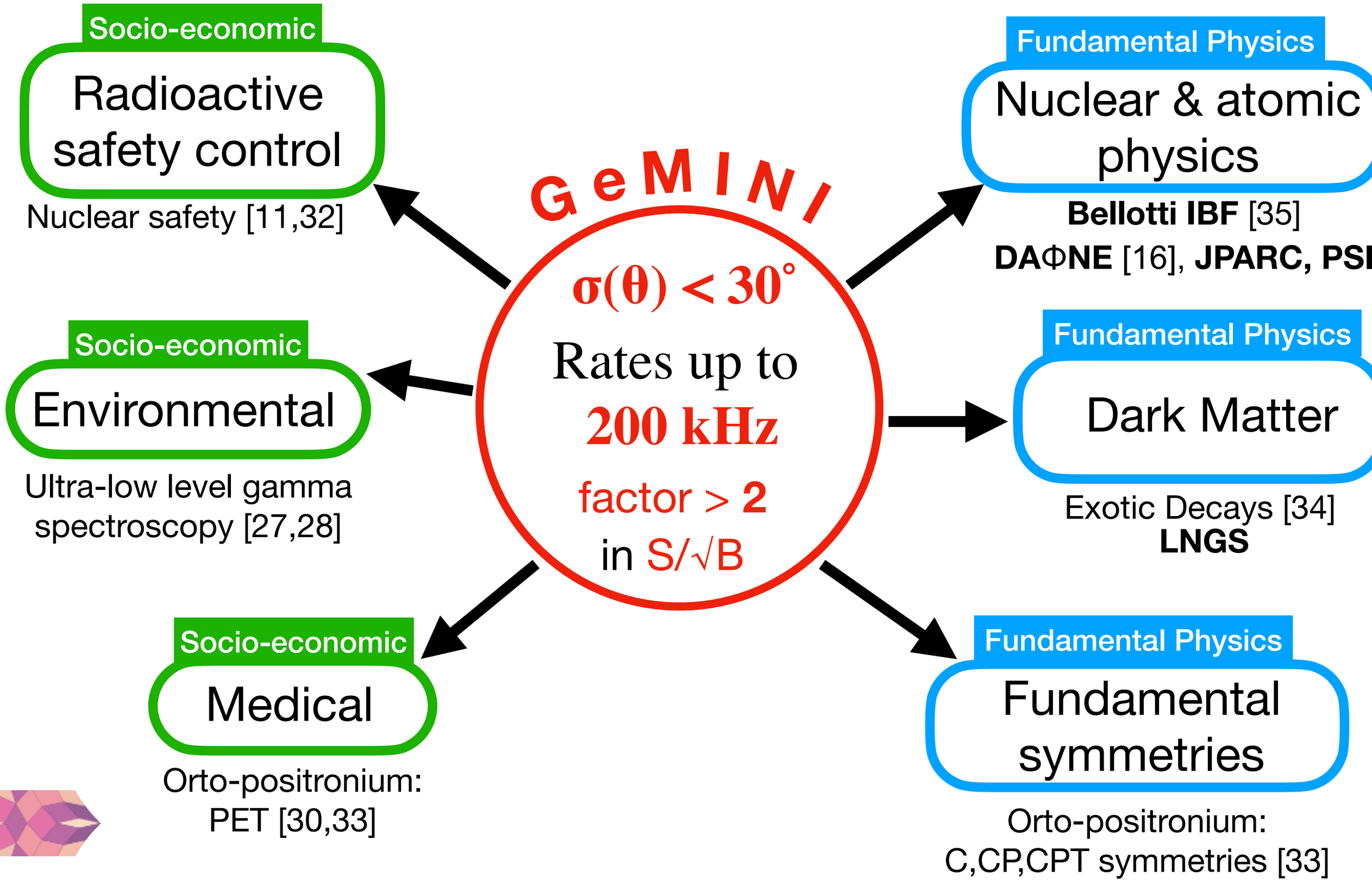
GeMINI
σ(θ) < 30°
Rates up to
200 kHz
factor > 2
in S/√B
Socio-economic
Radioactive safety control
Nuclear safety [11,32]
Socio-economic
Environmental
Ultra-low level gamma spectroscopy [27,28]
Socio-economic
Medical
Orto-positronium: PET [30,33]
Fundamental Physics
Nuclear & atomic physics
Bellotti IBF [35]
DAΦNE [16], JPARC, PSI
Fundamental Physics
Dark Matter
Exotic Decays [34]
LNGS
Fundamental Physics
Fundamental symmetries
Orto-positronium: C,CP,CPT symmetries [33]

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HIGHLIGHTS

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$\sigma(\theta) < 30^{\circ}$

Rates up to

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factor > **2**

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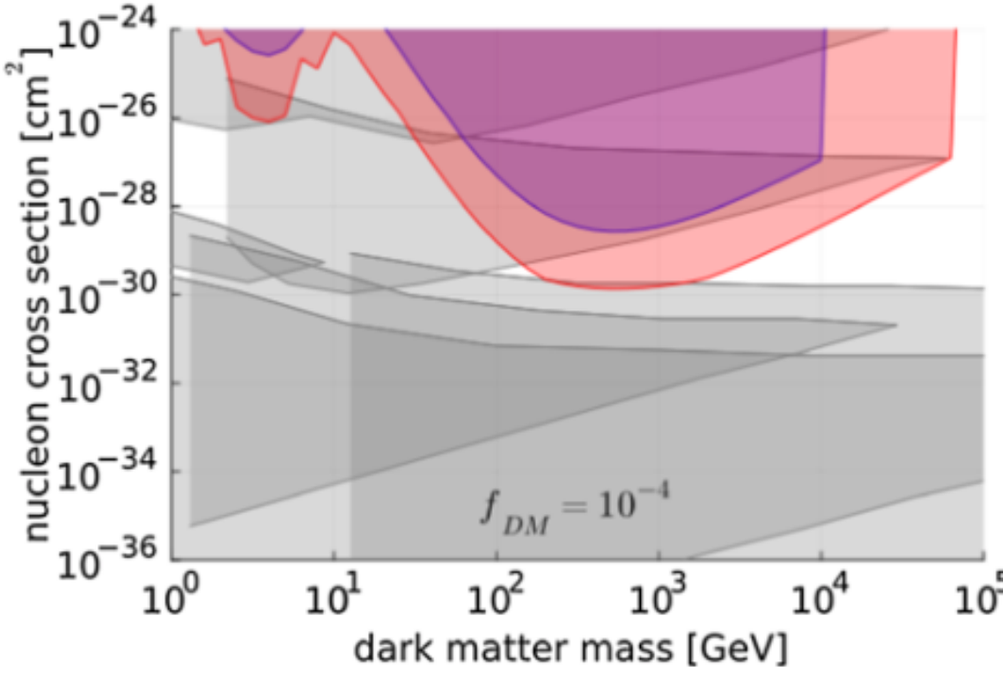

Search for the decay of 180m Ta with an ultra-low-background HPGe detector

Impact

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HIGHLIGHTS

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Bellotti IBF [35]
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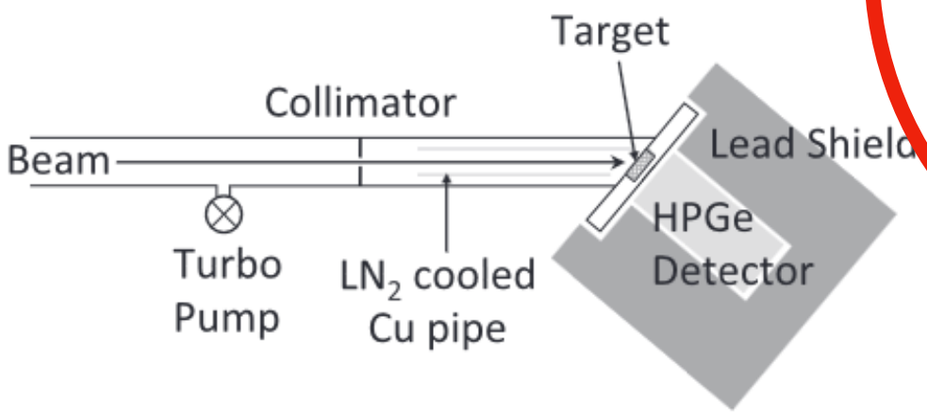

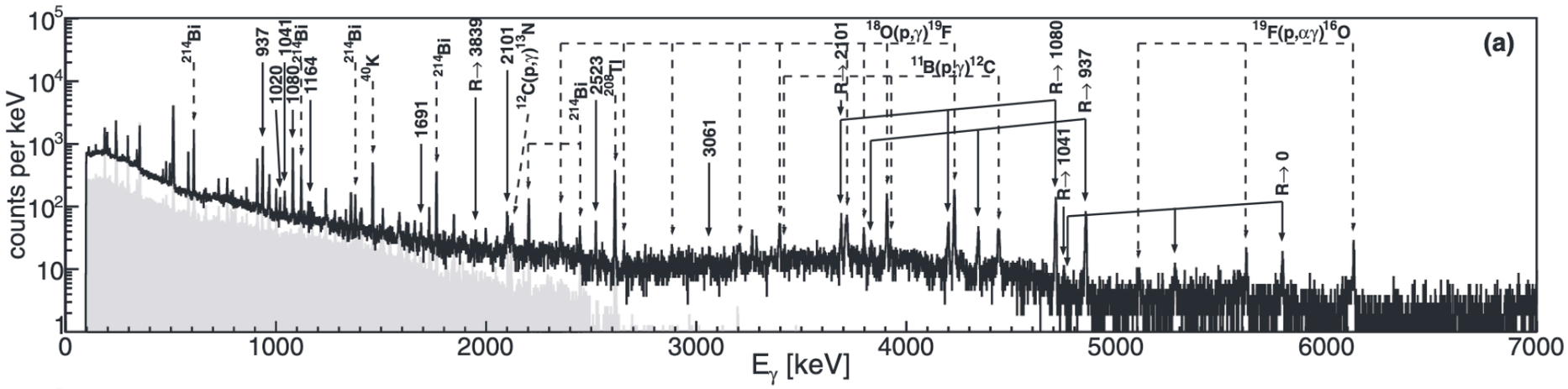

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Socio-economic

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Nuclear safety [11,32]

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Orto-positronium PET [30,33]

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factor > **2** in S/√B

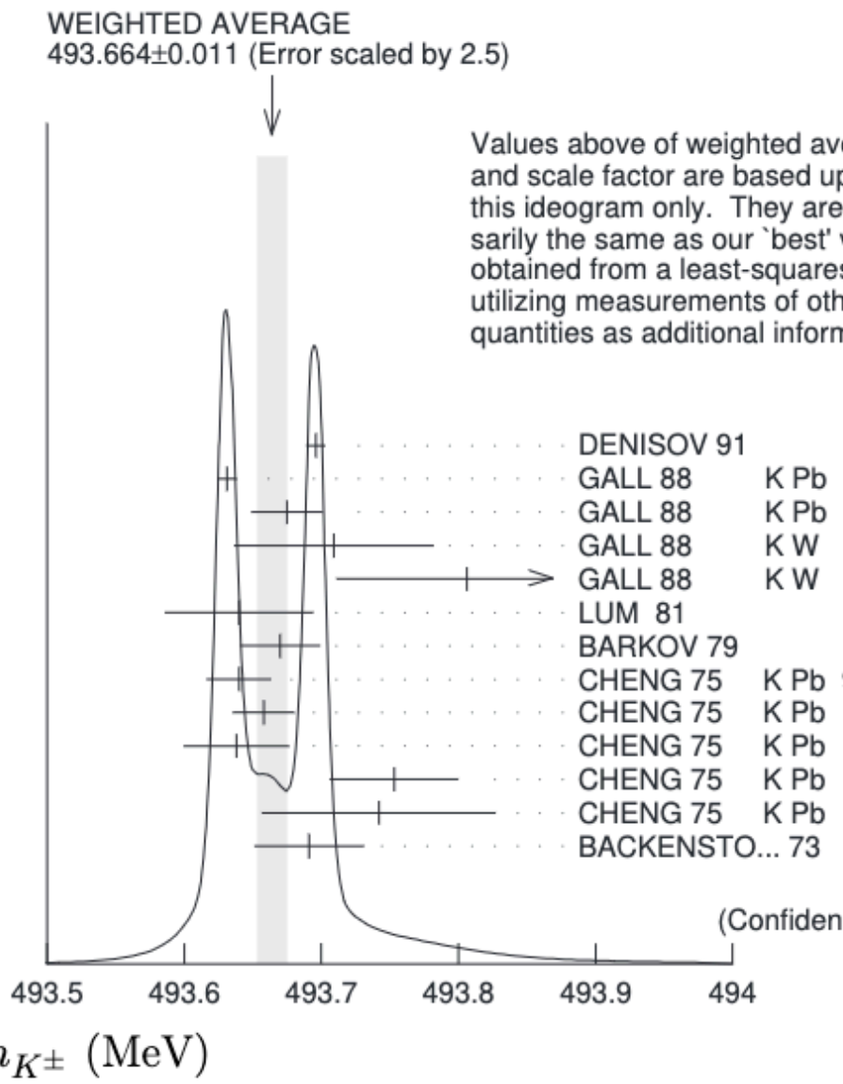
Fundamental Physics

Nuclear & atomic physics

Bellotti IBF [35]

DAΦNE [16], **JPARC, PSI**

Kaon mass measurement at DAΦNE [16] or **JPARC**



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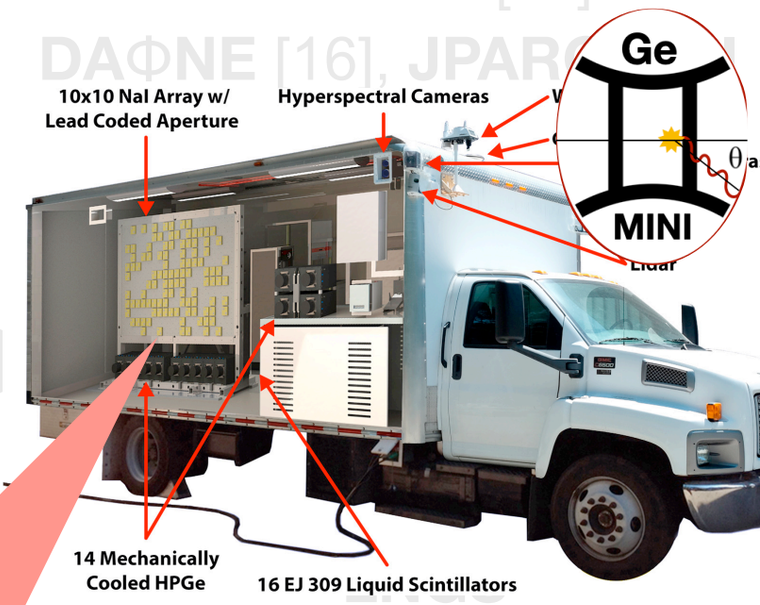
DAΦNE [16], JPARC



Socio-economic

Environmental

Ultra-low level gamma spectroscopy [28]

Medical

Orto-positronium: PET [30,33]

Fundamental Physics

Fundamental symmetries

Orto-positronium: C,CP,CPT symmetries [33]

Concluding Remarks

- **GeMINI** develops and **innovative** ultra fast on the edge **ML** approach on FGPA boards to unlock the true capabilities of HPGe detectors
- To achieve this, we leverage the best of two worlds: HEP and Underground physics
- As PI, I have years of experience in algorithm design, machine learning data analysis, as well as team leading and managerial skills; GeMINI can boost my future career
- GeMINI introduces a new technology, pathfinder to deeper understating of these excellent detectors, inside INFN and in concert with world-leading institution (University of Zurich and University of Zagreb)
- GeMINI will realize a demonstration through a preliminary measurement with radioactive sources and at the LNF accelerator complex (DAΦNE/BTF), **with high scientific impact**
- Within GeMINI, a wealth of experimental efforts will be able to benefit a great increase in sensitivity at a low cost

Many thanks for your attention!

Questions?

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Timeline & Financial requests

DT5810B	€12,392.54	€12,392.54	€0.00	Preventivo 1
“ Warranty	€1,022.07	€1,022.07	€0.00	Preventivo 1
DT5725S	€5,924.10	€0.00	€5,924.10	Preventivo 1
“ Warranty	€488.59	€0.00	€488.59	Preventivo 1
DT5725SB	€7,430.57	€7,430.57	€0.00	Preventivo 1
“ Warranty	€612.83	€612.83	€0.00	Preventivo 1
XpressVUP FPGA	€10,728.08	€10,728.08	€0.00	Preventivo 2
Low cost FPGA	€2,000.00	€2,000.00	€0.00	
Computer DAQ	€2,000.00	€2,000.00	€0.00	
Computer FPGA Bizon	€7,370.22	€7,370.22	€0.00	Preventivo 3
NVIDIA RTX	€2,000.00	€2,000.00	€0.00	
Subtotal Instrumentations	€51,968.99	€45,556.31	€6,412.69	
Cables and connectors	€2,000.00	€1,000.00	€1,000.00	
Mechanical structure	€2,000.00	€2,000.00	€0.00	
Cryogenic nitrogen	€2,000.00	€1,000.00	€1,000.00	
Software licence	€2,000.00	€1,000.00	€1,000.00	
Acustic panels	€2,000.00	€0.00	€2,000.00	
Suspensions	€1,215.12	€1,215.12	€0.00	Preventivo 4
Lead shielding	€2,000.00	€1,000.00	€1,000.00	
Radiative sources	€9,955.20	€5,000.00	€4,955.20	Preventivo 5
Subtotal Consumables	€23,170.32	€12,215.12	€10,955.20	
Missions	€15,000.00	€5,000.00	€10,000.00	
Maintenance HPGe	€2,000.00	€1,000.00	€1,000.00	
Final workshop	€2,000.00	€0.00	€2,000.00	
Total	**€94,139.31**	**€63,771.43**	**€30,367.89**	

Total cost : 94kEur

Project budget:

The total cost of the project is: 214 kEuro

The total request for the project is: **94 kEuro in 2 years**

The remaining part, **120 kEuro** is insured by already existing resources as in-kind contribution from the participating institutions, as:

- - Baltic Scientific Instruments coaxial p-type HPGe, model GCD-30185: 100 kEuro
- - Caen 1CH HV Programmable Power Unit: 5 kEuro
- - Caen N8315 NIM Power Supply 150W: 5 kEuro
- - Simulation software and person power: 10 kEuro

- **Instrumentation: 52 kEuro**

1. DT5810B Dual channel FAST Digital Detect or Emulator with channel correlation: 10.157,84 + 837,76 (warranty) + IVA22
2. DT5725S - 8 Ch. 14 bit 250 MS/s Digitizer, 640 kS/ch: 4855,82 + 400,48 (warranty) + IVA22
3. DT5725SB - 8 Ch. 14 bit 250 MS/s Digitizer, 5.12 MS/ch: 6090,63 + 502,32 (warranty) + IVA22
4. XpressVUP - LP9P Low-profile PCIe Network Processing FPGA Board Featuring 2 x 100G Ethernet: 8793,51 + IVA22
5. Low-cost FPGA board: 2kEuro
6. Computer DAQ: 2 kEuro
7. Computer for ML deployment on the FPGA and simulations: BIZON X5500: 7370,22 Euro
8. NVIDIA RTX series GPU: 2kEuro

Consumables: 23 kEuro

1. Cables and connectors: 2kEuro
2. Mechanical construction: 2kEuro
3. Cryogenic nitrogen: 2kEuro
4. Software licenses: 2kEuro
5. Suspensions for low-vibration base of HPGe (8x): 1.2kEuro
6. Acustic panels for microphonic effect reduction: 2kEuro
7. Lead shielding: 2kEuro
8. Radioactive sources for calibration: 10kEuro. All the radioactive sources will be purchased in collaboration with the LNF radio protection service

Travel expenses: 15 kEuro

Travel expenses for:

- - 2 missions to Zagreb for readout electronics: 2kEuro
- - 4 missions to Zurich for work on ML FPGA model: 6kEuro
- - Test at LNF: 2 persons for 10 days for test at DAFNE: 7kEuro

Other costs:

1. Maintenance HPGe: 2 kEuro
2. Organization of final workshop: 2kEuro

For the first year of the project, the request is of 64 kEuro, the following items being requested (the remaining items are for the second year of the project):

Instrumentation: 46 kEuro

1. DT5810B Dual channel FAST Digital Detect or Emulator with channel correlation: 10.157,84 + 837,76 (warranty) + IVA22
2. DT5725SB - 8 Ch. 14 bit 250 MS/s Digitizer, 5.12 MS/ch: 6090,63 + 502,32 (warranty) + IVA22
3. XpressVUP - LP9P Low-profile PCIe Network Processing FPGA Board Featuring 2 x 100G Ethernet: 8793,51 + IVA22
4. Computer DAQ: 2 kEuro
5. Computer for ML deployment on the FPGA and simulations: BIZON X5500: 7370,22 Euro

Consumables: 12 kEuro

1. Cables and connectors: 1kEuro
2. Mechanical construction: 2kEuro
3. Cryogenic nitrogen: 1kEuro
4. Software license: 1kEuro
5. Suspensions for low-vibration base of HPGe (8x): 1.2kEuro
6. Lead shielding: 1kEuro
7. Radioactive sources: 5kEuro

Travel expenses: 5 kEuro

Travel expenses for:

- - 2 missions to Zagreb for readout electronics: 2kEuro
- - 2 missions to Zurich for ML FPGA model: 3kEuro

Other costs: 1 kEuro

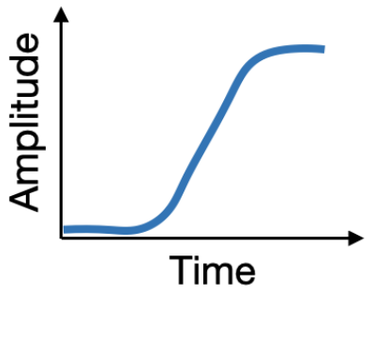
1. Maintenance HPGe: 1 kEuro

GeMINI activities

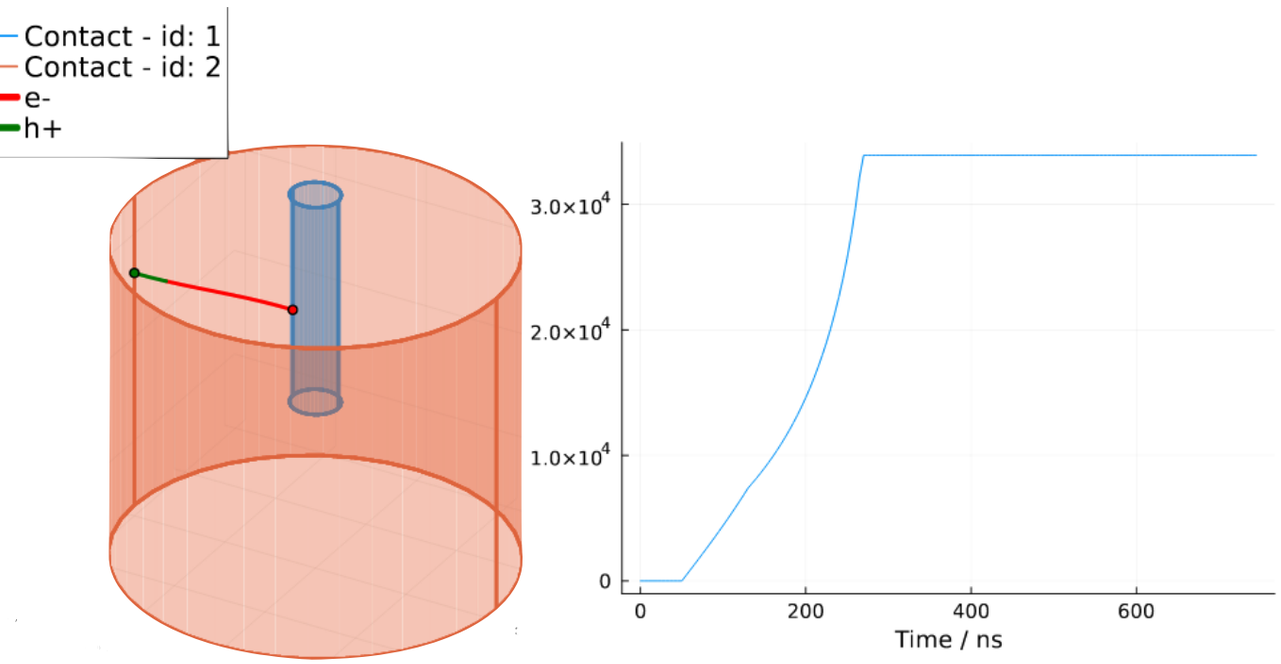
Ortec HPGe and preAmp

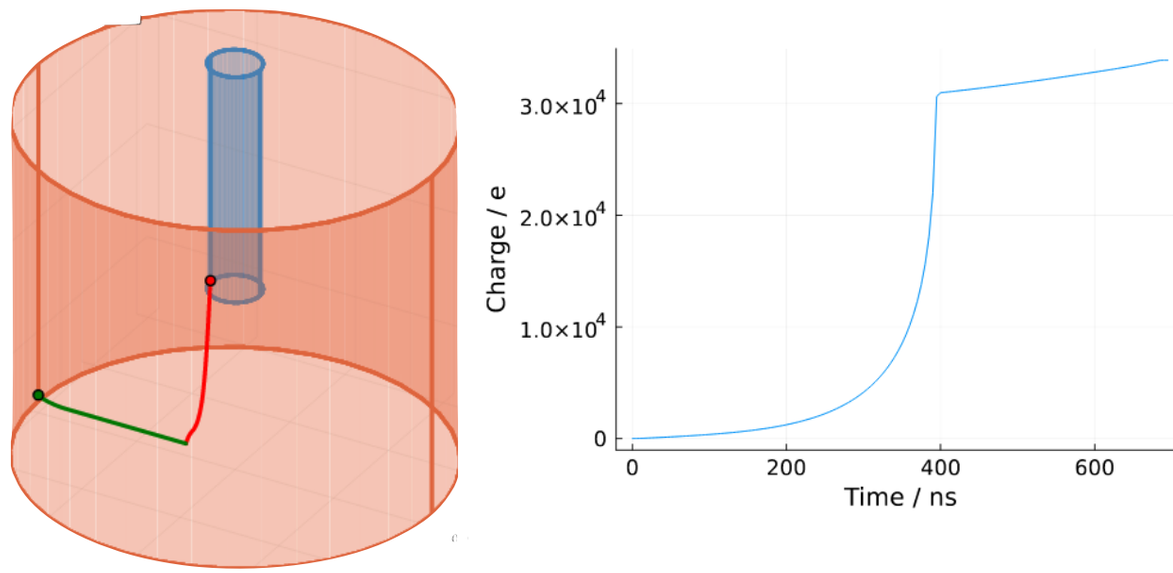
Digitizer

FPGA



- Understanding the underlying physics processes of pulse shapes: simulations





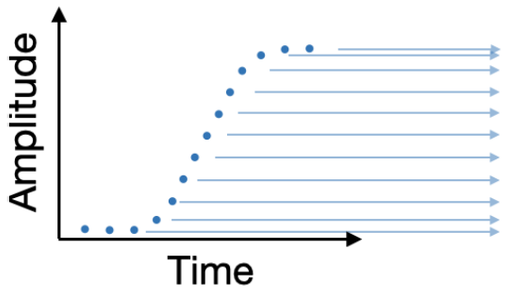
Simulation I made with SolidStateDetector.jl [20] of 100 KeV gamma hitting different regions of a coaxial HPGe

GeMINI activities

Ortec HPGe and preAmp

Digitizer

FPGA



- Train model based on data and simulation on **GPU**
- HPGe **test rig**: Virtex UltraScale+ VU9P (up to 200 kHz)
  - Xilinx Zynq XC7Z020 SoC (up to 5 kHz)
- Caen fast digital detector emulator DT5810
- Digitizer DT5725 (250 MS/s, 14 bits) 80MB/s optical link
- Leverage already **rich environment** from LHC experiments:
  - ML to FPGA framework hls4ml
  - Quantisation aware model (QKeras Pruning)

GeMINI activities

- System characterization at LNF with:
  - Radioactive sources (Ba-133 to benchmark against Gladen et al.) at various angles, spanning the entire operational range of the HPGe detector.
  - At the DAΦNE collider: high count rates and possibly measure of kaonic lead lines

Timeline & Financial requests

		YEAR 1						YEAR 2					
Work Packages	**Task**	**1-2**	**3-4**	**5-6**	**7-8**	**9-10**	**11-12**	**13-14**	**15-16**	**17-18**	**19-20**	**21-22**	**23-24**
WP1	T1.1		M1.1	D1.1									
	T1.2			M1.2	D1.2								
	T1.3						M1.3						
	T1.4												
	T1.5												
	T1.6												
	T1.7												D1.3
WP2	T2.1				M2.1								
	T2.2					M2.2							
	T2.3						D2.1						
	T2.4											M2.3	
WP3	T3.1												
	T3.2												
	T3.3							M3.1/D3.1					
	T3.4										M3.2		
WP4	T4.1											M4.1	D4.1
	T4.2												D4.2

WP1: Detector setup and operation

WP2: Front-end and FPGA setup

WP3: Simulation and ML

WP4: Dissemination

Timeline & Financial requests

WP1

GeMINI is organized in 3 scientific workpackages (WP) targeting the scientific activity of the project. WP4 will coordinate the scientific dissemination.

WP1: Detector setup and operation

Tasks:
T1.1: Set up a BSI coaxial p-type monolithic HPGe detector, from the in-kind contribution of Prof. Bosnar's group. The liquid nitrogen and support required for the setup will be provided by the LNF. Additionally, we will install the Data Acquisition (DAQ) system.
T1.2: Characterization of the HPGe with Zagreb University front-end electronics, in order to check the performance of the detector and as reference point for subsequent studies.
T1.3: With the assistance of the LNF, realization of a testing setup that allowing the precise position of the radioactive source at various angles around the HPGe detector.
T1.4: Data taking campaign at various angles, involving radioactive source that span the entire energy range of the HPGe detector.
T1.5: Data taking campaign at various angles and energies, with first test of the FPGA and demonstration of the ability of the detector system to predict the direction of the gamma radiation.
T1.6: Test of function and limitations using the low cost Xilinx Zynq FPGA board.
T1.7: Test of the detector system at radiation dense environment of the DAFNE collider. Test using a low-vibration base for microphonic noise suppression

Milestones:

M1.1: Acquisition of a calibration spectrum with the HPGe
M1.2: Acquisition of pulses at multiple angles and energies
M1.3: Integrated detector system ready, with FPGA providing real-time data

Deliverables:

D1.1: Report on detector performance of calibration spectrum
D1.2: First report on system capabilities in terms of angular precision and latency
D1.3: Second report on system capabilities in terms of angular precision and latency in radiation dense environment

Timeline & Financial requests

WP2

WP2: Front-end and FPGA setup

Tasks:

T2.1: Setup of a test rig equipped with the Xilinx FPGA, together with an high-end computer with the necessary GPU (NVIDA RTX).
T2.2: Installation of the high-frequency sampling digitizer (CAEN DT 5730/5725) in the acquisition chain of the detector and relative front-end and power supply.
T2.3: Installation of the FPGA onto the front-end, creation of a new DAQ system incorporating the output of the FPGA (directionality prediction).
T2.4: Installation of the second FPGA (low-cost Xilinx Zynq XC7Z020 SoC), necessary rearrangement of the DAQ and front-end.

Milestones:

M2.1: Characterization test of the FPGA, load of the first algorithm
M2.2: Demonstration of the ability of the FPGA to receive the digitized pulse
M2.3: Demonstration of the ability of the second FPGA to receive the digitized pulse

Deliverables:

D2.1: Report on the system layout

Timeline & Financial requests

WP3

WP3: Simulation and ML

Tasks:

T3.1: Realization of simulation campaign with the precise geometry of the HPGe.
T3.2: Development of a ML CNN model for shape discrimination. Training on simulated data. The architecture used in the original study [12] will be employed as a benchmark.
T3.3: Model training on data taking campaign.
T3.4: Development of a more complex architecture based on Transformers, train on simulated and real data.

Milestones:
M3.1: Determination of the angular resolution from data and simulations
M3.2: Prediction from Transformer model

Deliverables:

D3.1: Characterization of the angular resolution at various rates and energies, report.

WP4

WP4: Dissemination

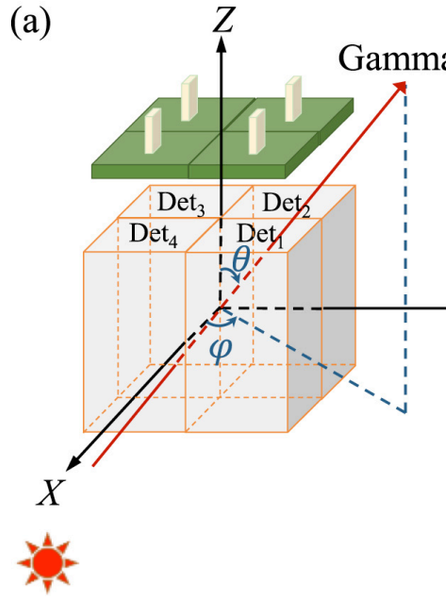
T4.1: Article preparation
T4.2: Project workshop

Milestones:
M4.1: Article circulated to the authors

Deliverables:
D4.1: Article submitted
D4.2: Organization of a final workshop

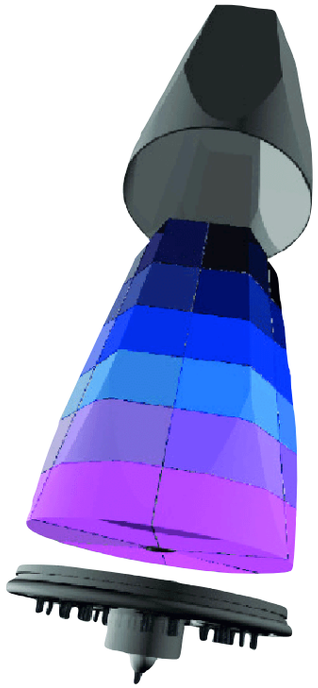
State of the art & GeMINI Innovative key-features

- Determining the gamma ray direction is a challenging task, so far achieved through **multiple detection volumes** or readout channels



2D CsI(Tl) array [3]

- Inexpensive scintillator can be employed in larger numbers in arrays, however they offer poor performances in terms of energy resolution with respect to HPGe, unusable where precise spectral properties have to be determined.



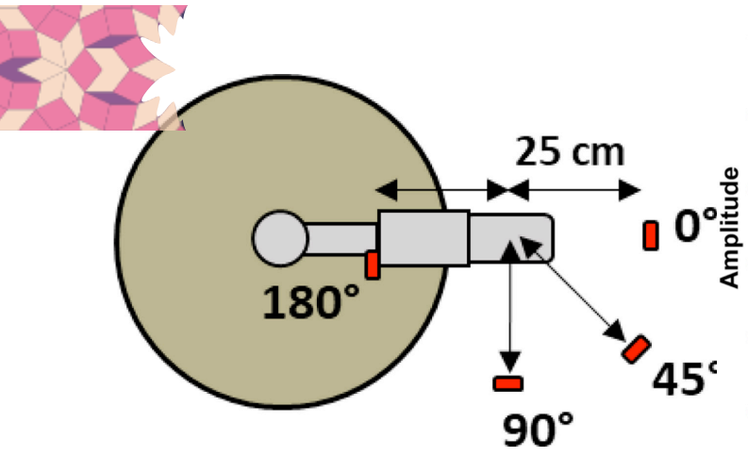
AGATA Segmented HPGe

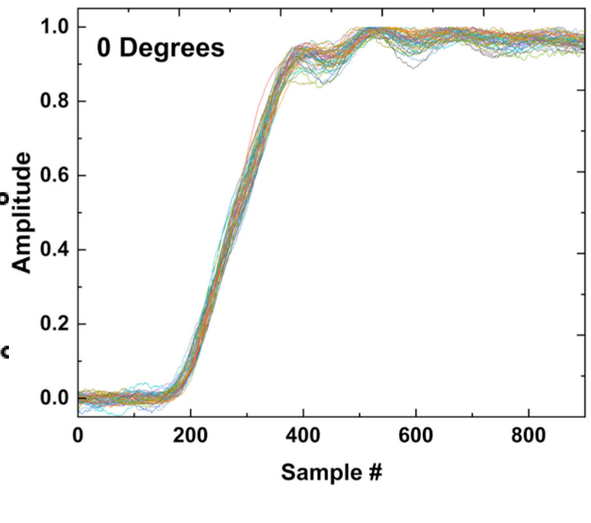
- HPGe detectors can be manufactured with **segmented** geometries of the contact electrodes to obtain spatially resolved readouts.

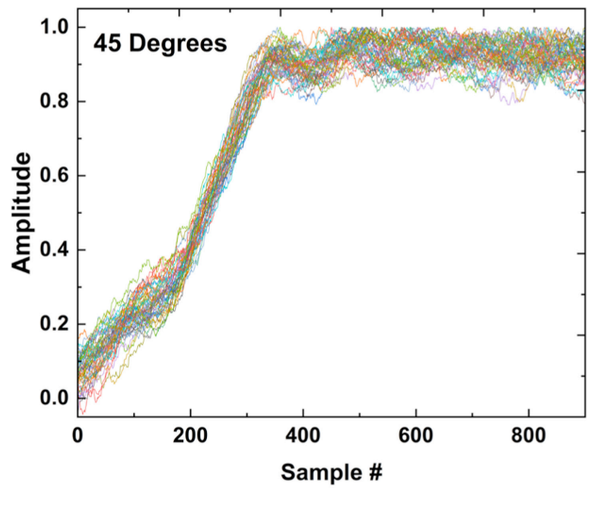
- However, due to **high cost**, **complexity** in setup operation and interpretation relegate them in **large collaboration** (e.g. AGATA/GRETINA/GRETA). Moreover capacitive coupling known to partially degrade spectroscopic response.

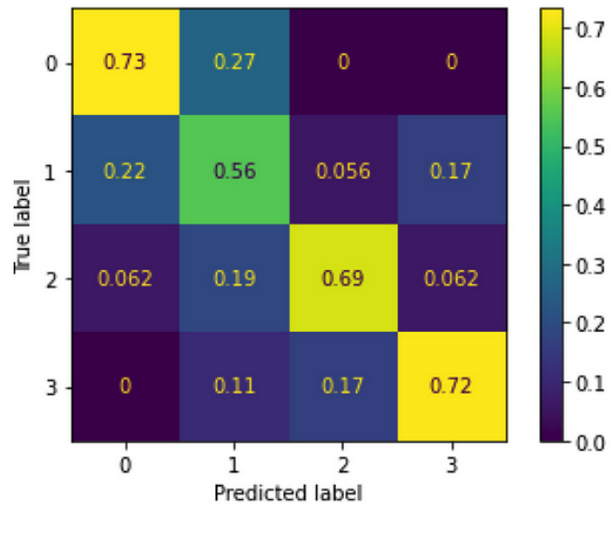
State of the art & GeMINI Innovative key-features

- Idea of inferring the gamma radiation direction from pulse shapes in HPGe's pioneered by Gladen et. al. [12]

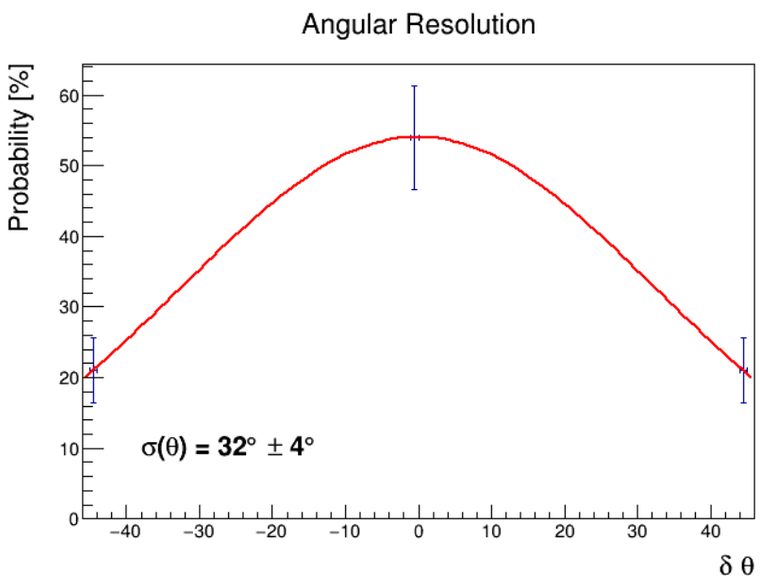








- They use a hybrid algorithmic-ML approach, claiming a 70% efficiency in direction classification.



However

- **Missing**: general method to infer direction
  - As a **function** of gamma's **energies**
  - As a function of detector **rates**
- Missing **angular resolution**
- Can exploit modern ML approach to pulse shape analysis (CNN, (V)AEs) [13-15]
- Have to run the analysis **offline**

PI: Fabrizio Napolitano

What qualifies me for GeMINI

- M.Sc. and PhD in particle physics at Universität Heidelberg (DE)
- Now assegnista senior @ INFN-LNF
- Coordinator of 2 laurea thesis
- Co-responsible for SIDDHARTA-2's Luminometer detector system at DAΦNE
- SIDDHARTA-2 Run Coordinator
- Responsible for online monitoring of SIDDHARTA-2 experiment
- Responsible analysis for VIP-2 experiment at LNGS
- Organizing a **workshop** on **Machine Learning** next week in Trento (financed by ECT* after competitive selection)

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INFN

ALPACA: MODERN ALGORITHMS IN MACHINE LEARNING AND DATA ANALYSIS: FROM MEDICAL PHYSICS TO RESEARCH WITH ACCELERATORS AND IN UNDERGROUND LABORATORIES

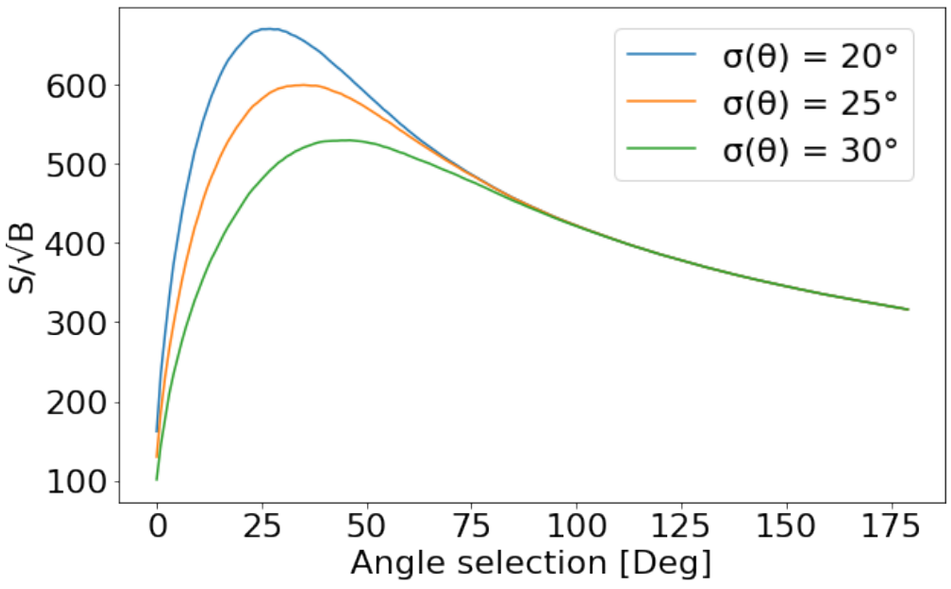
Organizers

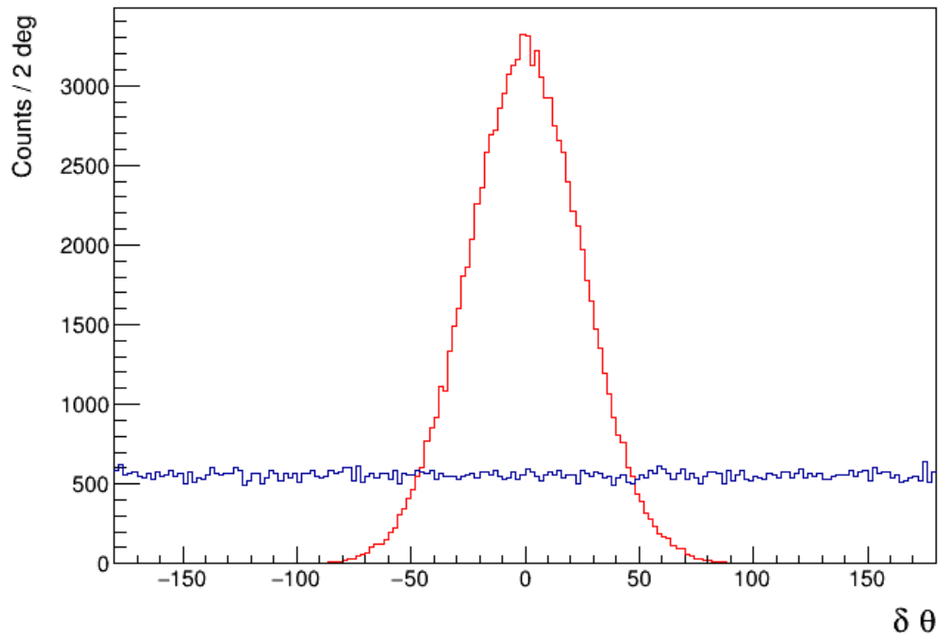
Fabrizio Napolitano (INFN - Laboratori Nazionali di Frascati, Italy)

Fabrizio.Napolitano@lnf.infn.it

Impact

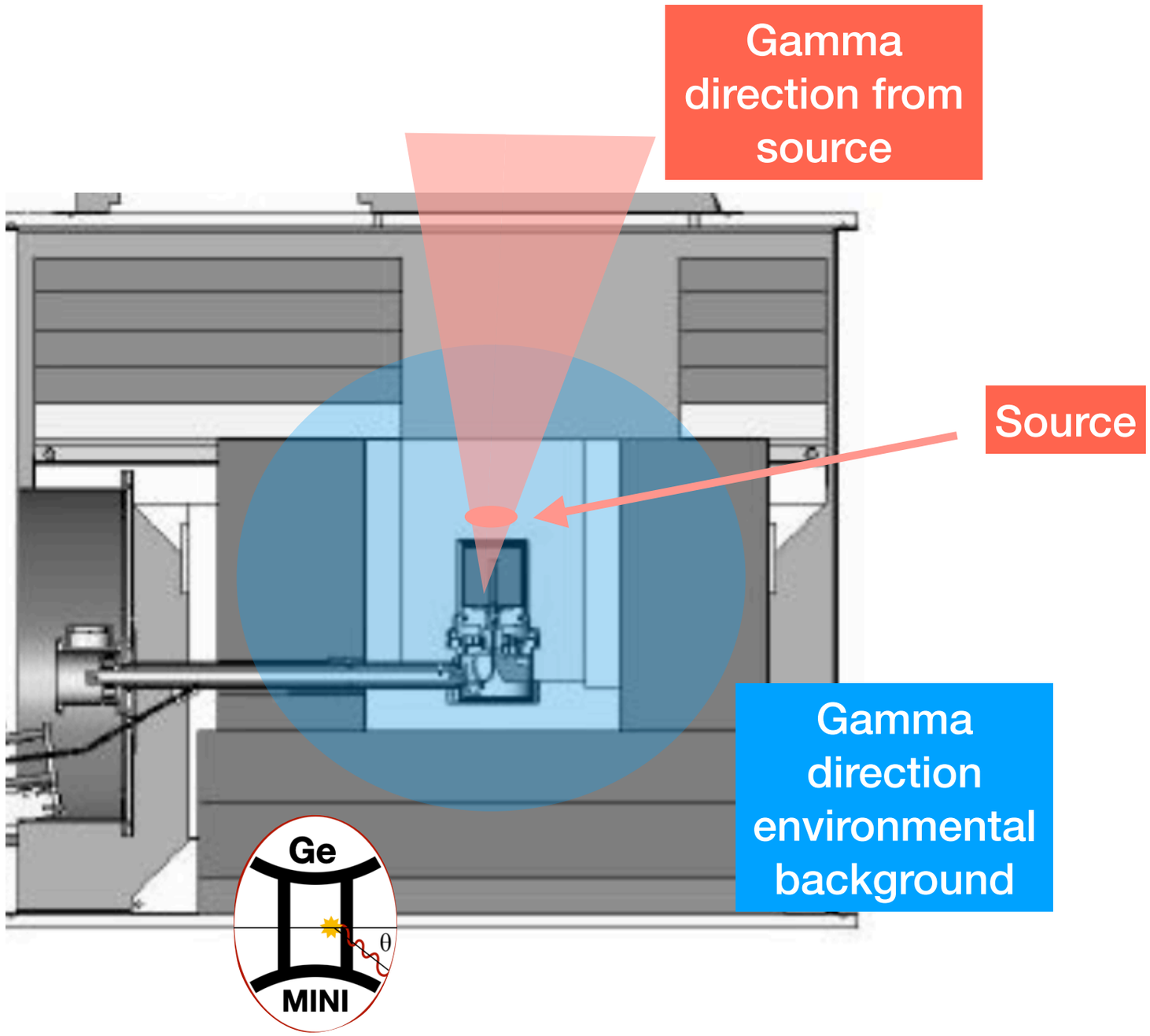
- **GeMINI** will allow gain on S/√B of a factor of two with respect to a system without angular selection
  - Even greater if background predominant from one direction (e.g. beam-induced background)
- Benefit where a HPGe is used to measure a source from a point-like source
  - **DAΦNE**: kaon mass measurements
  - Searches for new physics at LNGS
  - Ultra low level gamma spectroscopy: **STELLA** [27] and Gator [28] facilities
  - High rate, high precision measurement of rare nuclear processes: **Bellotti Ion Beam Facility** (LUNA)
  - Orto-positronium annihilation





Impact

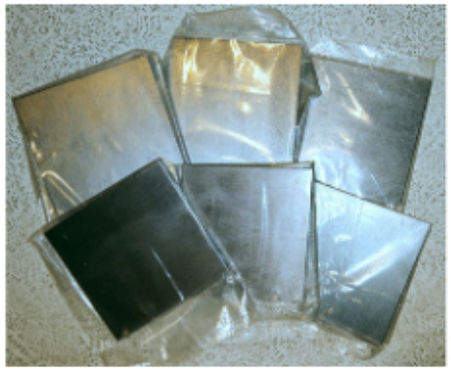
Example: Ultra low level gamma spectroscopy: STELLA [27]

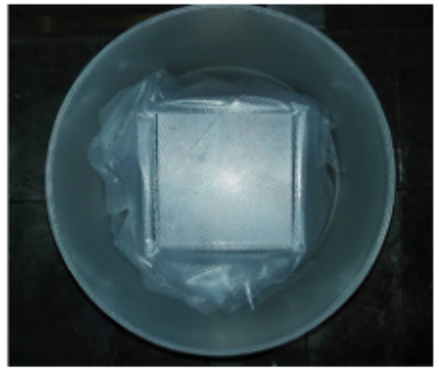


Impact

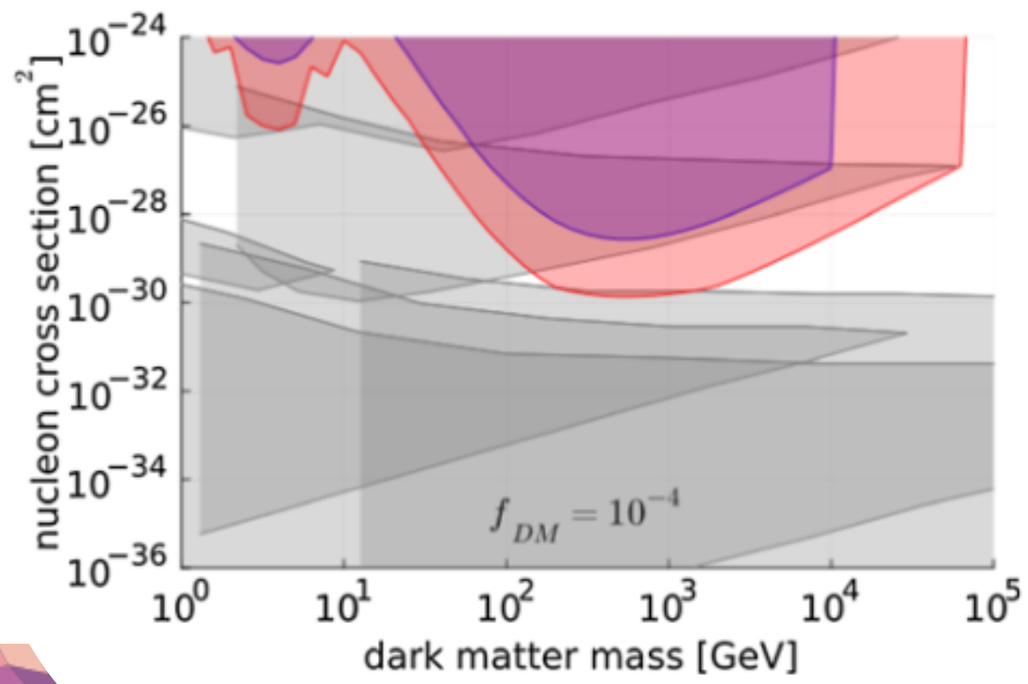
Example: Ultra low level gamma spectroscopy: STELLA [27]

Search for the decay of 180m Ta with an ultra-low-background HPGe detector





Implication to Dark Matter and Beyond the standard model



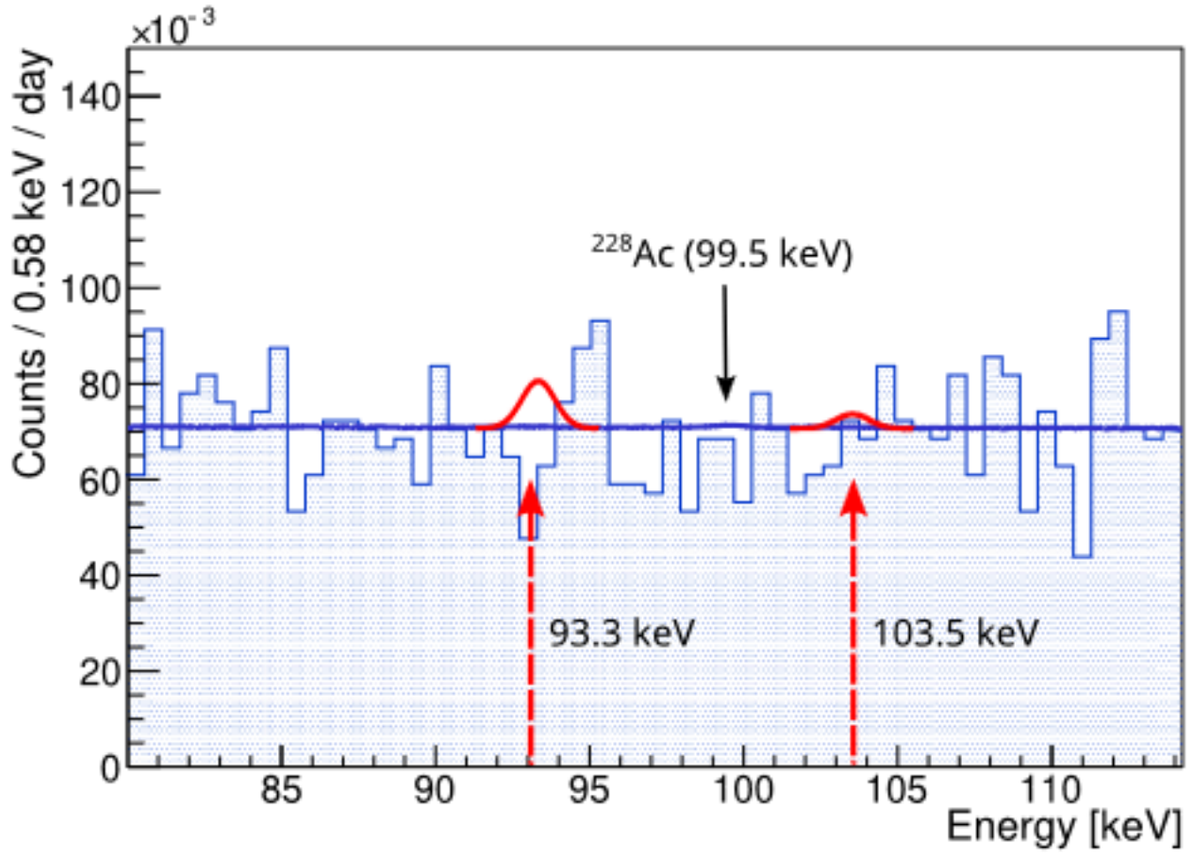
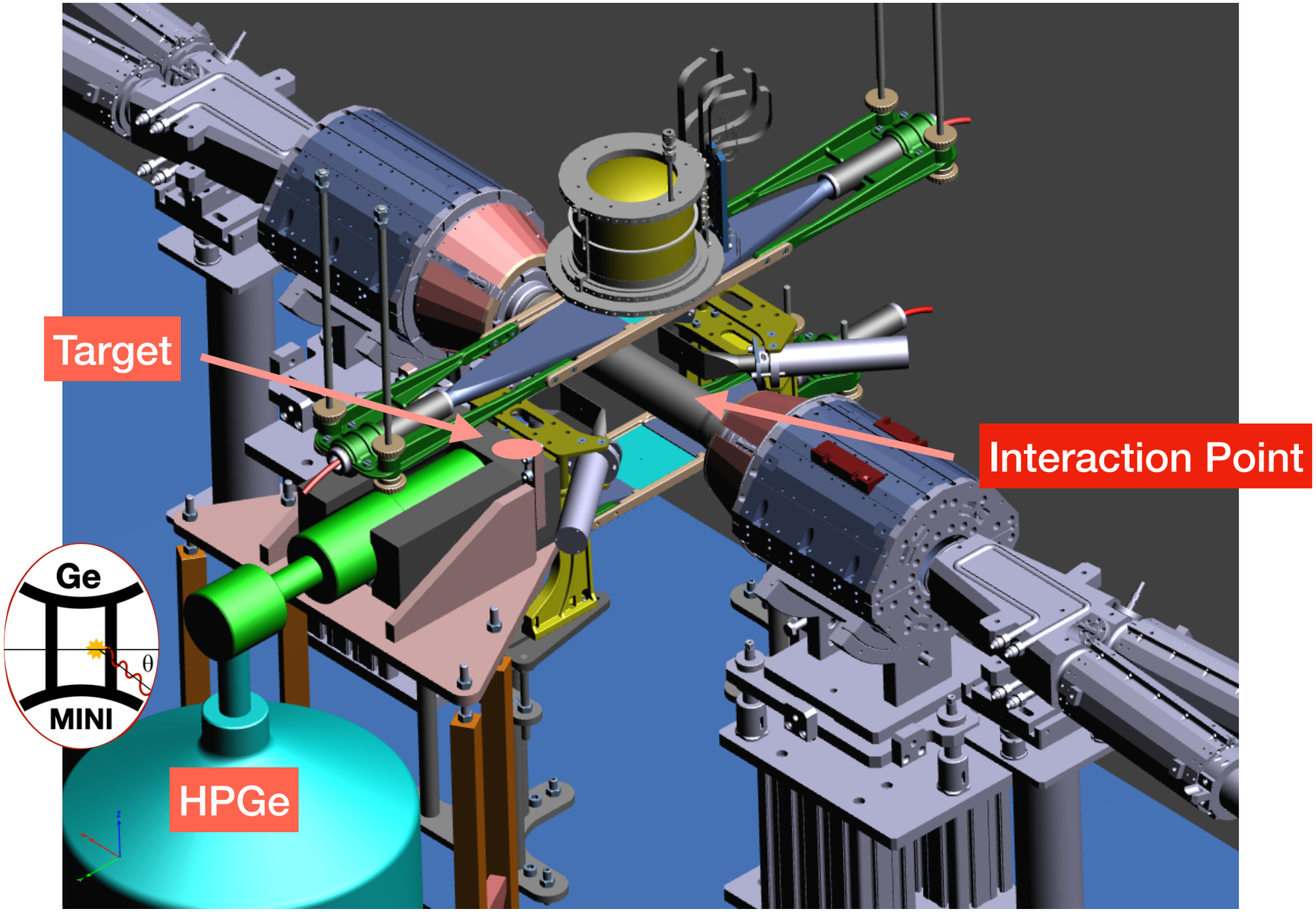


Fig. 6 Fit window for the γ de-excitation/IC decay branch of ^{180m}Ta. The blue line represents the best fit associated to the posterior mode; the red lines correspond to a signal peak with associated yield from the 90% quantile of the inverse-half-life parameter posterior. The background and signal FEPs are indicated by the solid and dashed arrows, respectively

https://doi.org/10.1140/epjc/s10052-023-12087-5

Impact

Example: DAΦNE kaon mass measurement



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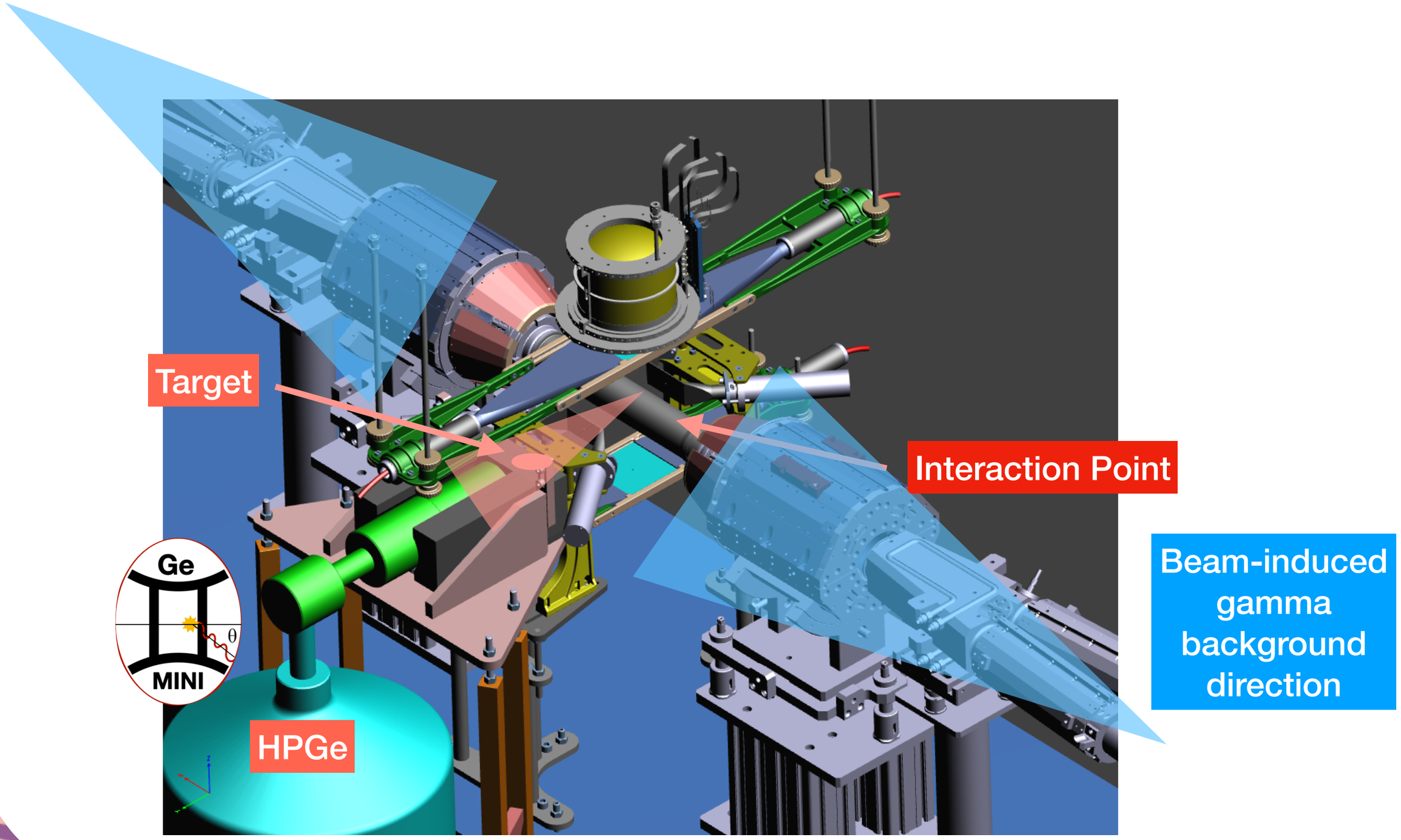
Example: DAΦNE kaon mass measurement



Impact

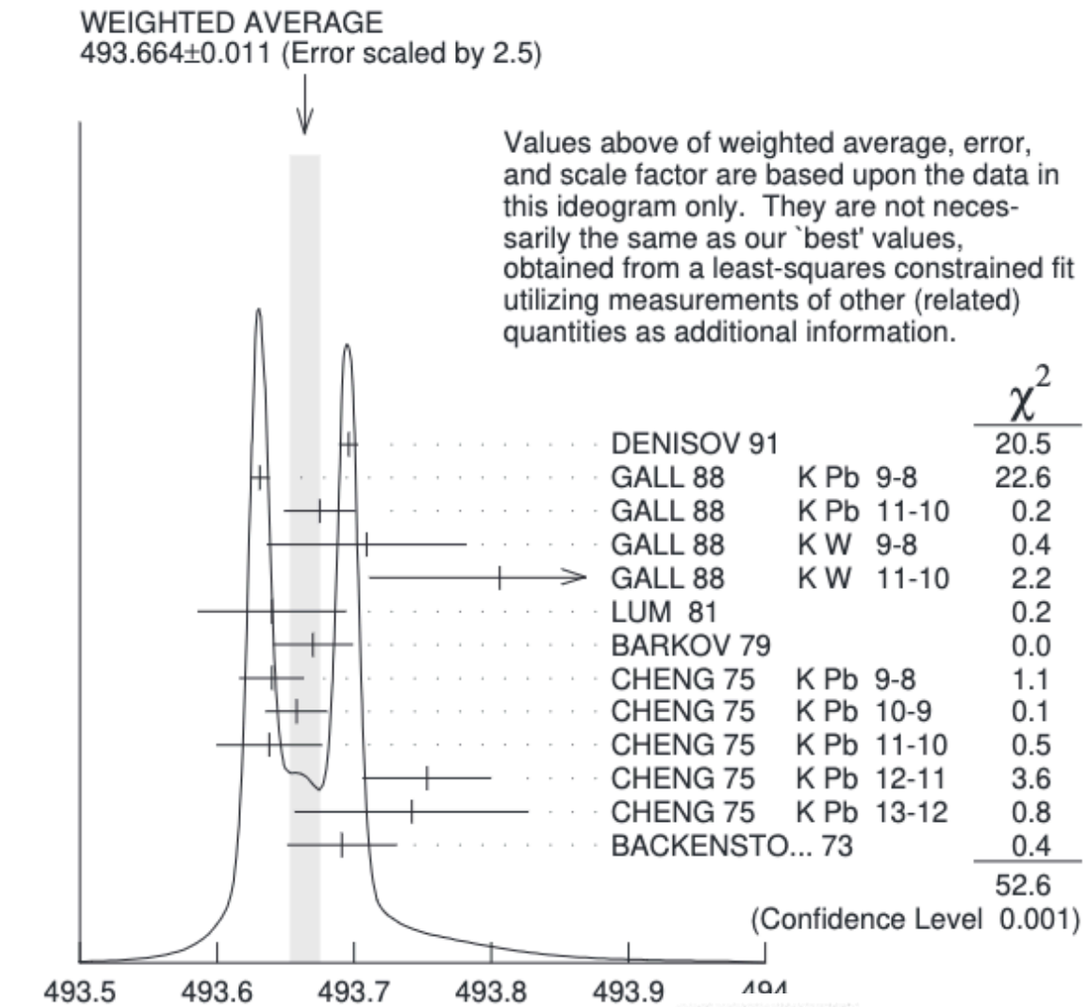
Example: DAΦNE kaon mass measurement

Kaon mass discrepancy

The kaonic Neon measurement to determine the K^- (K^+) mass

Less systematic uncertainty with respect to DENISOV 91 and GALL 88 measurements, thanks to the use of a low Z gas (Ne) target

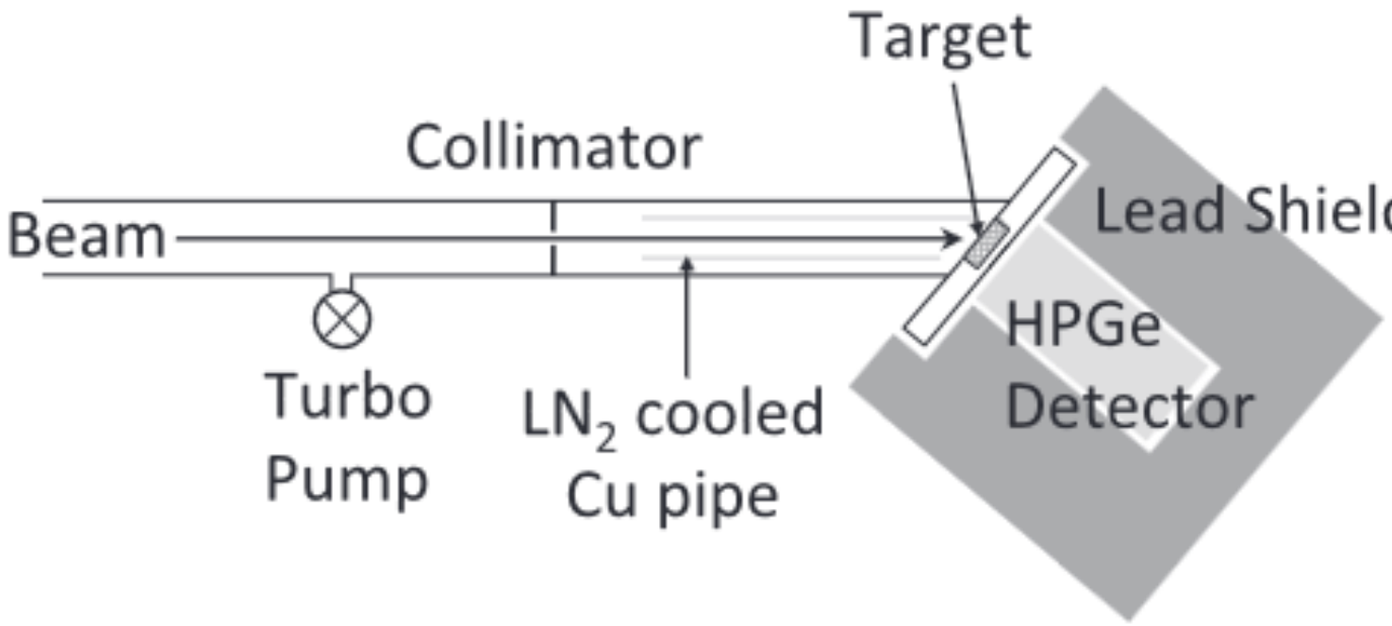
It could solve the kaon mass discrepancy issue → Impact on the charmonium spectrum and on all processes in which charged kaons are involved



VALUE (MeV)	*DOCUMENT ID*		*TECN*	*CHG*
493.677±0.016 OUR FIT Error includes scale factor of 2.8.				
493.677±0.013 OUR AVERAGE Error includes scale factor of 2.4 below.				
493.696±0.007	[1] DENISOV	91	CNTR	–
493.636±0.011	[2] GALL	88	CNTR	–
493.640±0.054	LUM	81	CNTR	–
493.670±0.029	BARKOV	79	EMUL	±
493.657±0.020	[2] CHENG	75	CNTR	–
493.691±0.040	BACKENSTO...	73	CNTR	–

Impact

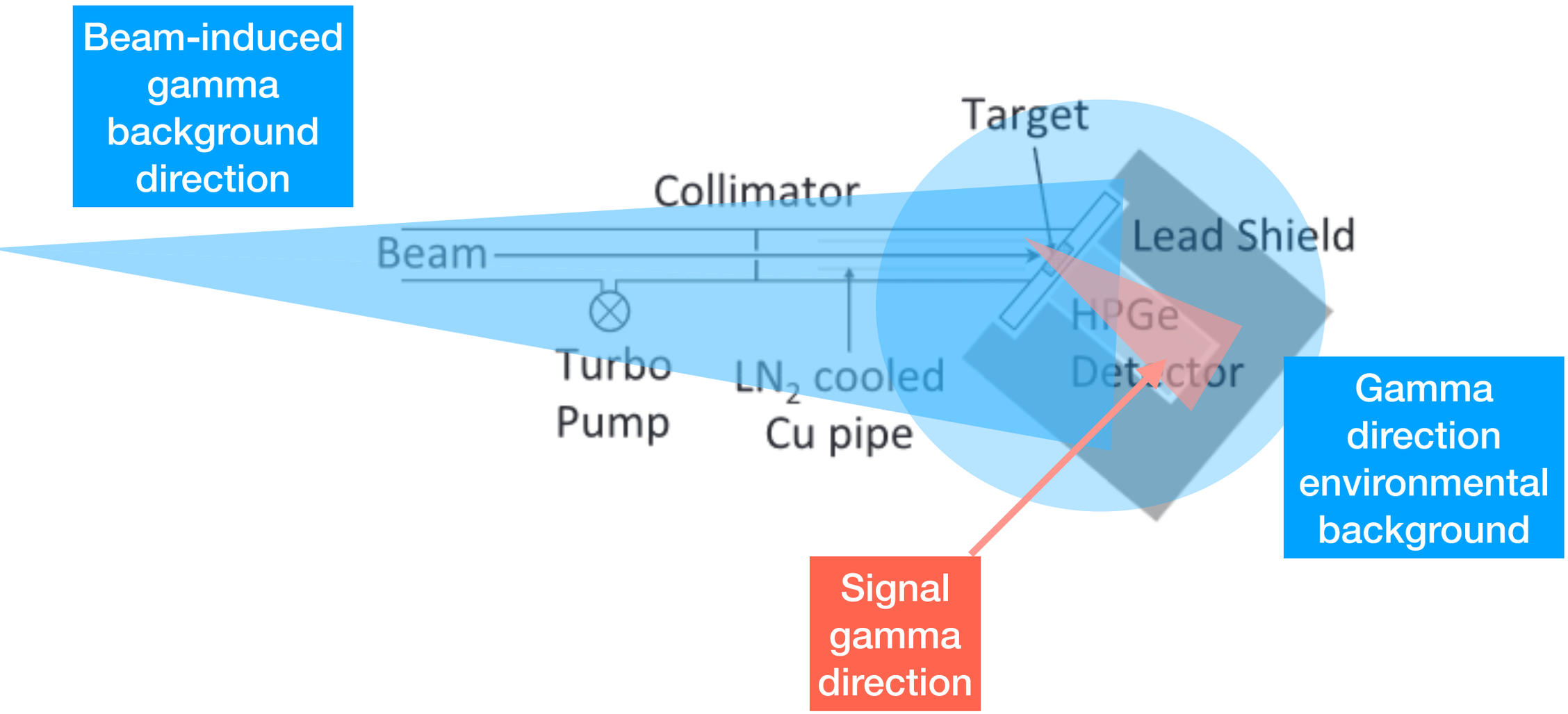
Example: LUNA / Bellotti Ion Beam Facility (LNGS)



https://journals.aps.org/prc/pdf/10.1103/PhysRevC.89.015803

Impact

Example: LUNA / Bellotti Ion Beam Facility (LNGS)



https://journals.aps.org/prc/pdf/10.1103/PhysRevC.89.015803

Impact

Example: LUNA / Bellotti Ion Beam Facility (LNGS)

$^{17}O(p,\gamma)^{18}F$ reaction relevant for explosive hydrogen burning

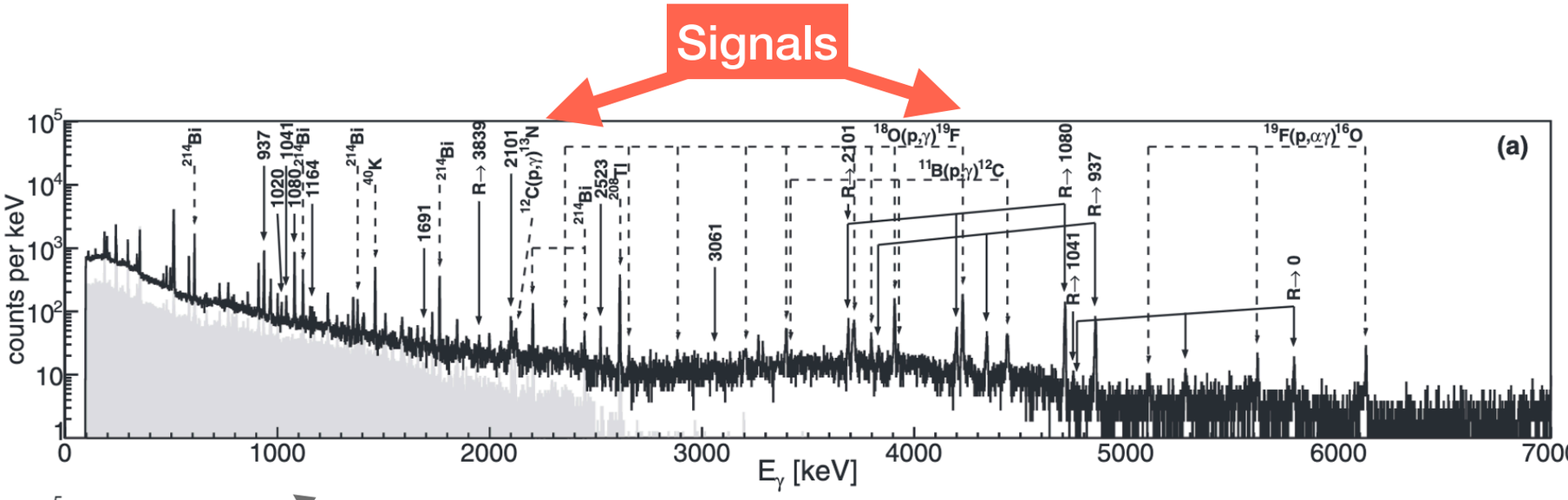


FIG. 5. (a) Sample spectrum of an on-resonance measurement at energy $E_{\text{c.m.}} = 183\,\text{keV}$. (b) Sample spectrum for an off-resonance measurement at $E_{\text{c.m.}} = 250\,\text{keV}$. In gray is the time-normalized room background with 10 cm of lead surrounding the detector.

Lower energy: background

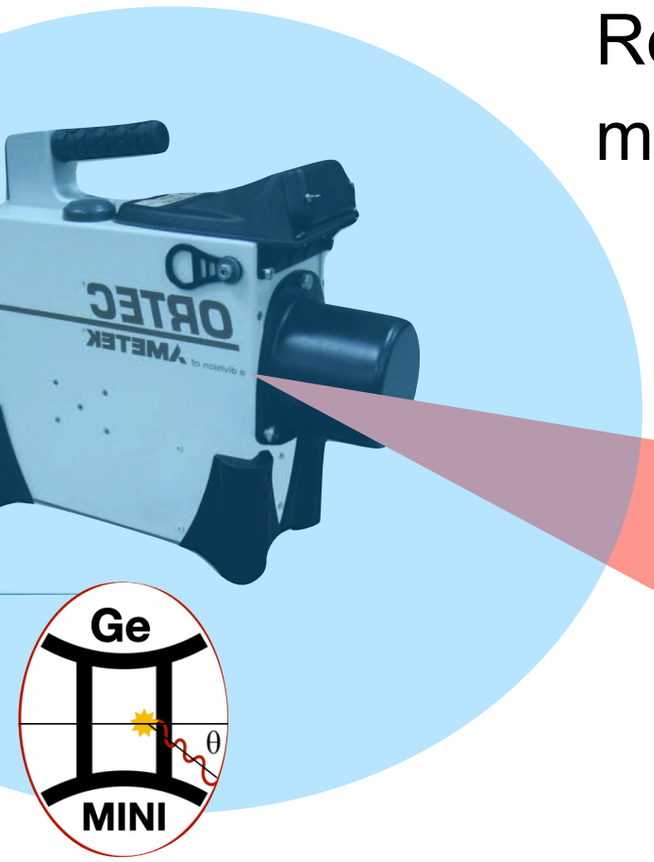
- Can greatly improve the otherwise incompressibile signal-to-background

https://journals.aps.org/prc/pdf/10.1103/PhysRevC.89.015803

Impact

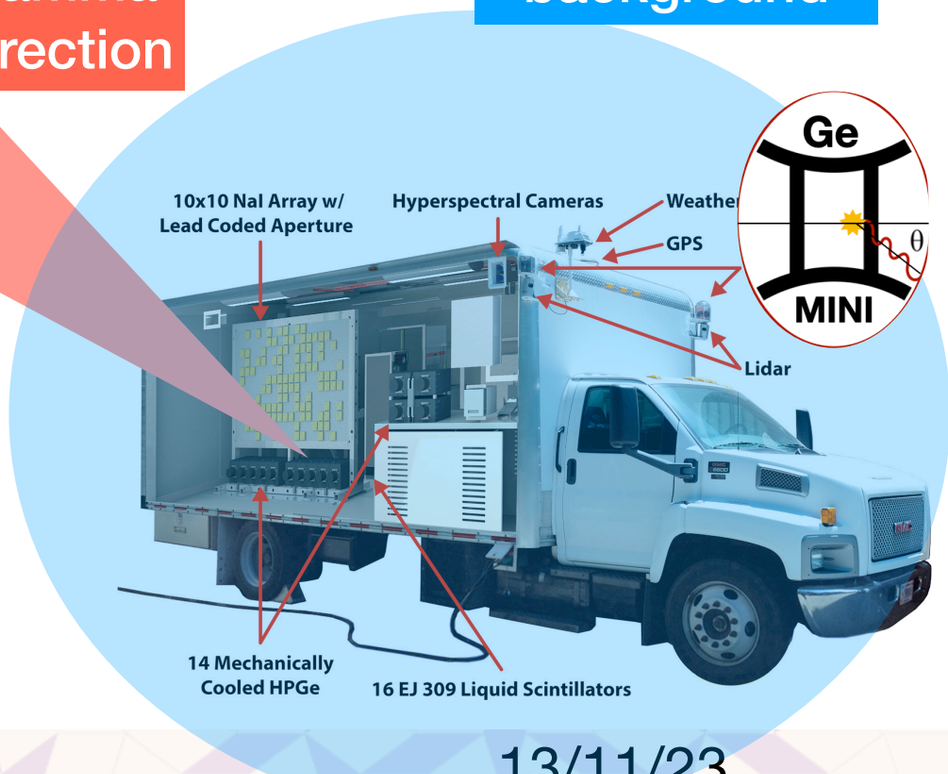
- Environment and nuclear safety:

Real time directionality would allow quicker and more effective source identification



Signal gamma direction

Gamma direction environmental background

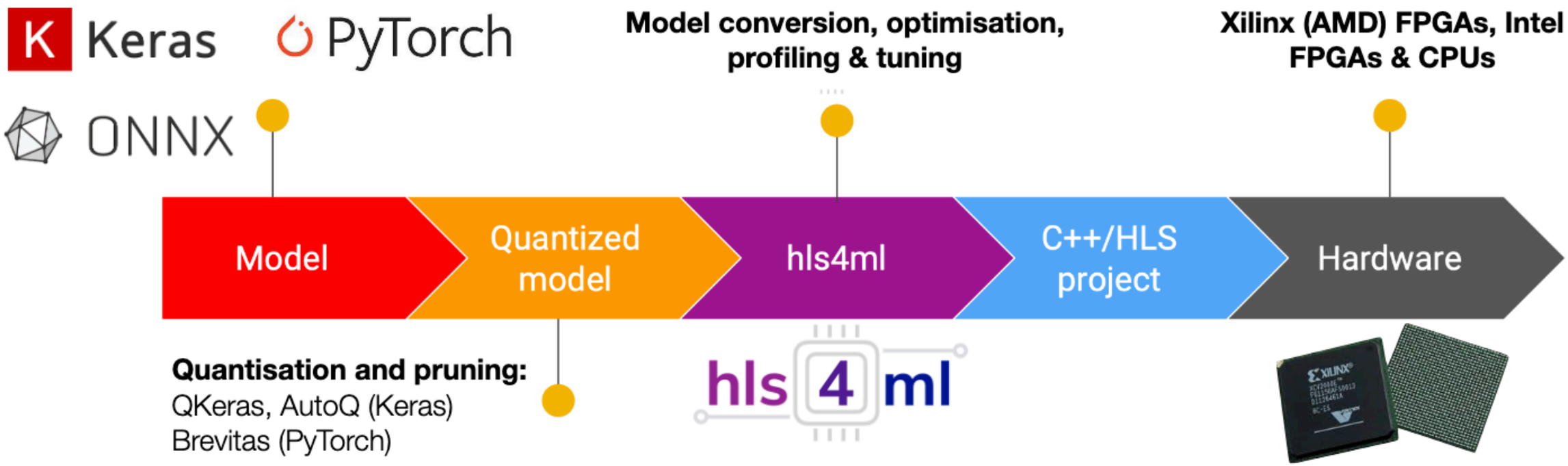


- High demand of quick & precise detectors for threats discovery

hls4ml

[1] https://fastmachinelearning.org/hls4ml

› Open-source Python API & command line tool that translates trained NNs to synthesizable FPGA firmware



› Implementations of common ingredients - layer types, activation functions

› Novel ingredients for fast, efficient inference - binary/ternary NNs, network optimisations

FPGA performances

Table 3. Accuracy, resource consumption and latency for the BF and BP models quantized to a bit width of 14, the Q and QP models quantized to a bit width of 7 and the heterogeneously quantized AQ and AQP models. The numbers in parentheses correspond to the total amount of resources used.

FPGA: Xilinx Virtex UltraScale+ VU9P							
Model	Accuracy	DSP (%)	LUT (%)	FF (%)	BRAM (%)	Latency (cc)	II (cc)
BF 14-bit	0.87	93.23 (6377)	19.36 (228 823)	3.40 (80 278)	3.08 (66.5)	1035 (5.2 μs)	1030
BP 14-bit	0.93	48.85 (3341)	12.27 (145 089)	2.77 (65 482)	3.08 (66.5)	1035 (5.2 μs)	1030
Q 7-bit	0.94	2.56 (175)	12.77 (150 981)	1.51 (35 628)	3.10 (67.0)	1034 (5.2 μs)	1029
QP 7-bit	0.94	2.54 (174)	9.40 (111 152)	1.38 (32 554)	3.10 (67.0)	1035 (5.2 μs)	1030
AQ	0.88	1.05 (72)	4.06 (48 027)	0.64 (15 242)	1.5 (32.5)	1059 (5.3 μs)	1029
AQP	0.88	1.02 (70)	3.28 (38 795)	0.63 (14 802)	1.4 (30.5)	1059 (5.3 μs)	1029

Table 4. Resource consumption and latency for the QP 7-bit model on a Xilinx Zynq XC7Z020 SoC. A clock frequency of 100 MHz is used.

FPGA: Xilinx Zynq XC7Z020 SoC							
	DSP	LUT	FF	BRAM (18 kb)	Latency (cc)	II (cc)	frame/s
Available	220	53 200	106 400	280	—	—	—
Used	213 (96.82%)	48 259 (90.71%)	35 118 (33.01%)	122 (43.57%)	17 085 (171 μs)	16 385	2831

https://iopscience.iop.org/article/10.1088/2632-2153/ac0ea1/pdf

FPGA performances

Pruning

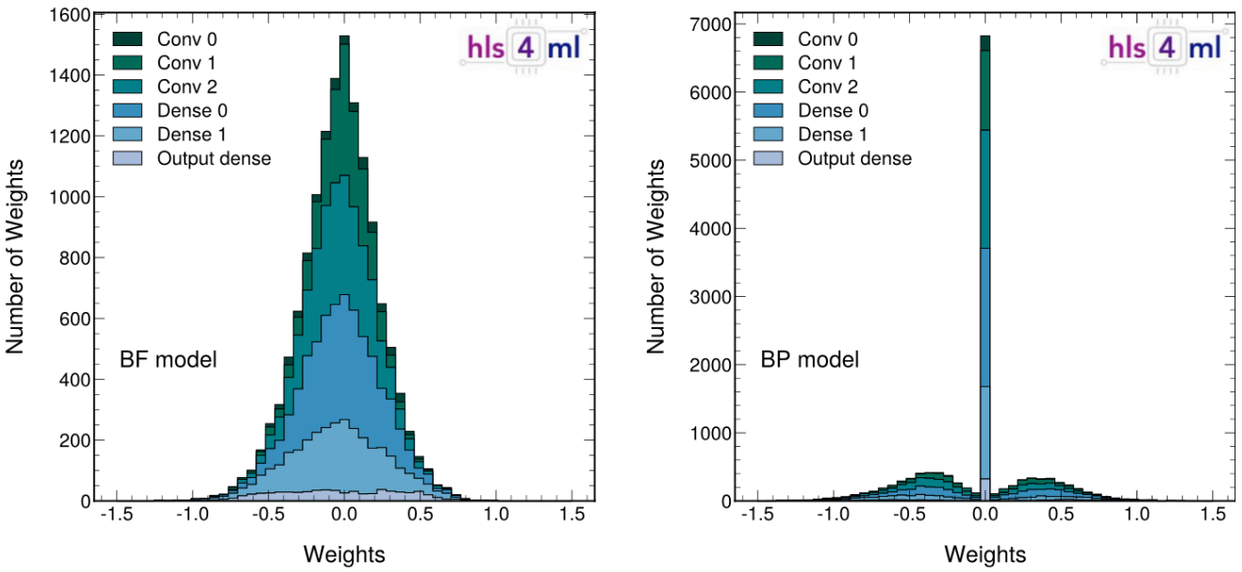


Figure 5. The weights per layer for the the BF model (left) and the baseline pruned (BP) model (right). The BP model is derived by starting from the BF model and repeating the training while applying a pruning procedure with a target sparsity of 50% for each layer.

Quantisation

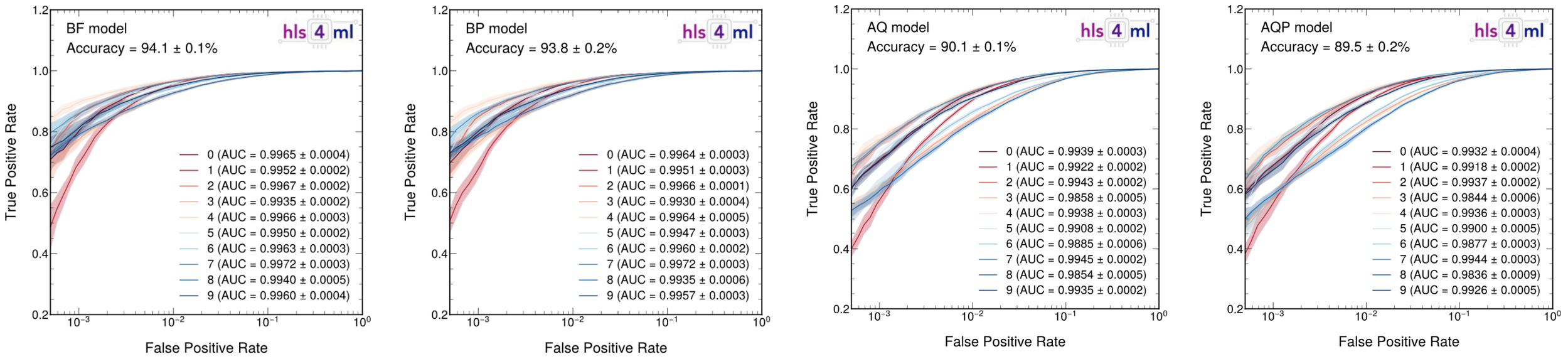


Figure 6. ROC curves of FPR versus true positive rate (TPR) for the BF model (left) and the BP model (right). Training is performed using k-fold cross validation, with $k = 10$. For each digit, the solid line corresponds to the mean and the band to the standard deviation across the ten folds. The area under the curve (AUC) is reported in the legend and is defined as the mean across the ten folds with an uncertainty given by the standard deviation.

https://iopscience.iop.org/article/10.1088/2632-2153/ac0ea1/pdf

FPGA performances

Pruning Quantisation

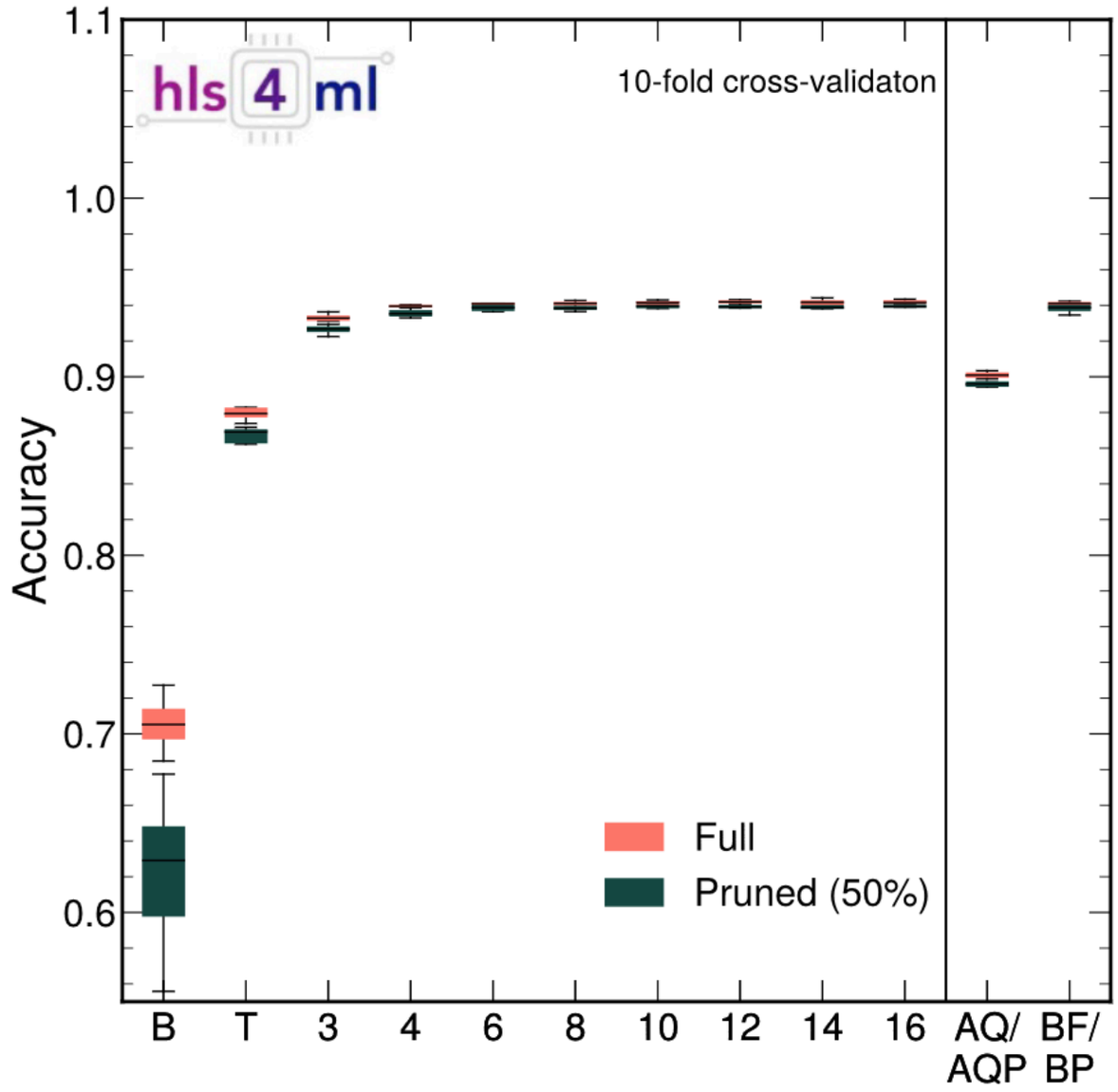
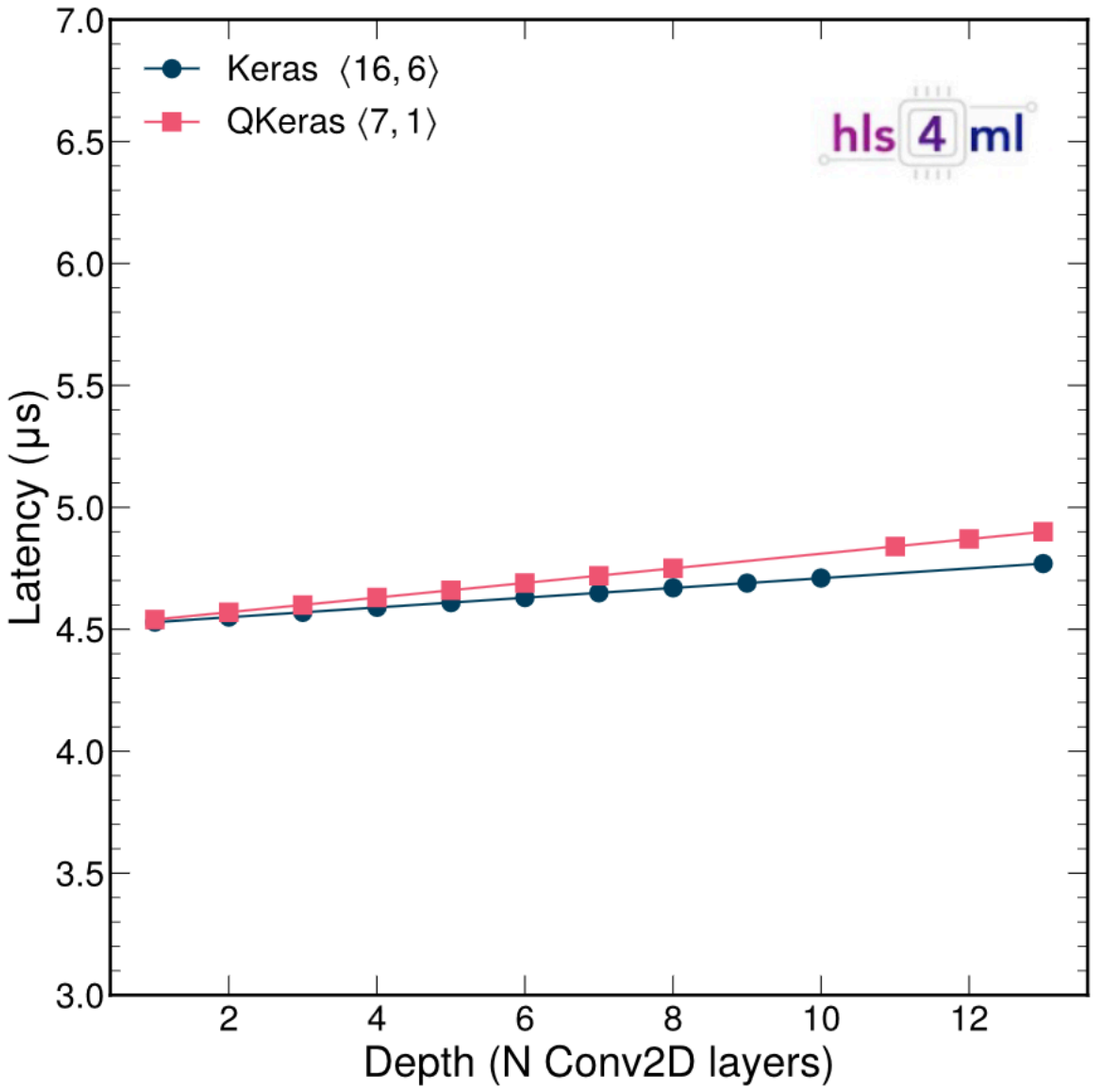
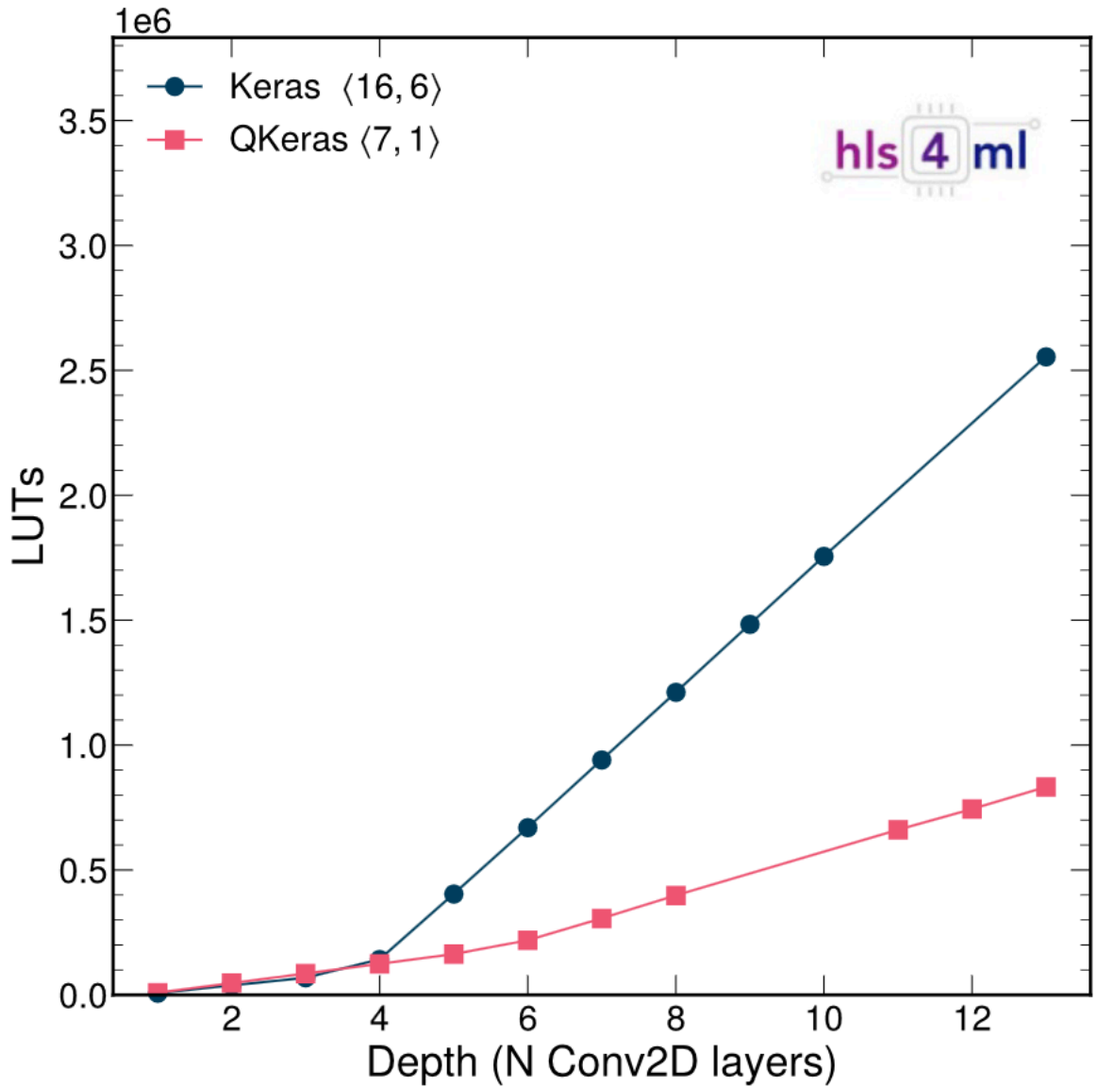


Figure 11. Accuracy for unpruned (red) and pruned (yellow) models for binary (B) and ternary (T) precisions, homogeneously quantized models with bit widths between 3 and 16, the heterogeneously quantized models AQ and AQP, compared to the BF and BP models. The black line represents the median, and the box extends from the lower to upper quartile values across the ten folds.

https://iopscience.iop.org/article/10.1088/2632-2153/ac0ea1/pdf

FPGA performances





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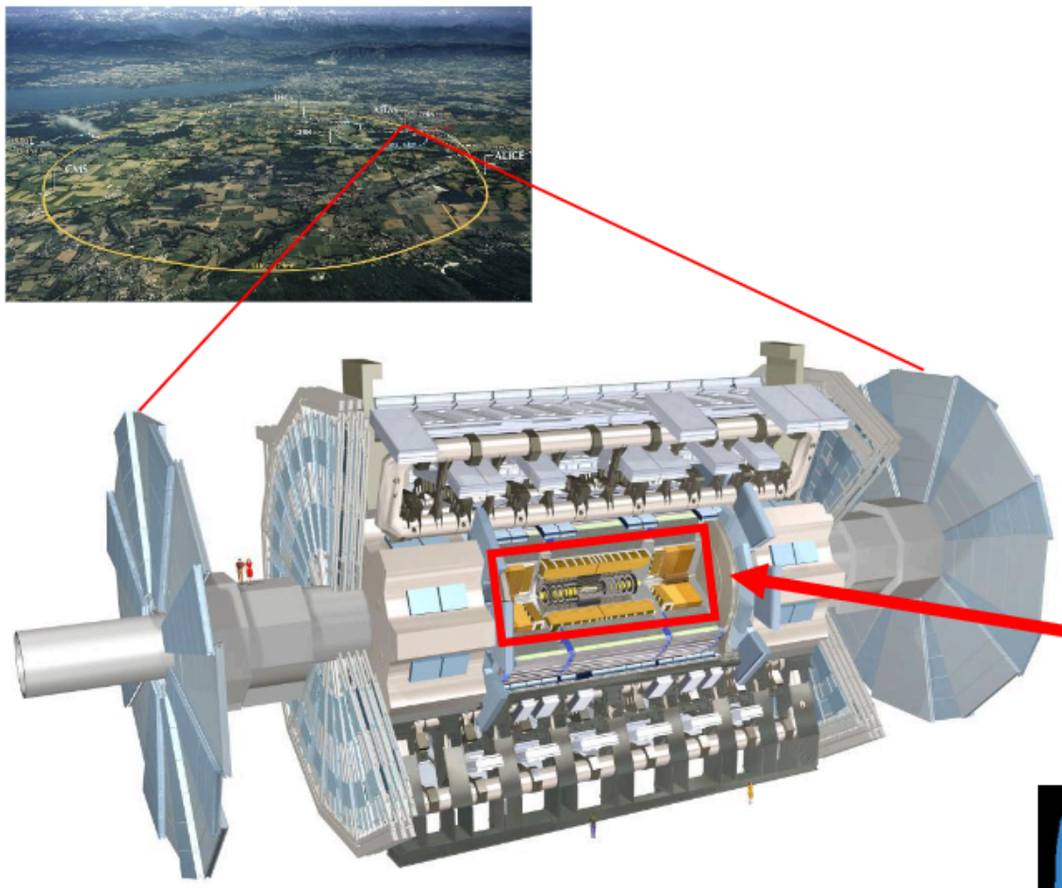
Case study: ATLAS LAr calorimeter ML FPGA



Large Hadron Collider (LHC):

- Proton bunches collide with **25 ns** spacing (40 MHz)
- 2029: Start of **High Luminosity LHC** (HL-LHC) with up to ~7 x nominal luminosity

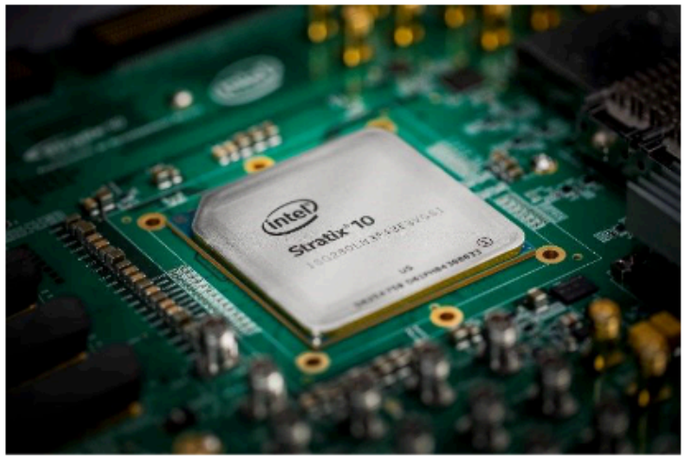
ATLAS Detector

- From ~20 collisions to up to ~200 collisions per bunch crossing (BC) → pileup increases
- 2026-2028: Phase-II upgrade
- Readout electronics of **Liquid-Argon (LAr) calorimeter** need to be improved

https://iopscience.iop.org/article/10.1088/2632-2153/ac0ea1/pdf

LAr Calorimeter Readout

FPGA – Field Programmable Gate Array



[7]

- Real-time signal processing
- Installation of **556 high-performance FPGAs**
- 250 Tbps total data rate
- 150 ns maximum latency

Optimal Filtering Algorithm (OF)

- calculates deposited energy per cell

$$y_t = \sum_{n=0}^{M-1} x_{t-n} \cdot a_n$$

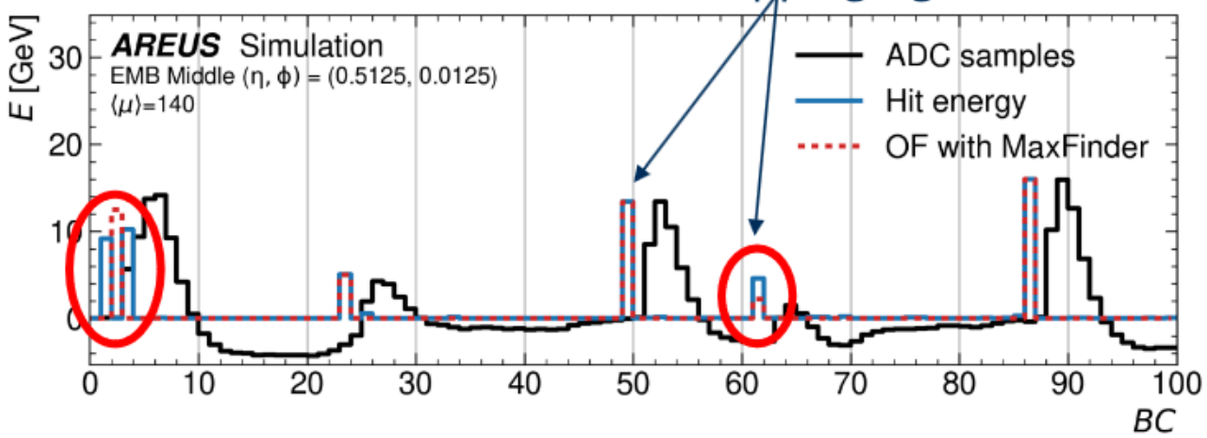
y_t ... OF output for bunch crossing (BC) t
$\vec{x}$ ... input ADC samples
a_n ... OF coefficients
M ... OF filter depth

- Trigger system applies additional maximum finder

Trigger

Event Readout

- Good in energy resolution **but**
- Weak in reconstruction of overlapping signals



- Development of Neural Networks to improve energy reconstruction
- Keep parameters low (≈ 100) due to FPGA resource limits

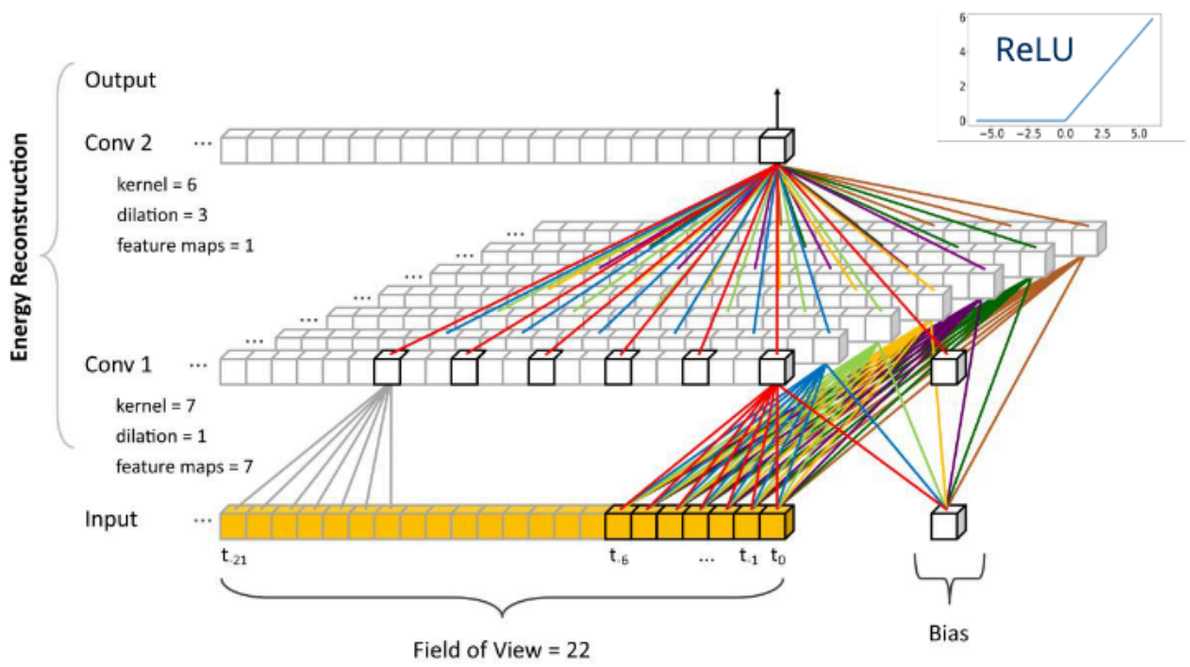
https://indico.cern.ch/event/1283970/contributions/5554340/attachments/2720393/4726092/FastML_for_Science_2.pdf

Case study: ATLAS LAr calorimeter ML FPGA

Convolutional Neural Networks for LAr Readout

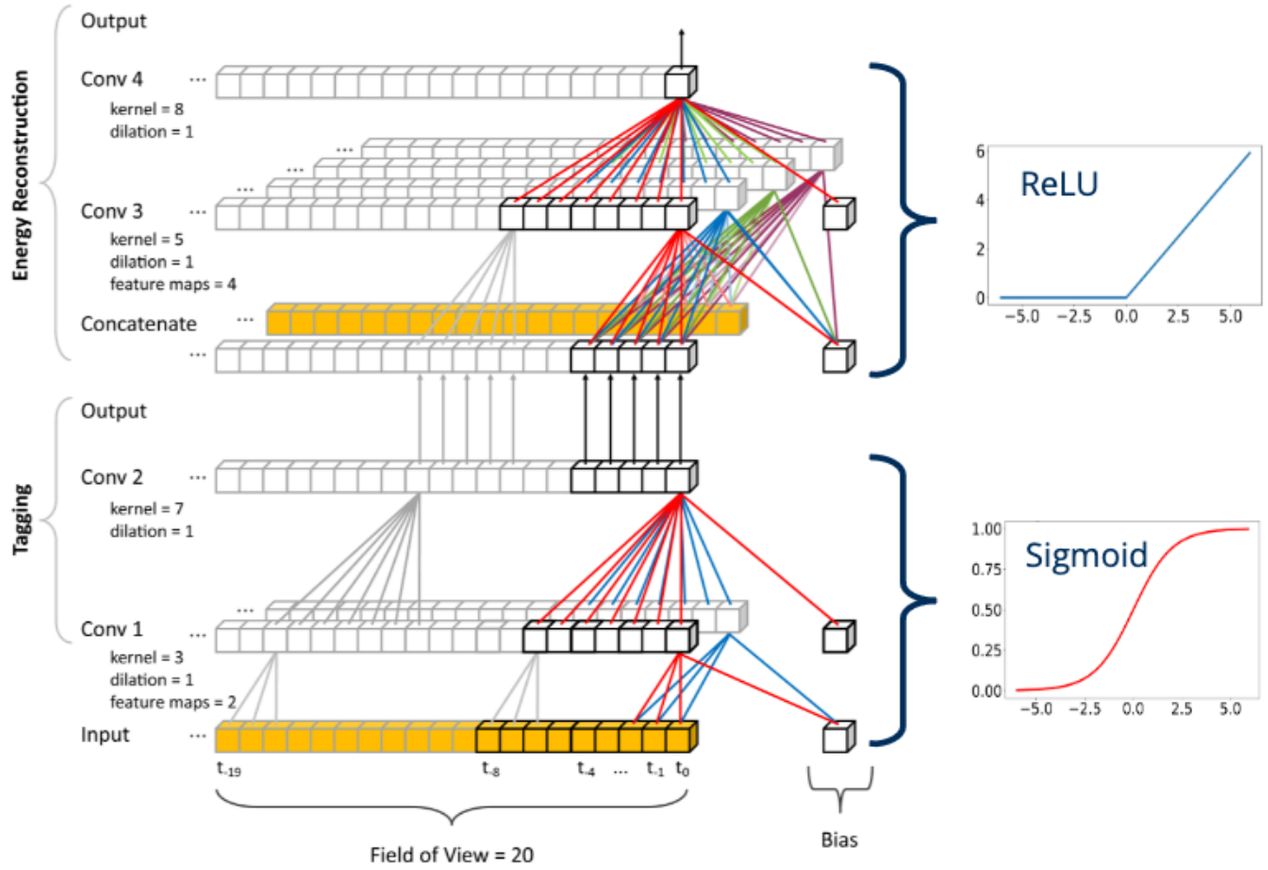
Plain 2-layered CNN (2-Conv CNN)

- Dilation enables larger Field of View (FoV)
- ReLU activation functions
- Output: reconstructed energy



4-layered CNN with Tagging (4-Conv CNN)

- Sigmoid and ReLU activation functions
- Intermediate output tags signal overlaps
- Output: reconstructed energy

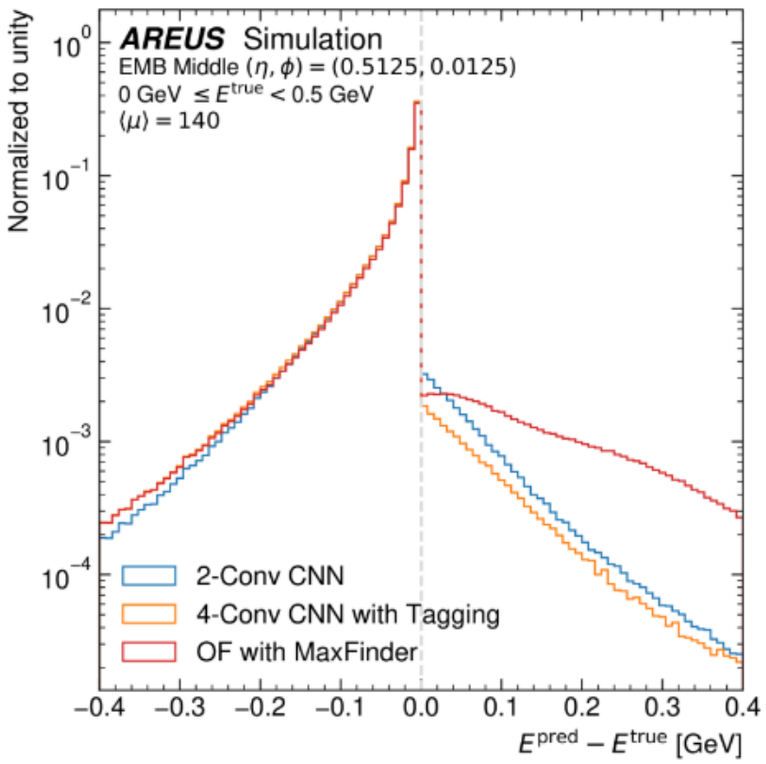


https://indico.cern.ch/event/1283970/contributions/5554340/attachments/2720393/4726092/FastML_for_Science_2.pdf

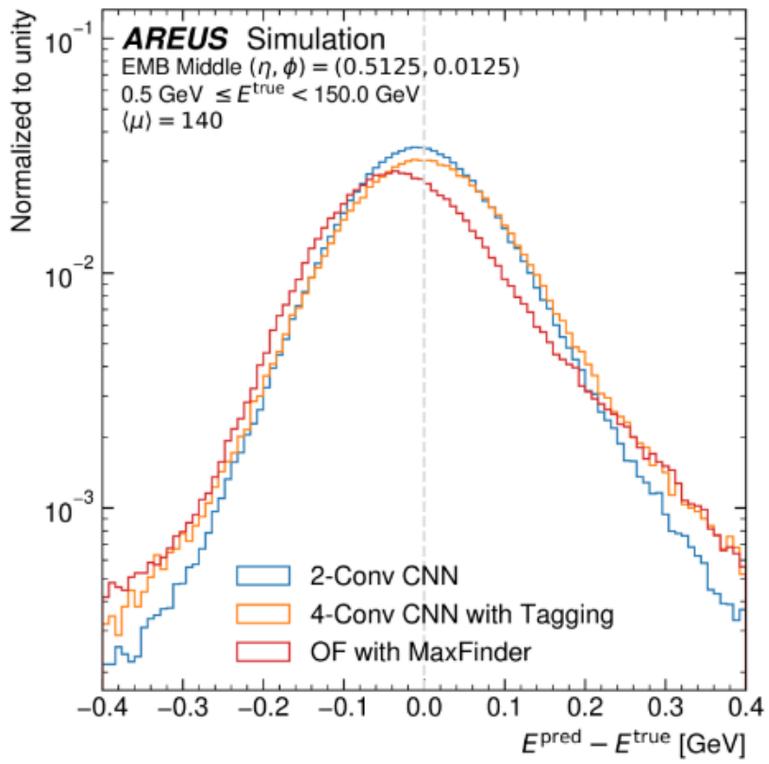
Case study: ATLAS LAr calorimeter ML FPGA

Performance Evaluation: Energy Resolution

$E^{true} < 0.5$ GeV



$E^{true} > 0.5$ GeV



Optimal Filter:

- Larger deviation spread in low energy region
- Negative bias in high energy region

CNNs:

- Improvement in energy resolution, more symmetric and centered distribution
- Performance stable within large energy range

https://indico.cern.ch/event/1283970/contributions/5554340/attachments/2720393/4726092/FastML_for_Science_2.pdf

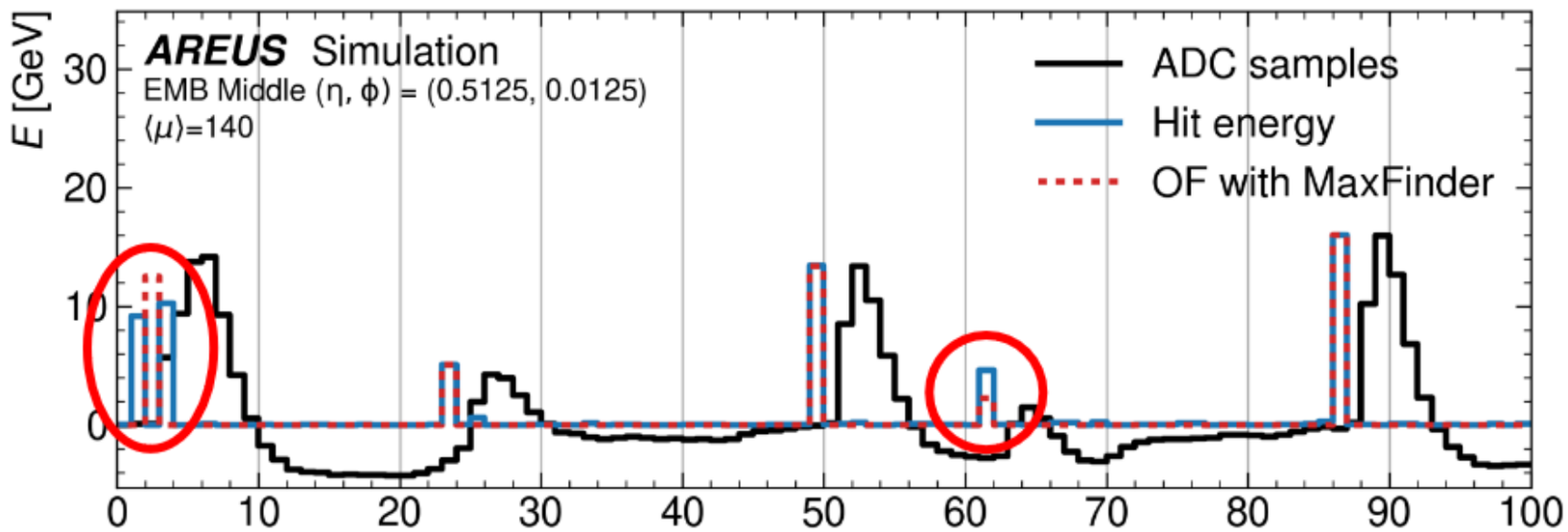
Case study: ATLAS LAr calorimeter ML FPGA

Performance Evaluation: Sequence Comparison

Example Sequence

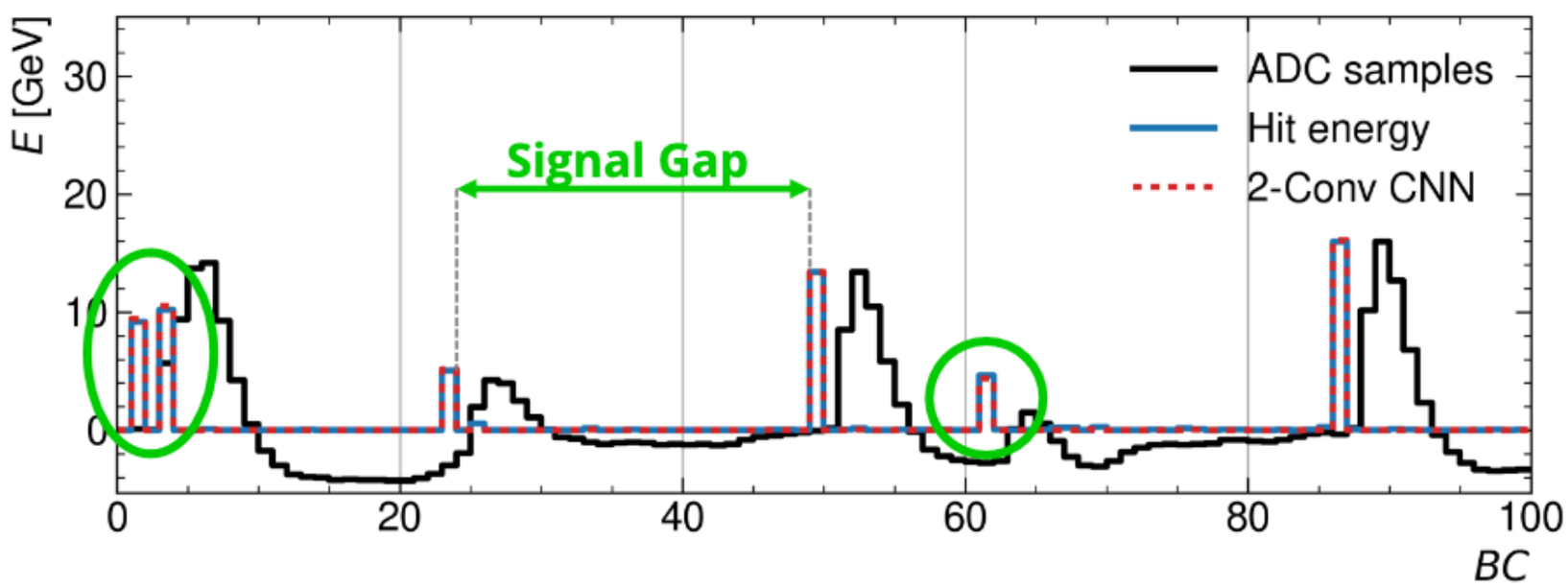
Optimal Filter

- Close signals cannot be resolved
- Signals within undershoot underestimated



CNN

- Optimized to reconstruct overlapping signals



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Case study: ATLAS LAr calorimeter ML FPGA

Firmware Implementation

ANN model from Keras

Keras

TensorFlow

[8,9]

Converter

FPGA

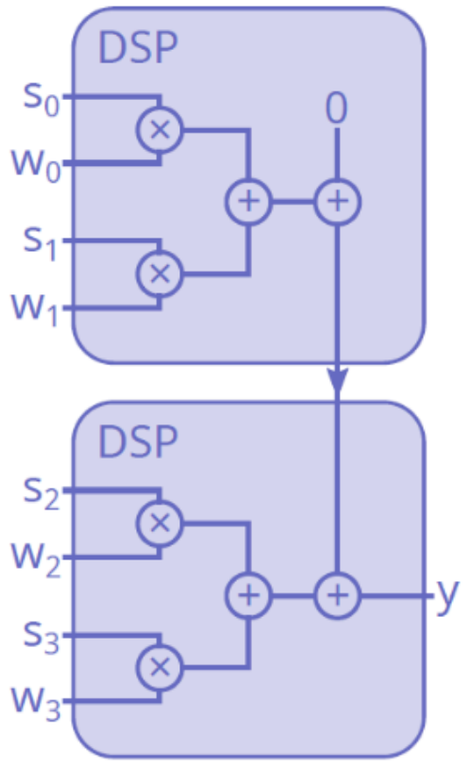
[7]

Integrated circuits, configurable by user after manufacturing

- **DSP:** digital signal processor
- **ALM:** adaptive logic module with lookup table

Firmware development on **Intel Stratix-10** FPGA

Final design uses **Intel Agilex**



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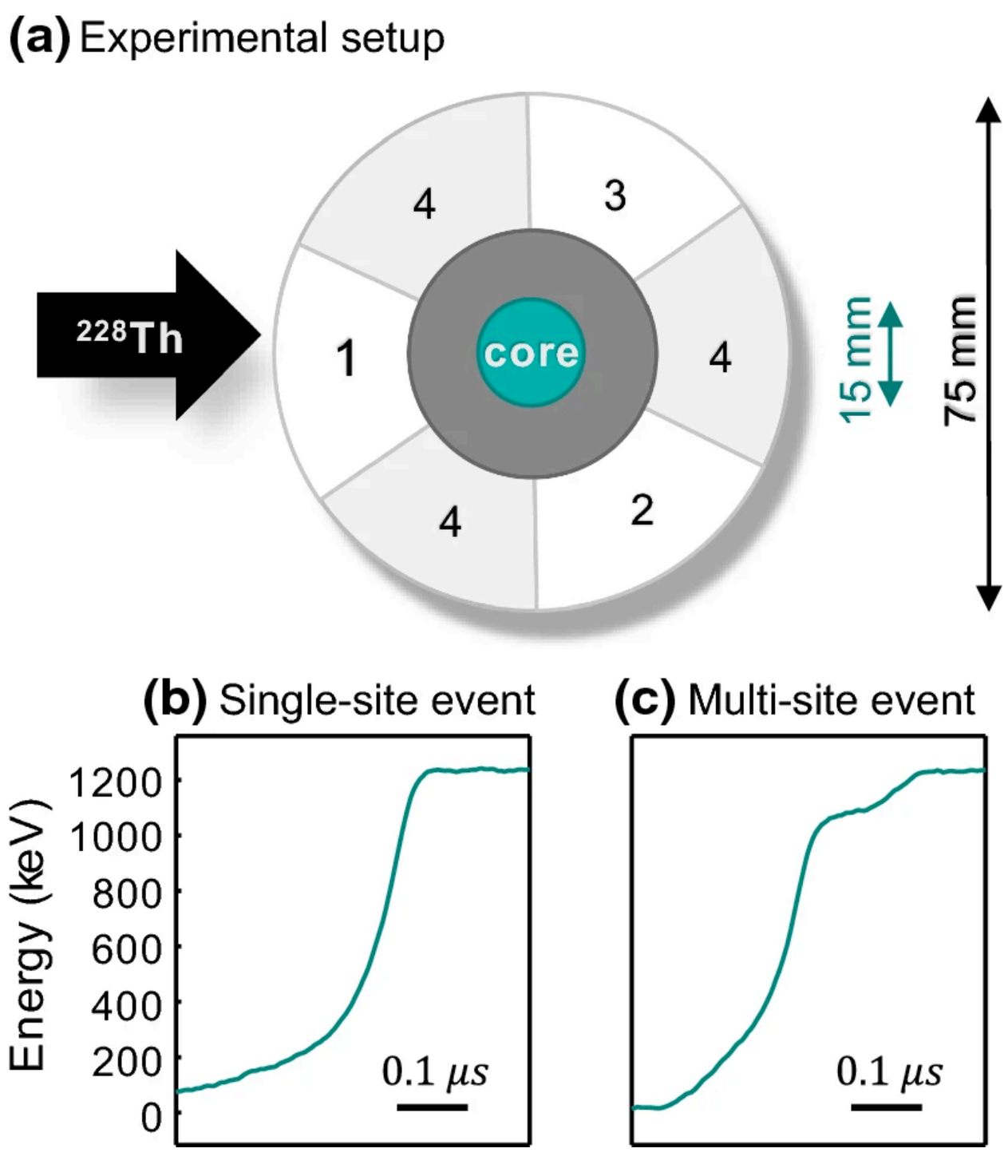
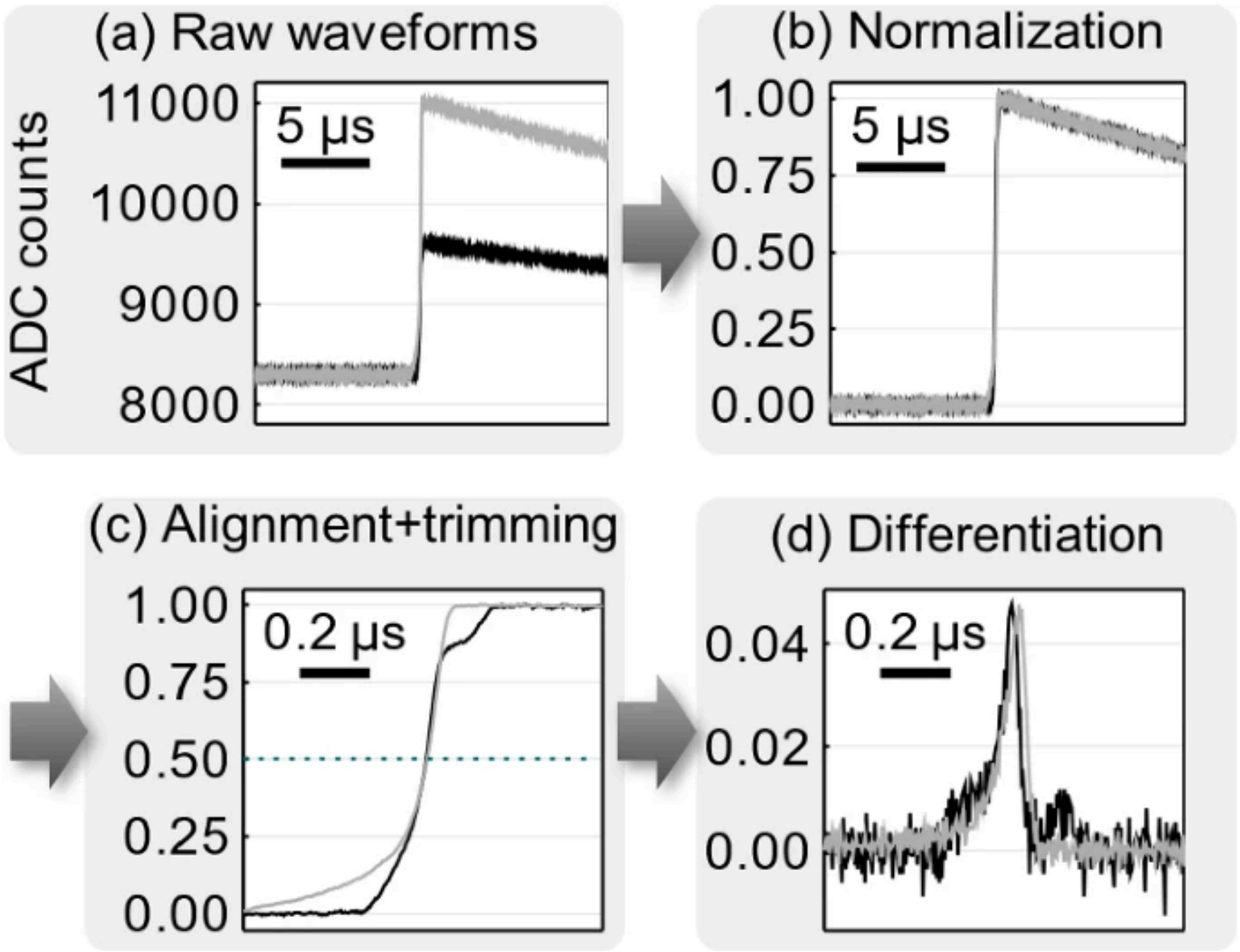
Case study: GERDA ML for pulse discrimination



https://link.springer.com/article/10.1140/epjc/s10052-019-6869-2

Case study: GERDA ML for pulse discrimination

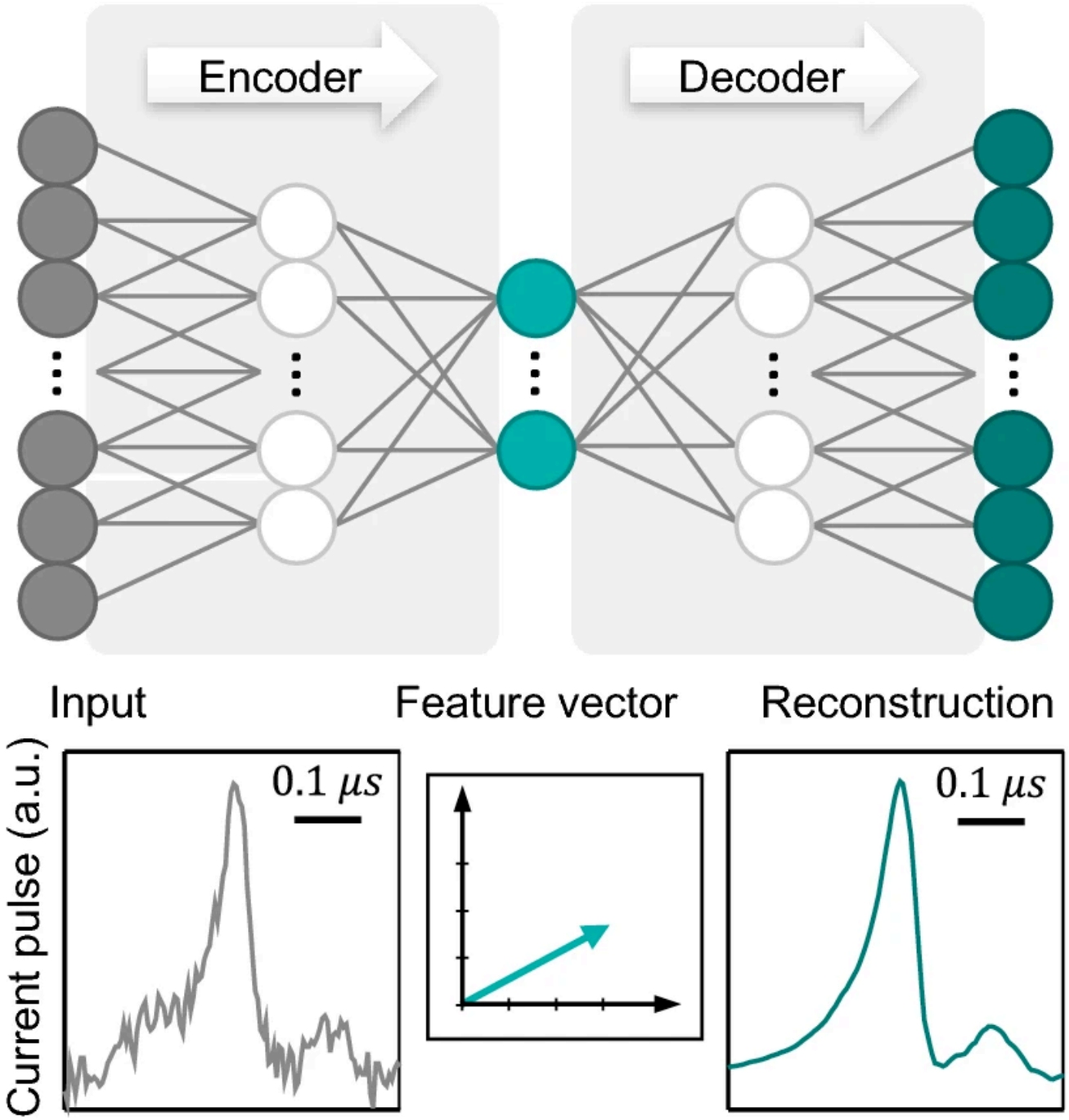
Fig. 3



Preprocessing steps shown for the MS event from Fig. 1b (black curve) and a presumed SS event with 2578 keV energy (gray curve). The raw waveforms (**a**) are baseline-subtracted and normalized to a charge of one (**b**), aligned at the intersection with a threshold of 0.5, then trimmed to a 1 µs long window (**c**), and finally differentiated yielding the current-signal (**d**)

https://link.springer.com/article/10.1140/epjc/s10052-019-6869-2

Case study: GERDA ML for pulse discrimination



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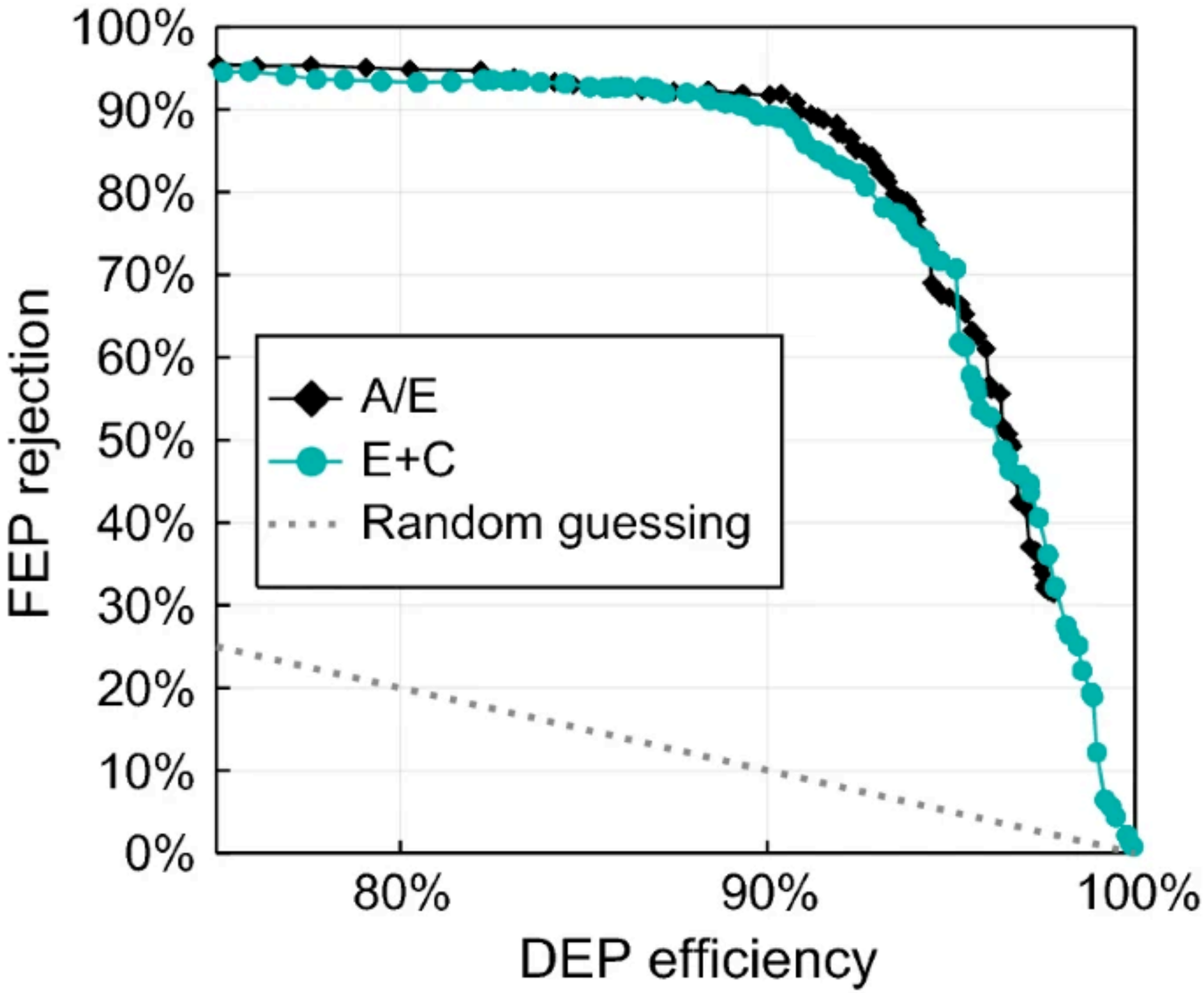
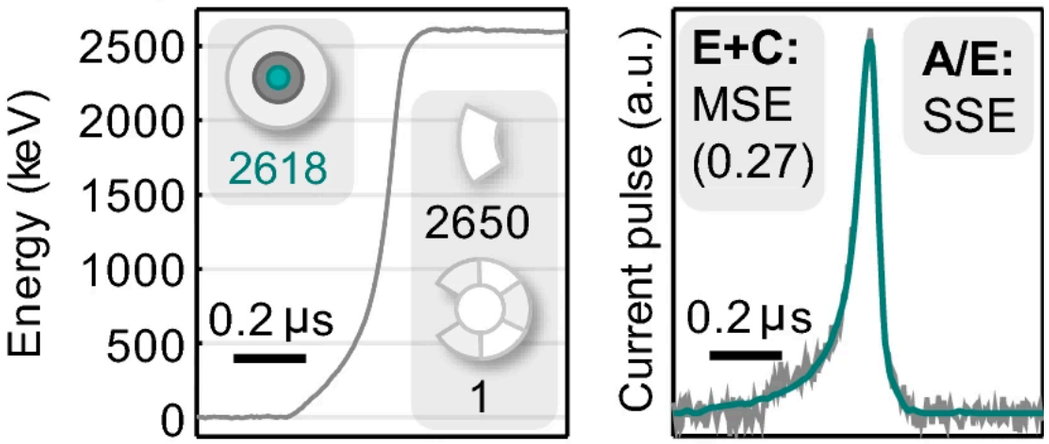
Case study: GERDA ML for pulse discrimination



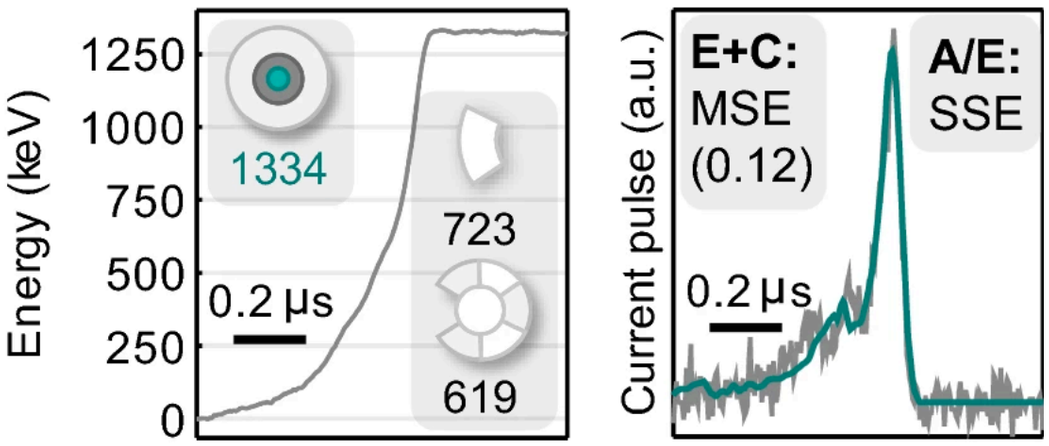
https://link.springer.com/article/10.1140/epjc/s10052-019-6869-2

Case study: GERDA ML for pulse discrimination

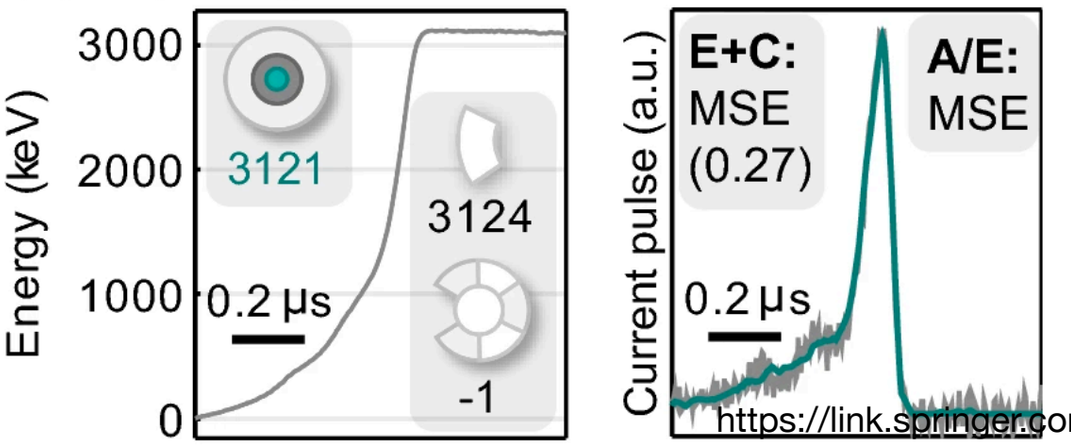
(a) Single-segment event with unusual shape

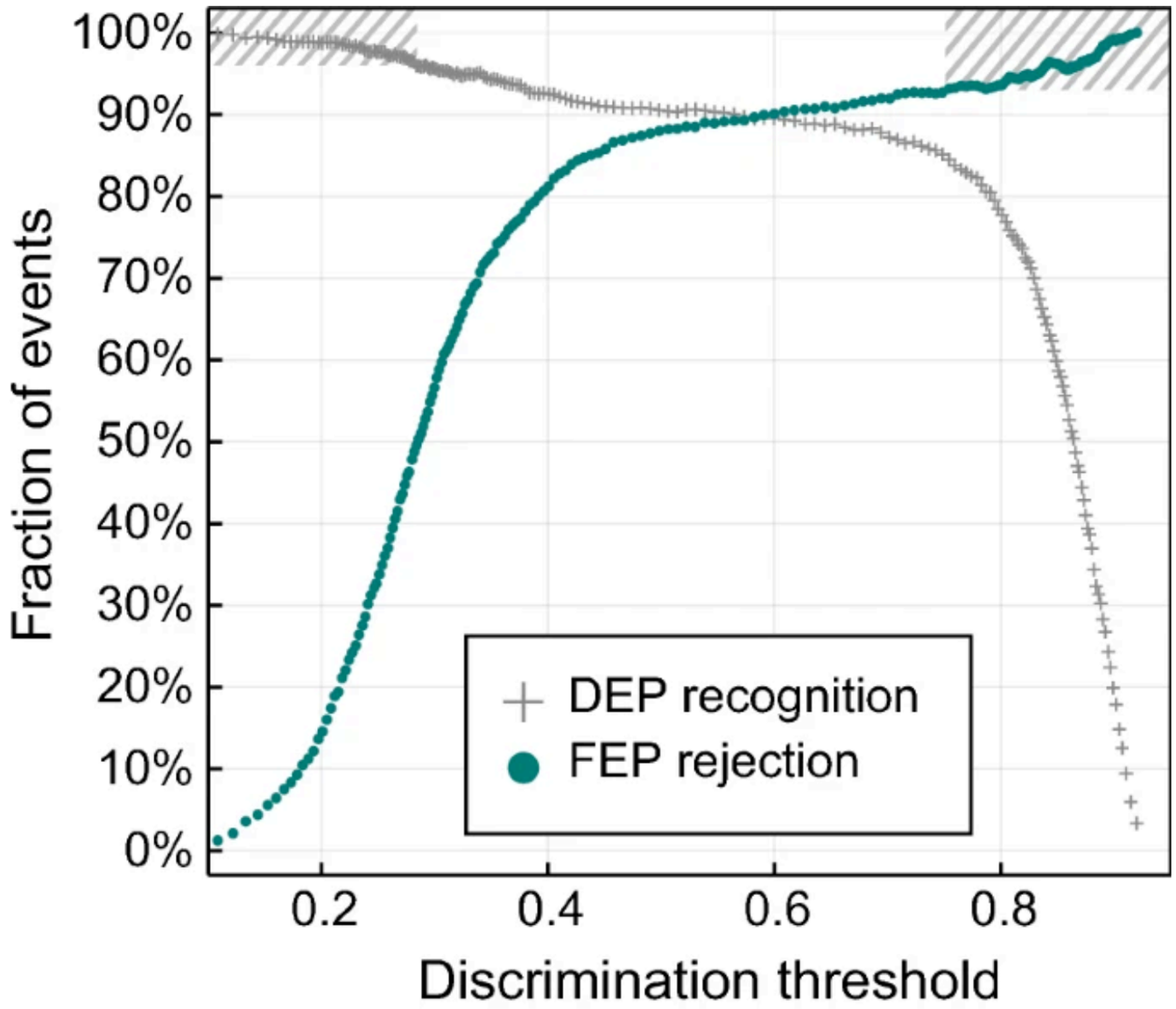


(b) MS event with high current-pulse peak



(c) Single-segment MS event

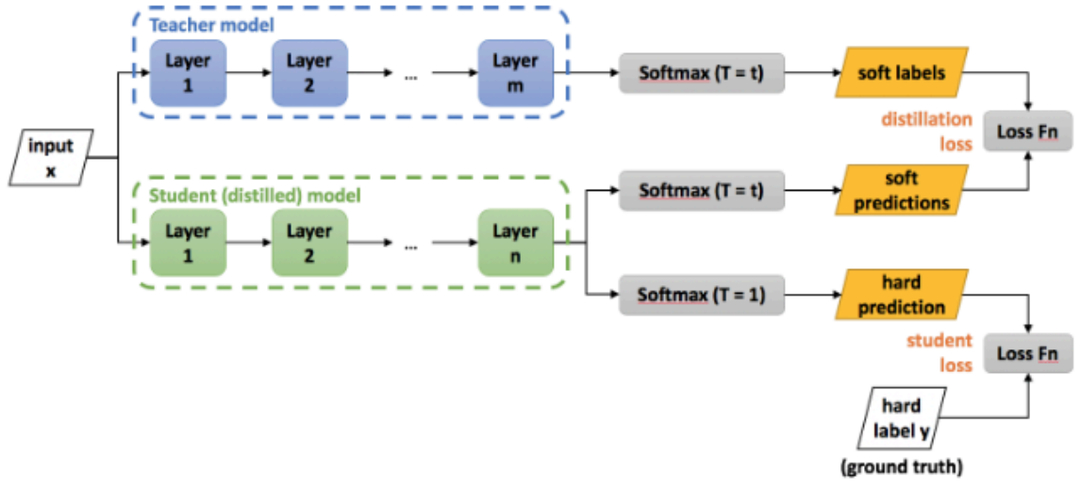




https://link.springer.com/article/10.1140/epjc/s10052-019-6869-2

Knowledge distillation

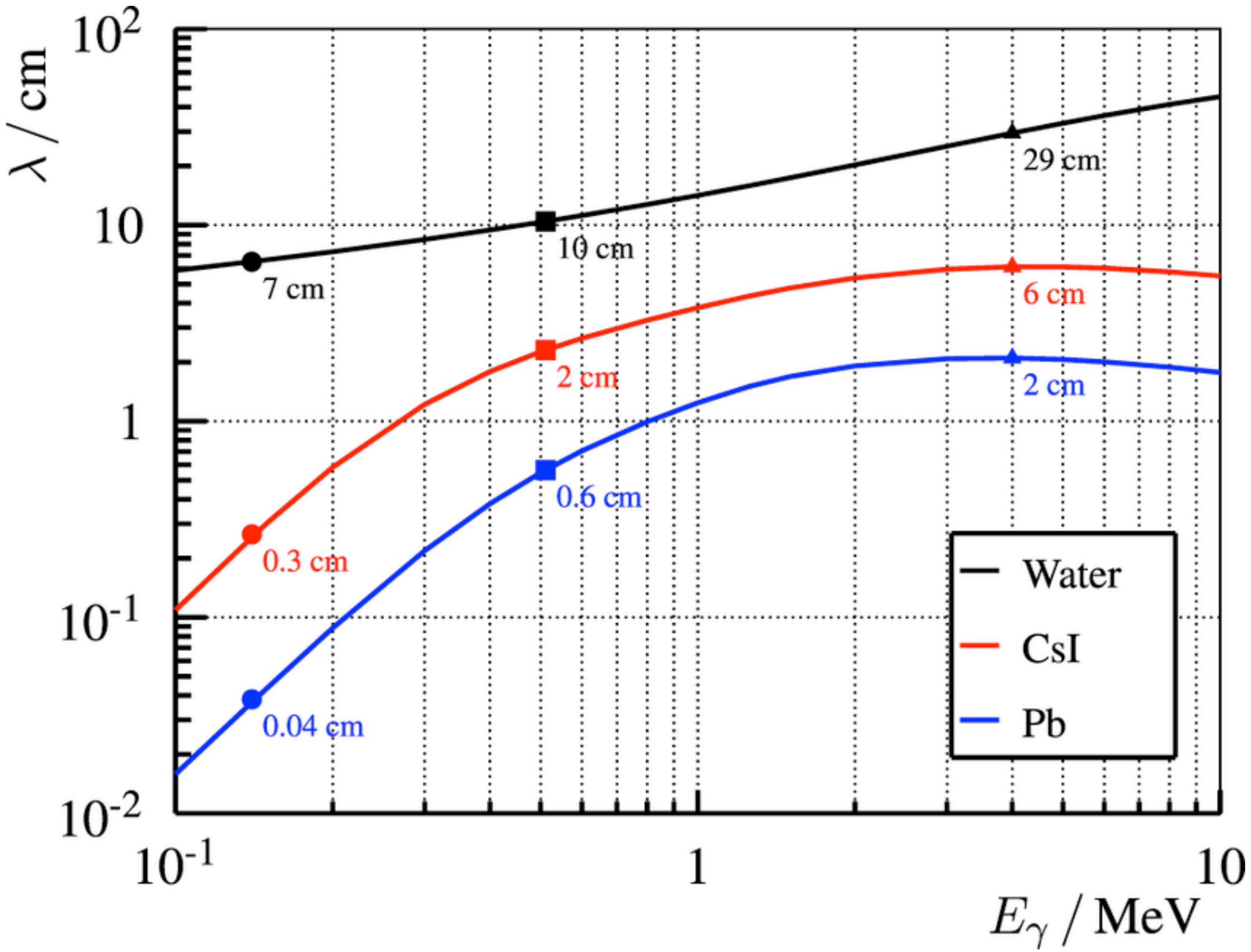
https://intellabs.github.io/distiller/knowledge_distillation.html



- Process of transferring knowledge from large model to smaller model *i.e reducing computations while maintaining performance*
- Larger Teacher model pre-trained
- Smaller Student model trained with loss function *dependent on the output of the teacher model: A*Loss(student, teacher) + B*Loss(student, truth)*
- Applied to both re-calibration & classification: both models 4 layers; precision AP (18,5)

Performance comparison	N Neurons / layer	N parameters total	FWHM			Classification: area under ROC curve
			$\Delta\varphi$ [rad]	$\Delta\eta$	Δp_T [GeV]	
Teacher	256	204 000	0.12	0.061	0.41	0.97
Student	8	419	0.12	0.063	0.42	0.96
Original baseline	16	1219	0.14	0.066	0.43	0.97

- Compression factor of ~200 with almost no performance loss: no pruning required


λ / cm
10²
10
1
10⁻¹
10⁻²
10⁻¹
1
10
Eγ / MeV
7 cm
10 cm
29 cm
6 cm
2 cm
2 cm
0.6 cm
0.3 cm
0.04 cm
Water
CsI
Pb

BALTIC SCIENTIFIC INSTRUMENTS

Bringing Solutions

Zagrebs HPGe

P-type HPGe Coaxial Detectors GCD

(Liquid Nitrogen cooled)

Application

Detection of Gamma-rays in nuclear energetics and environmental control, in industry and scientific research, in medicine and other applications.

Complete set

- HPGe coaxial detector
- Preamplifier with cooled input stage
- Different cryostat modifications are available

Accessories

- Digital or Analog-Digital Multichannel Analyzer
- Analytical software for quantitative and qualitative analysis
- Liquid nitrogen storage and filling system

Features

- 10% - 160% efficient HPGe coaxial detectors are available
- Energy range from 40 keV to 10 MeV with P-type HP(Ge) Detector
- Well type detectors are available
- High energy rate up to 200000 MeV/s
- Excellent peak symmetry
- Aluminum, Beryllium or Carbon-fiber input window
- Detection of radiation in any spatial orientation depending on cryostat modification
- Manufacture in variable cryostat design is possible
- HV supply protection if detector is warm
- High count rate indicator

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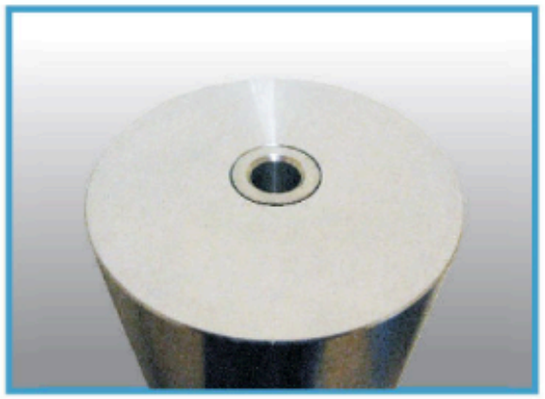
www.bsi.lv

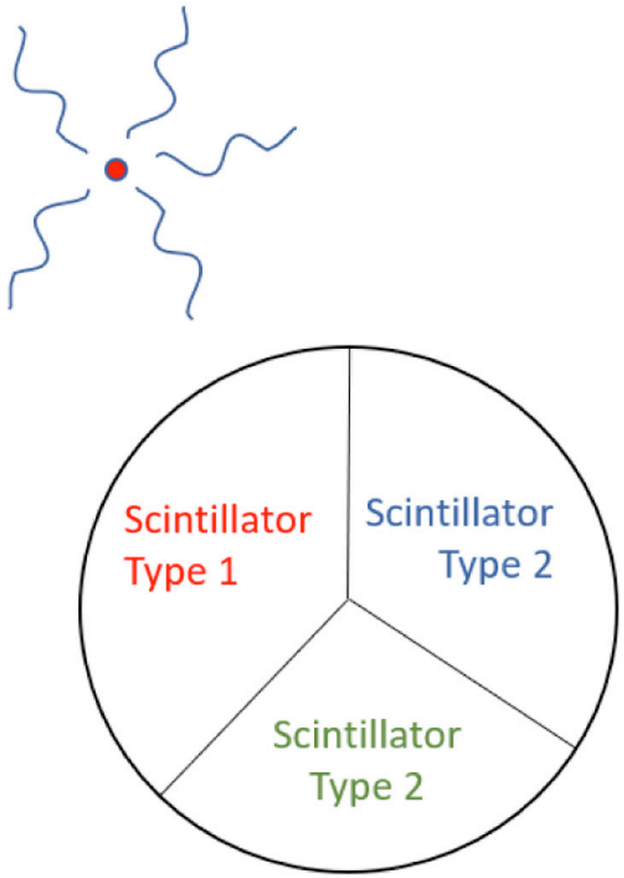
Specification

Model	Efficiency, %	Energy resolution 122 keV, (eV)	Energy resolution 1.33 MeV, (keV)	Peak/ Compton ratio	Peak Shape FW.1M FWHM	Peak Shape FW.02M FWHM
GCD - 10 175	10	825	1.75	41:1	1.9	2.65
GCD - 15 180	15	825	1.80	46:1	1.9	2.65
GCD - 20 180	20	850	1.80	51:1	1.9	2.65
GCD - 25 185	25	850	1.85	55:1	1.9	2.65
GCD - 30 185	30	875	1.85	58:1	1.9	2.65
GCD - 35 190	35	875	1.90	60:1	1.9	2.65
GCD - 40 190	40	895	1.90	62:1	1.9	2.65
GCD - 50 190	50	895	1.90	64:1	1.9	2.65
GCD - 60 200	60	1000	2.00	68:1	2.0	3.00
GCD - 70 200	70	1000	2.00	73:1	2.0	3.00
GCD - 80 210	80	1000	2.10	77:1	2.0	3.00
GCD - 100 220	100	1000	2.10	81:1	2.0	3.00
GCD - 120 220	120	1000	2.10	83:1	2.0	3.00
GCD - 140 220	140	1100	2.20	86:1	2.0	3.00
GCD - 160 220	160*	1150	2.20	88:1	2.0	3.00

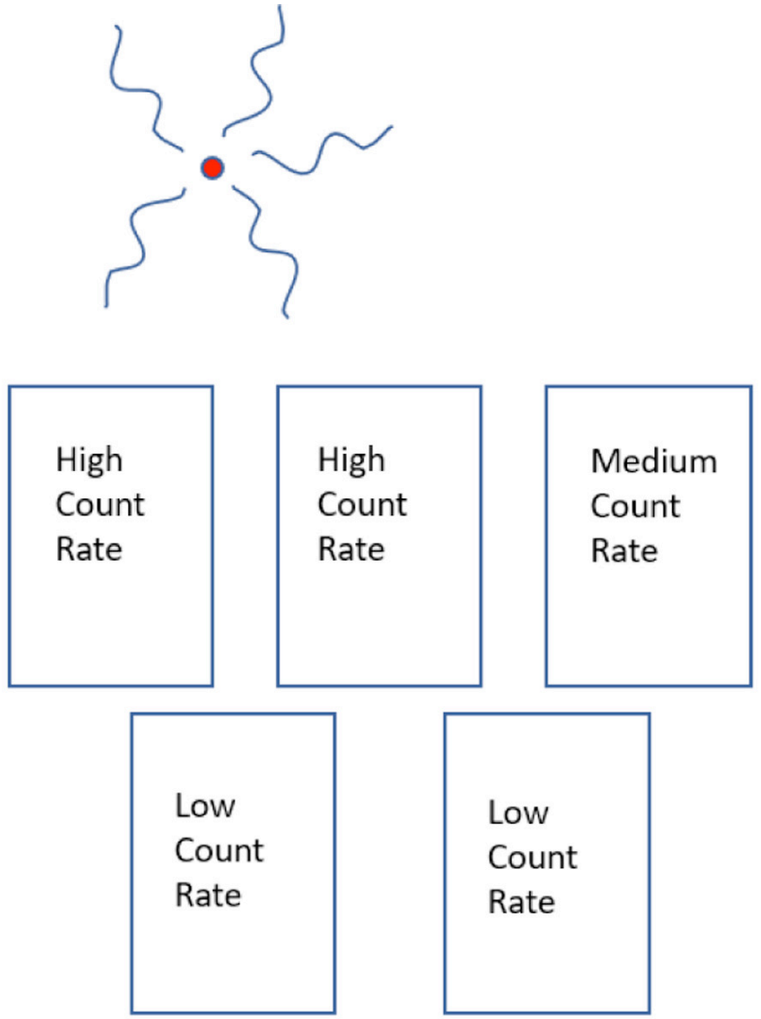
* Detectors with higher efficiency are available

Plenty of cryostat geometries available

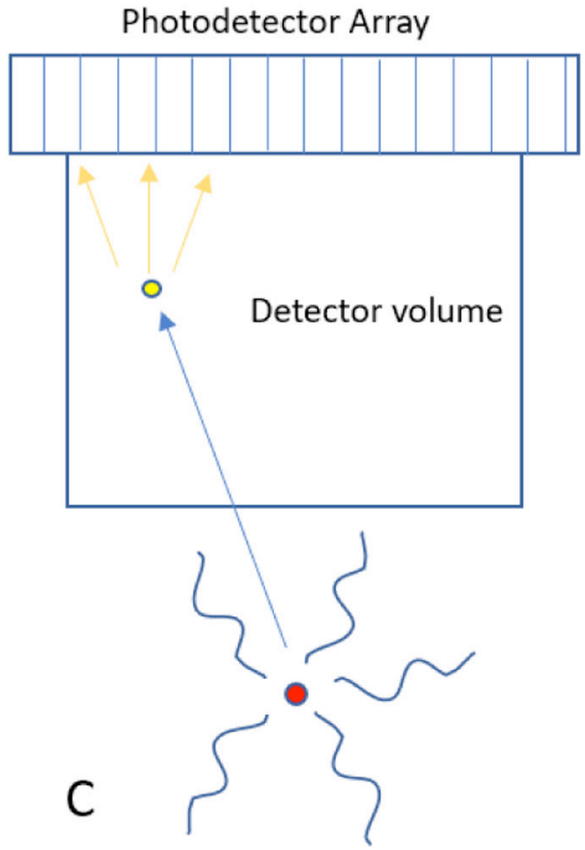




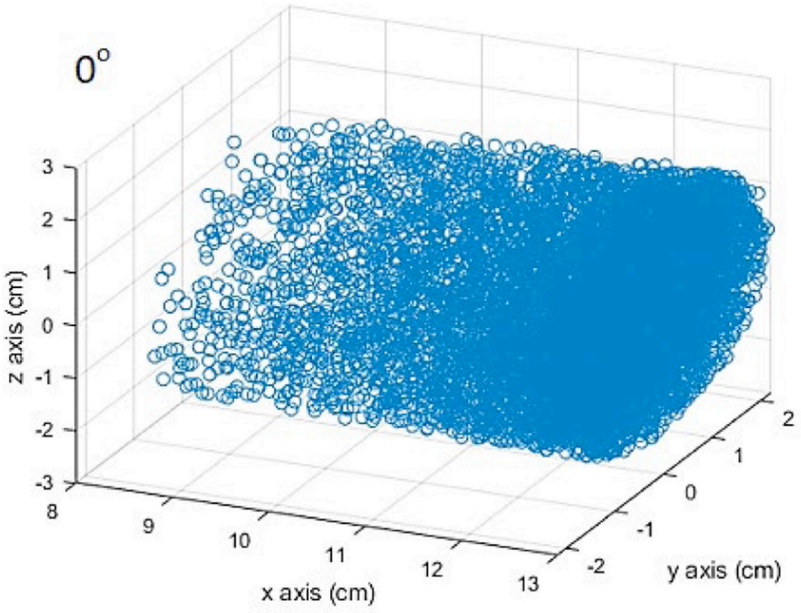
A

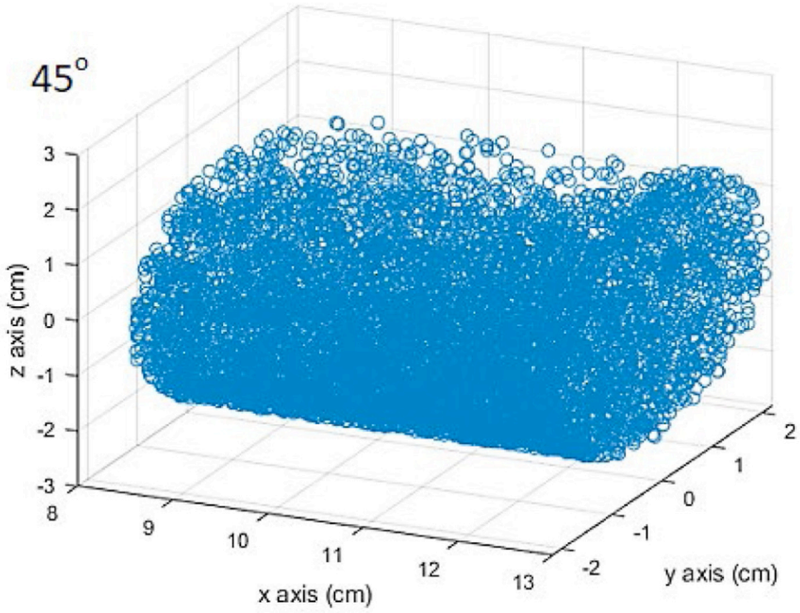


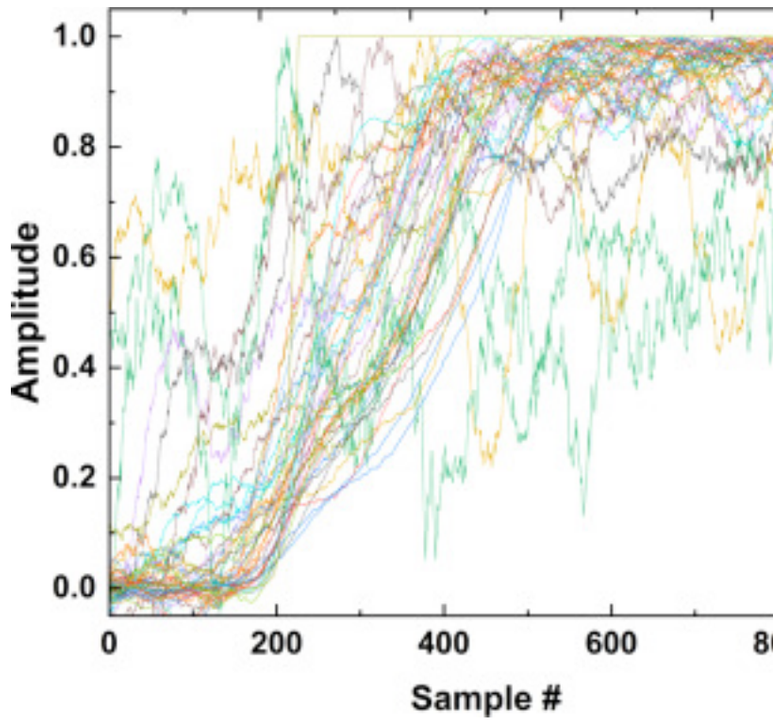
B Separated scintillator volumes

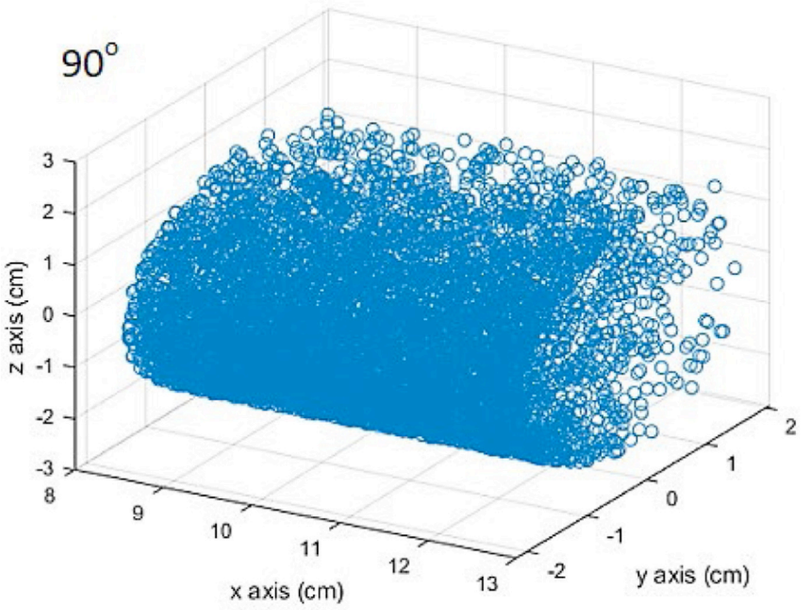


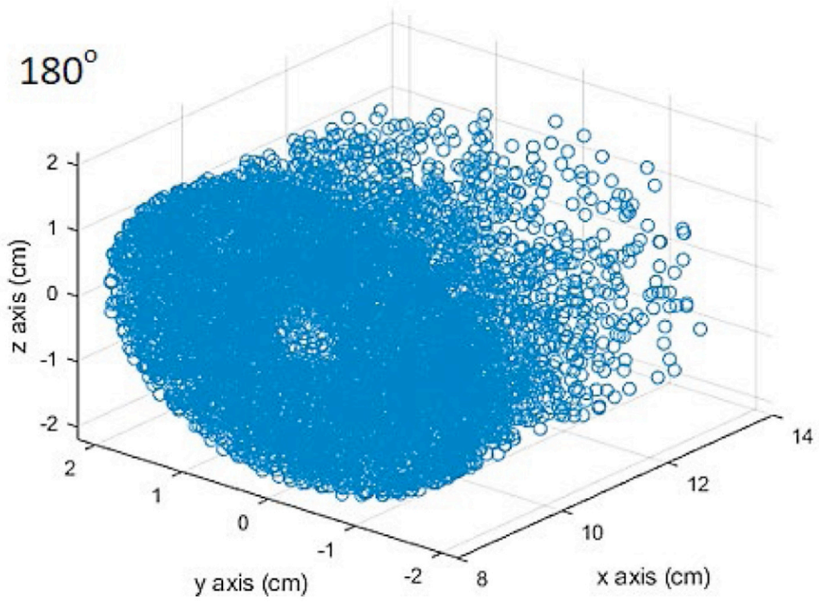
C

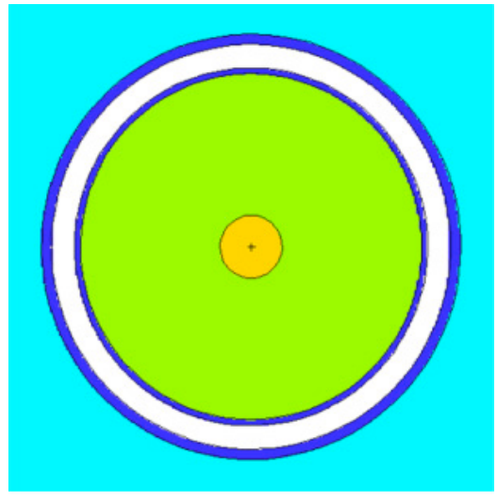

0°
z axis (cm)
x axis (cm)
y axis (cm)

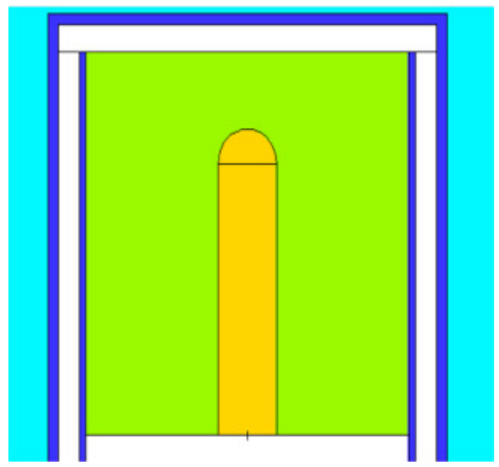

45°
z axis (cm)
x axis (cm)
y axis (cm)

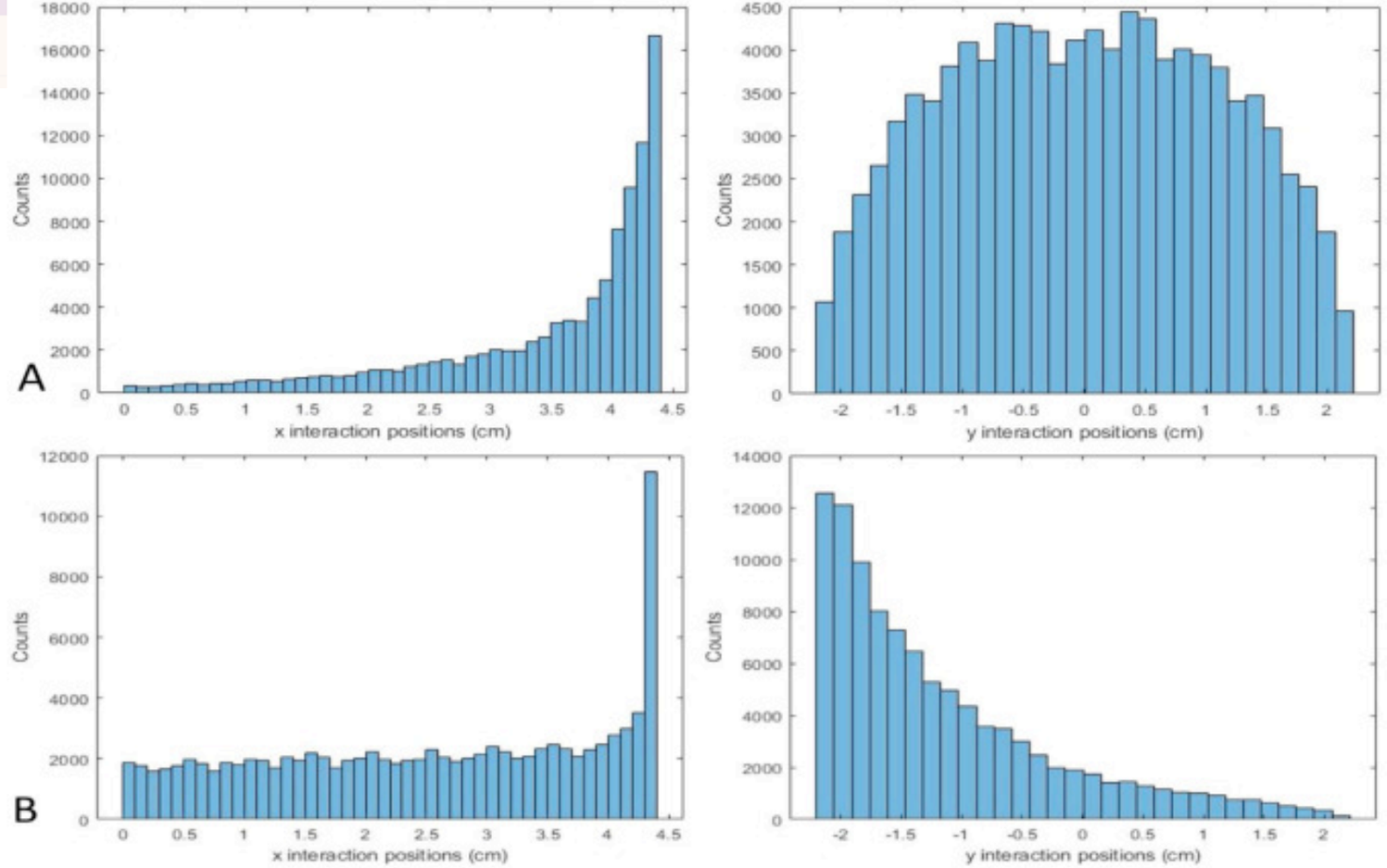

Amplitude
Sample #

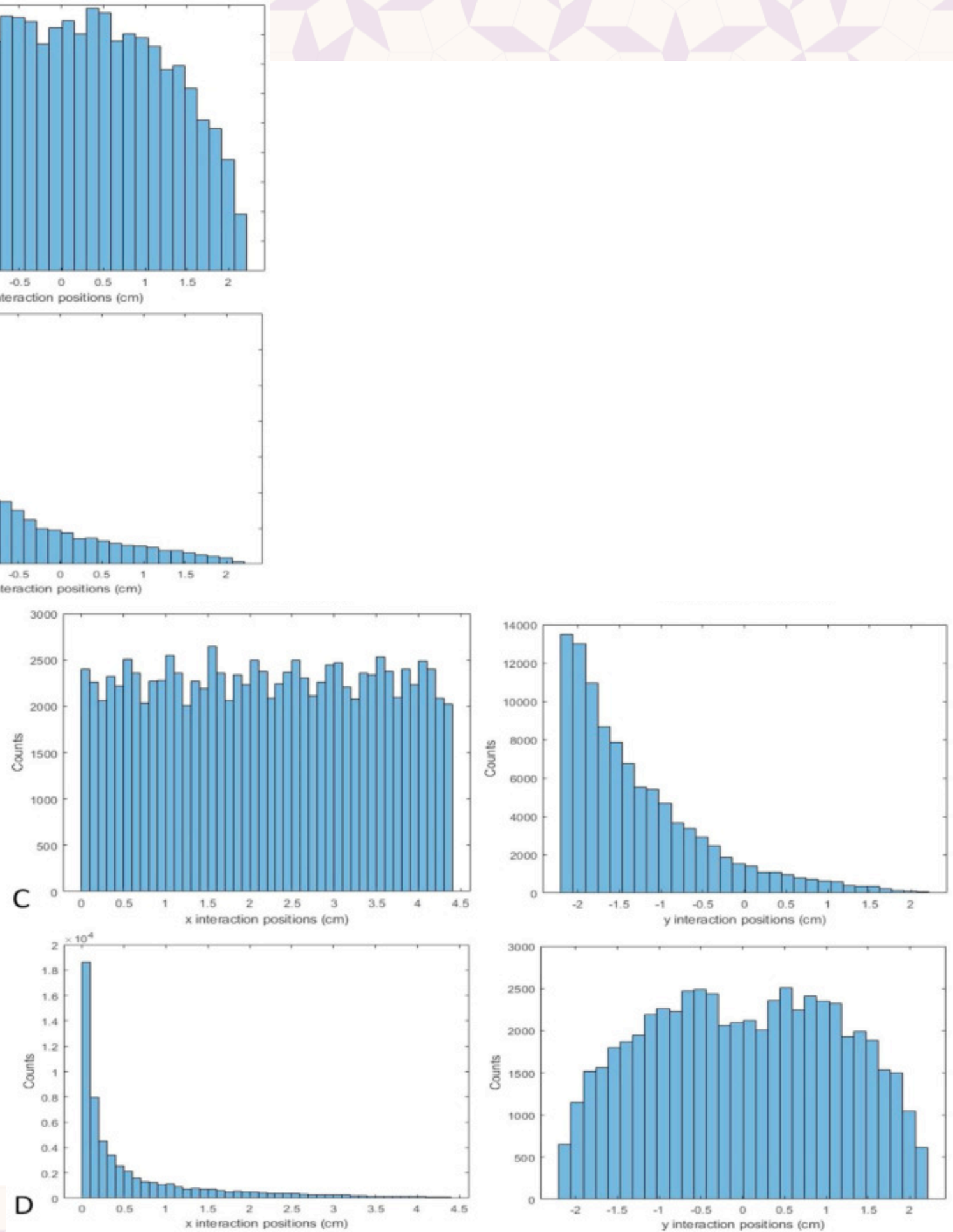

90°
z axis (cm)
x axis (cm)
y axis (cm)


180°
z axis (cm)
y axis (cm)
x axis (cm)

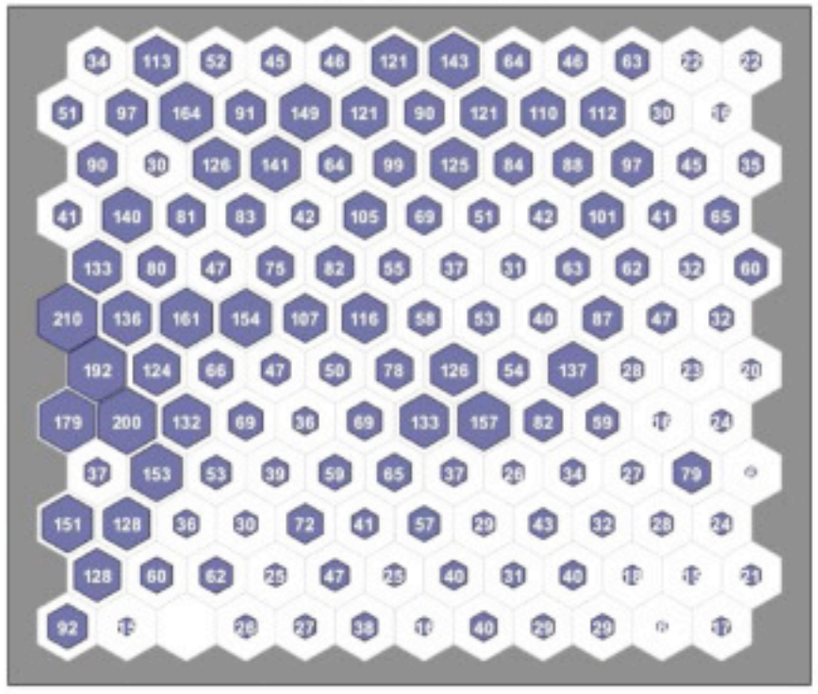




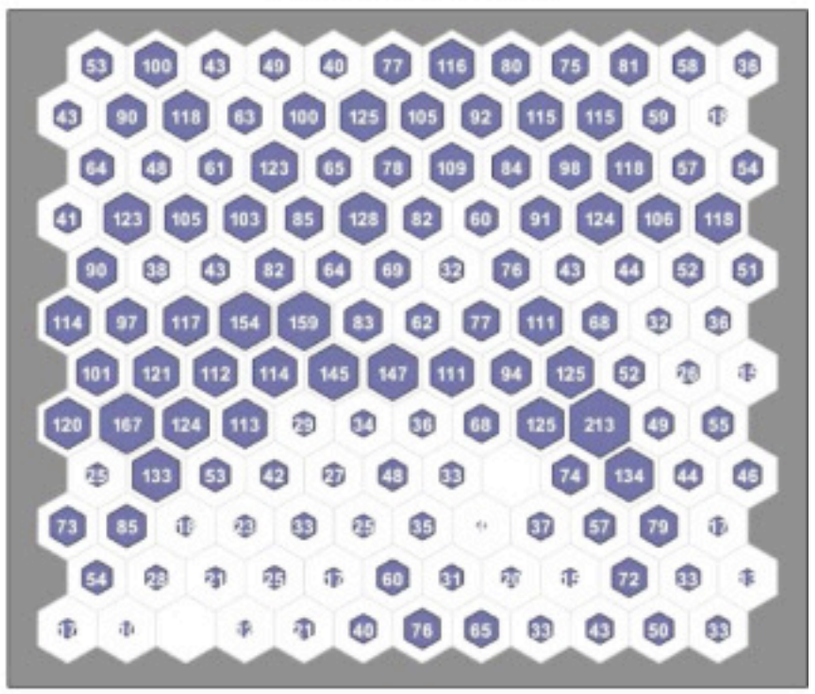

A
Counts
x interaction positions (cm)
y interaction positions (cm)
B
Counts
x interaction positions (cm)
y interaction positions (cm)


C
Counts
x interaction positions (cm)
y interaction positions (cm)
D
Counts
x interaction positions (cm)
y interaction positions (cm)

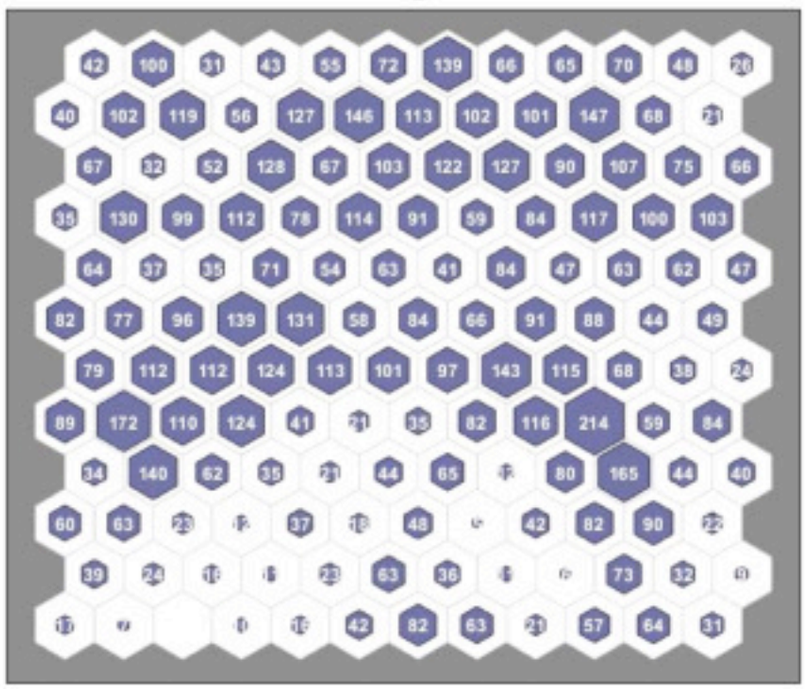
0 Degrees



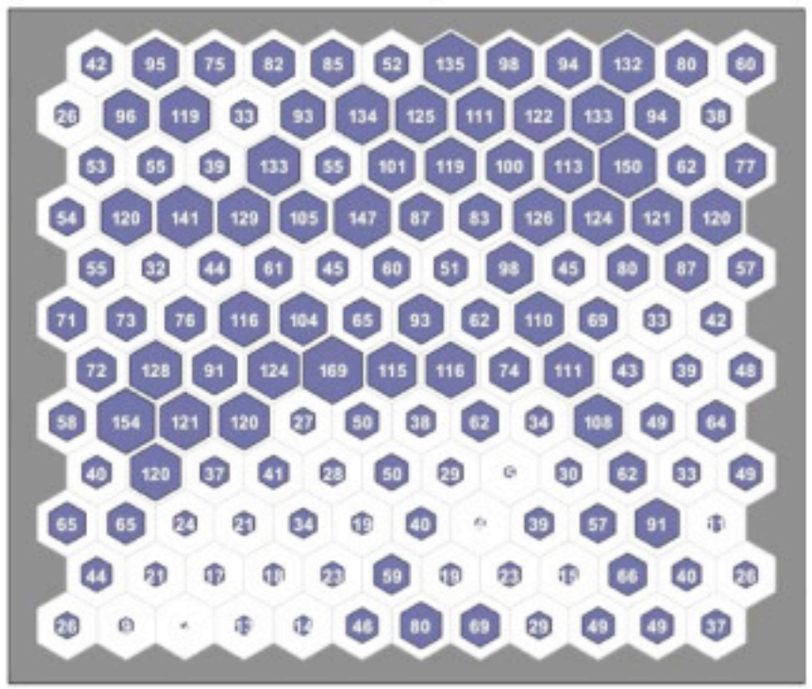
45 Degrees

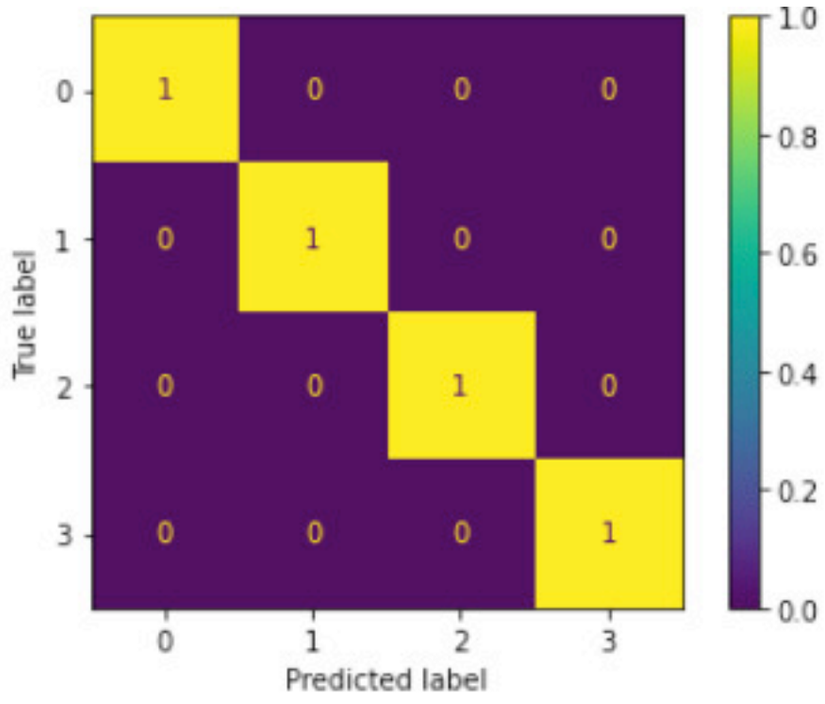


90 Degrees

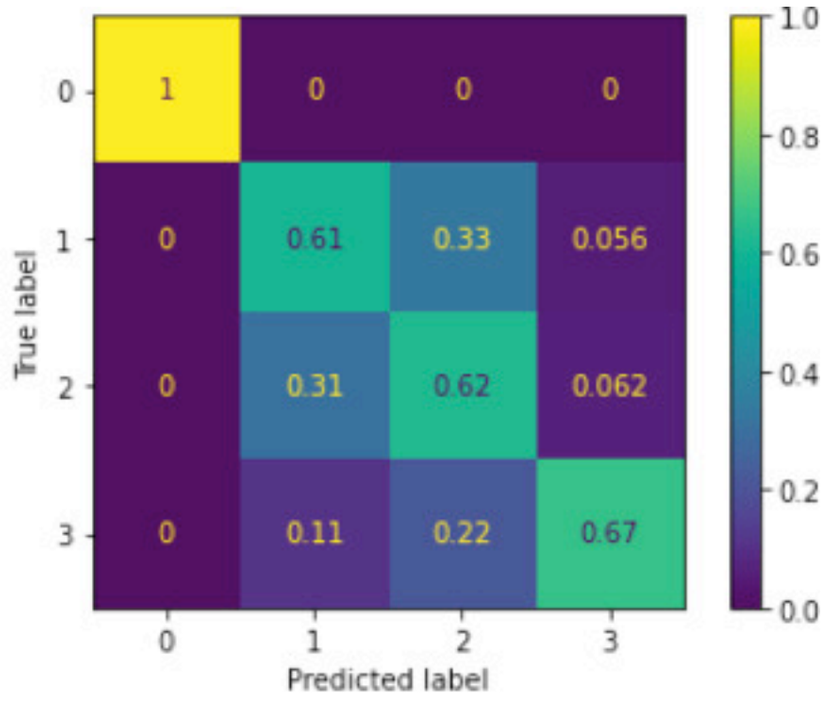


180 Degrees

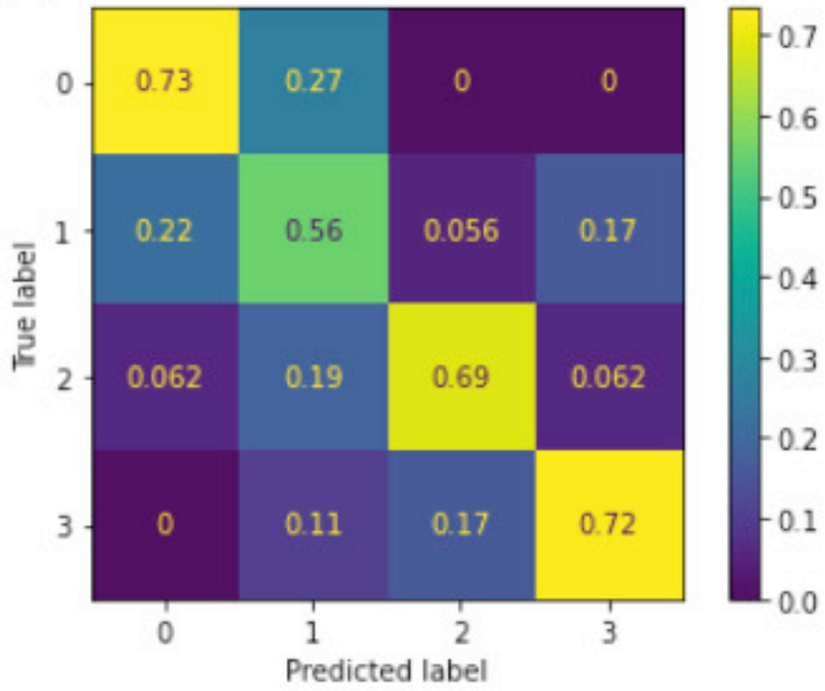



True label
Predicted label
0
1
2
3
1
0
0
0
0
1
0
0
0
0
1
0
0
0
0
1
1.0
0.8
0.6
0.4
0.2
0.0

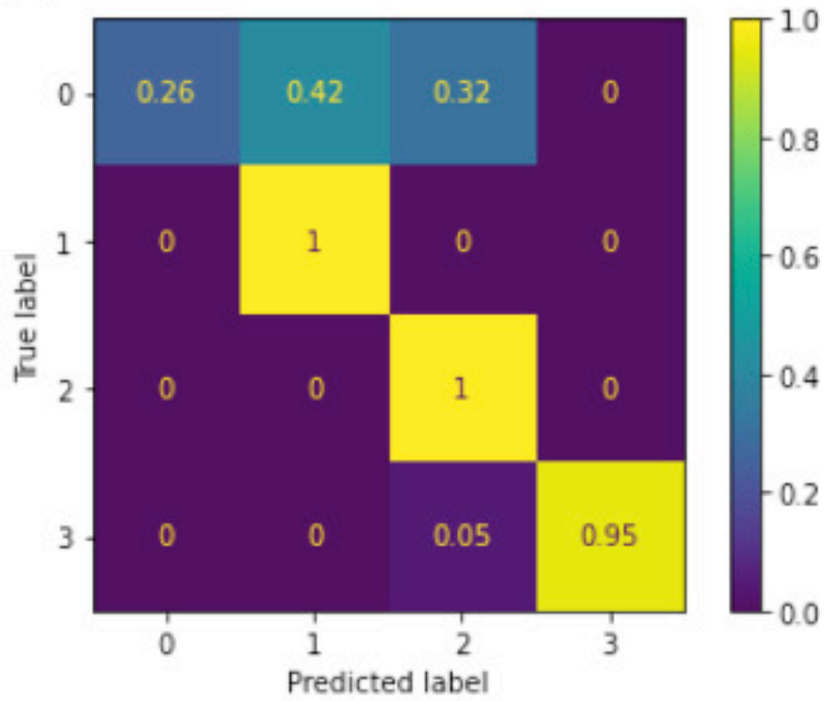
(a)


True label
Predicted label
0
1
2
3
1
0
0
0
0
0.61
0.33
0.056
0
0.31
0.62
0.062
0
0.11
0.22
0.67
1.0
0.8
0.6
0.4
0.2
0.0

(b)


True label
Predicted label
0
1
2
3
0.73
0.27
0
0
0.22
0.56
0.056
0.17
0.062
0.19
0.69
0.062
0
0.11
0.17
0.72
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

(c)


True label
Predicted label
0
1
2
3
0.26
0.42
0.32
0
0
1
0
0
0
0
1
0
0
0
0.05
0.95
1.0
0.8
0.6
0.4
0.2
0.0

(d)